

From 5G to 6G Networks: A Survey on AI-Based Jamming and Interference Detection and Mitigation

POONAM LOHAN¹ (Member, IEEE), BURAK KANTARCI¹ (Senior Member, IEEE),
MOHAMED AMINE FERRAG² (Senior Member, IEEE), NORBERT TIHANYI² (Member, IEEE),
AND YI SHI³ (Senior Member, IEEE)

¹School of Electrical Engineering and Computer Science, University of Ottawa, Ottawa, ON K1N 6N5, Canada

²Artificial Intelligence and Digital Science Research Center, Technology Innovation Institute, Abu Dhabi, UAE

³Bradley Department of Electrical and Computer Engineering, Virginia Tech, Blacksburg, VA 24061, USA

CORRESPONDING AUTHOR: B. KANTARCI (e-mail: burak.kantarci@uottawa.ca)

This work was supported in part by the Natural Sciences and Engineering Research Council (NSERC) Discovery Program; in part by the NSERC CREATE TRAVERSAL Program; and in part by the Innovation for Defence Excellence and Security (IDEaS) Program from the Department of National Defence (DND).

ABSTRACT Fifth-generation and Beyond (5GB) networks are transformational technologies to revolutionize future wireless communications in terms of massive connectivity, higher capacity, lower latency, and ultra-high reliability. To this end, 5GB networks are designed as a coalescence of various schemes and enabling technologies such as unmanned aerial vehicles (UAV)-assisted networks, vehicular networks, heterogeneous cellular networks (HCNs), Internet of Things (IoT), device-to-device (D2D) communication, millimeter-wave (mm-wave), massive multiple-input multiple-output (mMIMO), non-orthogonal multiple access (NOMA), re-configurable intelligent surface (RIS) and Terahertz (THz) communications. Due to the scarcity of licensed bands and the co-existence of multiple technologies in unlicensed bands, interference management is a pivotal factor in enhancing the user experience and quality of service (QoS) in future-generation networks. However, due to the highly complex scenarios, conventional interference mitigation techniques may not be suitable in 5GB networks. To cope with this, researchers have investigated artificial intelligence (AI)-based interference management techniques to tackle complex environments. Existing surveys either focus on conventional interference management methods or AI-based interference management only for a specific scheme or technology. This survey article complements the existing survey literature by providing a detailed review of AI-based intentional-interference management such as jamming detection and mitigation, and AI-enabled unintentional-interference mitigation techniques from the standpoints of UAV-assisted networks, vehicular networks, HCNs, D2D, IoT, mmWave-MIMO, NOMA, and THz communications. While identifying and presenting the AI-based techniques for interference management in 5G and beyond networks, this article also points out the challenges, open issues, and future research directions to adopt AI-enabled techniques to curtail the effects of interference in 5GB and towards 6G networks.

INDEX TERMS 5G, 6G, artificial intelligence, machine learning, jamming detection, jamming mitigation, interference mitigation, physical layer.

I. INTRODUCTION

THE THIRD-GENERATION partnership project (3GPP) introduced Release 15 of fifth-generation (5G) mobile communication in June 2018 with three main use cases, namely, enhanced mobile broadband (eMBB), massive machine type communication (mMTC), and ultra-reliable

low latency communication (URLLC) which laid the basis for worldwide deployment of commercial 5G networks [1].¹ In Release 15, a new 5G air interface, known as new

¹Table 1 provides the list of abbreviations used in this survey, accompanied by their corresponding explanations.

TABLE 1. List of abbreviations.

Abbreviation	Explanation	Abbreviation	Explanation	Abbreviation	Explanation
3GPP	Third Generation Partnership Project	GAN	Generative Adversarial Network	NSA	Non-Standalone
5G	Fifth-Generation	GMM	Gaussian Mixture Model	OAMP	Orthogonal Approximate Message Passing
6G	Sixth-Generation	GNN	Graph Neural Network	OFDM	Orthogonal Frequency Division Multiplexing
ABS	Almost Blank Subframe	GNSS	Global Navigation Satellite System	OMA	Orthogonal Multiple Access
ADMM	Alternating Direction Method for Multiplier	GRU	Gated Recurrent Unit	PA	Power Allocation
AI	Artificial Intelligence	GUE	Ground User Equipment	PD-NOMA	Power-Domain NOMA
AN	Artificial Noise	HBF	Hybrid Beamforming	PDR	Packet Delivery Ratio
AoA	Angle of Arrival	HCN	Heterogeneous Cellular Network	PPO	Proximal Policy Optimization
AoD	Angle of Departure	HetNet	Heterogeneous Network	PSO	Particle Swarm Optimization
AoI	Age of Information	HMM	Hidden Markov Model	PU	Primary User
AP	Access Point	I/Q	In-phase/Quadrature	QoS	Quality of Service
APC	Affinity Propagation Clustering	IA	Interference Alignment	RAN	Radio Access Network
APIM	Affinity Propagation Interference Management	ICIC	Inter-Cell Interference Coordination	RB	Resource Block
AR	Augmented Reality	IF	Isolation Forest	RedCAP	Reduced Capability
ASE	Aerial Spectrum Efficiency	IoD	Internet of Drone	RF	Random Forest
BER	Bit Error Rate	IoT	Internet of Things	RIS	Reconfigurable Intelligent Surface
BLE	Bluetooth Low Energy	IoTG	IoT Gateway	RL	Reinforcement Learning
BPR	Bad Packet Ratio	IoV	Internet of Vehicles	RMA	Random Multiple Access
BS	Base Station	ISI	Inter Symbol Interference	RNN	Recurrent Neural Network
BW	Bandwidth	IVD	Inter-Vehicle Distance	RSMA	Rate Split Multiple Access
CA	Carrier Aggregation	KMPA	K-means Position Adjustment	RSSI	Received Signal Strength Indicator
CCI	Co-Channel Interference	LLM	Large Language Model	RSU	Road Side Unit
CD-NOMA	Code-Domain NOMA	LoS	Line-of-Sight	SA	Spectrum Allocation
CNN	Convolutional Neural Network	LSTM	Long Short-Term Memory	SAC	Soft Actor-Critic
CRE	Cell range Expansion	LTE	Long-Term Evolution	SARSA	State-Action-Reward -State-Action
CSI	Channel State Information	MAB	Multi-Arm Bandit	SCMA	Sparse Code Multiple Access
CTI	Cross Technology Interference	MAC	Medium Access Control	SDR	Software Defined Radio
D2D	Device-to-Device	MALQL	Multi-agent Layered Q-Learning	SE	Spectral Efficiency
D3QN	Dueling Double Deep Q-Network	MARL	Multi-Agent Reinforcement Learning	SFW	Spatial Frequency Wideband
DBN	Dynamic Bayesian Network	MASC	Multiple Actor Shared Critic	SIC	Successive Interference Cancellation
DBSCAN	Density-based spatial clustering of application with noise	MD	Modulation	SINR	Signal to Interference plus Noise Ratio
DDPG	Deep Deterministic Policy Gradient	MDP	Markov Decision Process	SL	Supervised Learning
DDQN	Double Deep Q-Network	MEC	Mobile Edge Computing	SPS	Semi-persistent Scheduling
DL	Deep Learning	MFG	Mean Field Game	SS	Spectrum Sensing
DNN	Deep Neural Network	MFRL	Mean Field RL	SU	Secondary User
DSA	Dynamic Spectrum Access	MISO	Multiple Input Single Output	SVM	Support Vector Machine
DSC	Drone Small Cell	ML	Machine Learning	TD3	Twin Delayed DDPG
DSRC	Direct Short Range Communication	mMIMO	Massive Multiple Input Multiple Output	THz	Tera Hertz
DT	Decision Tree	MLD	Maximum likelihood Detector	UAV	Unmanned Aerial Vehicle
EE	Energy Efficiency	MMSE	Minimum Mean Square Error	UE	User Equipment
eMBB	Enhanced Mobile Broadband	mMTC	Massive Machine Type Communication	URLLC	Ultra-Reliable Low-Latency Communication
EOD	Ergodic Outage Duration	mmWave	Millimeter Wave	V2I	Vehicle to Infrastructure
EPM	Embedded Positioning Method	MRT	Maximum Ratio TX	V2N	Vehicle to Networks
ESN	Echo State Network	MUD	Multi-user Detection	V2P	Vehicle to Pedestrians
FDD	Frequency Division Duplex	NB-IoT	Narrow Band IoT	V2V	Vehicle to Vehicle
FL	Federated Learning	NE	Nash Equilibrium	V2X	Vehicle-to-Everything
FN	False Negative	NLoS	Non Line-of-Sight	VNF	Virtual Network Function
FP	False Positive	NN	Neural Network	VR	Virtual Reality
FrP	Fractional Programming	NOMA	Non-Orthogonal Multiple Access	XR	Extended Reality
GA	Genetic Algorithm	NR	New Radio	ZF	Zero Forcing

radio (NR) is identified which supports both non-standalone (NSA) and standalone operations. NSA operation in 5G communication relies on long-term evolution (LTE) for

mobility handling and initial access, while 5G NR is self-sufficient for standalone operation. NR key features include high-frequency operation (FR1: 410 MHz- 7.125 GHz;

FR2: 24.25 GHz- 52.6 GHz), spectrum flexibility, ultra-lean design to reduce interference and enhance energy performance, low latency support, forward compatibility, and advanced antenna technologies with beamforming design, among others [2]. Following this, 3GPP has introduced Releases 16 and 17 on evolving 5G technology to address new vertical use cases and enhance network performance. Release 16 addresses enhanced support of the critical Internet of Things (IoT) and URLLC, vehicle-to-everything (V2X) communications, relay-assisted communications, device-to-device (D2D) communications, positioning services, and communication in unlicensed spectrum. Similarly, Release 17 focuses on addressing support for reduced capability (RedCap) user equipment (UE), non-terrestrial networks such as Unmanned aerial vehicle (UAV)-assisted communications, multicast and broadcast services, and operations in beyond 52 GHz spectrum band [3].

Besides, 3GPP recently endorsed Release 18, introducing the features of 5G-advanced evolution [5], which focuses on enhanced network performance regarding coverage, capacity, and energy efficiency (EE) with new use cases such as eXtended Reality (XR). In Release 19, scheduled for 2024, this evolution continues and focuses on enhanced mobility support along with all objectives presented in Release 18. In Release 19, it is anticipated that massive antennas will play a crucial role in further improvement of network performance. To support eMBB use cases, 5G NR utilizes different frequency ranges, duplex schemes, multiple input multiple output (MIMO), and multi-carrier operations. In 5G Advanced, eMBB performance would be further enhanced. LTE and NR dynamic spectrum sharing enhancements in Release 18 facilitate an efficient switch from LTE to NR.

5G-Advanced Release 19 forms the basis for the sixth-generation (6G) communication system. After initial investigation and studies, the standardization process of the new 6G system is scheduled to align with Release 21. As per Ericsson's perspective and the 3GPP's initiative, the release of 5G-advanced is scheduled in 2024, and the fundamental components of 6G specification in Release 21 are scheduled for 2028, followed by ongoing 6G evolution [4]. 6G will bring together evolutionary and revolutionary wireless advances. 6G will provide an evolving path for fixed and mobile broadband characterized by widespread coverage, reduced latency, increased capacity, and higher throughput while supporting mission-critical services through improved reliability and availability. Furthermore, 6G will empower enhanced services and new use cases such as compute-AI services, spatiotemporal services, and immersive communication in the digital and virtual worlds, exploiting the full potential of the metaverse, including boundless XR, virtual reality (VR), and augmented reality (AR). 6G will derive the innovation and development for devices, experiences, and use cases that emerge around 2030 and beyond. The communication capabilities and use cases of 5G, 5G-advanced, and vision for 6G with evolutionary and revolutionary advances are shown in Fig. 1. The timeline,

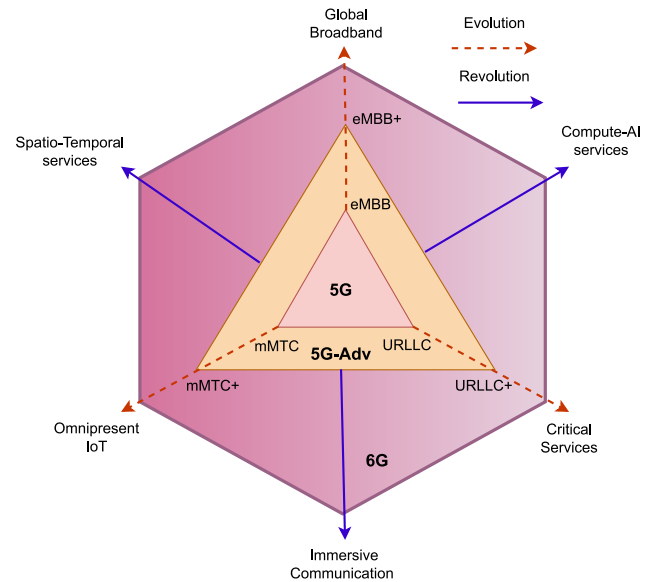


FIGURE 1. Communication capabilities of 5G, 5G-advanced, and 6G vision.

including the 3GPP 5G Releases and anticipated time plan for 6G advancements, is represented in Fig. 2.

Various advanced techniques and technologies are being investigated to fulfill the ever-evolving and revolutionary demands of modern communication facilitated by 5GB networks. Network densification through deploying heterogeneous cellular networks (HCNs) (incorporating a mix of macro, small, micro, and pico cells), spectrum sharing, beamforming, and massive MIMO are technical solutions that enhance coverage capacity, spectral efficiency, and communication reliability. However, effective interference management is essential to ensure their successful implementation and operation. Interference can result in dropped calls, reduced data rates, and deteriorated overall user experience. To meet the high standards of users' expectations in 5GB networks, interference management is crucial for ensuring secure and reliable connectivity. In the following subsection, we explore the rationale for using artificial intelligence (AI) and machine learning (ML)-based approaches for interference management in 5GB and towards 6G networks.

A. RESEARCH MOTIVATION: ROLE OF AI/ML IN 5G-ADVANCED AND 5G EVOLUTION TOWARDS 6G

The complexity of continuously evolving wireless communication technologies beyond 5G is increasing as they support diverse deployment scenarios and applications. Integrating AI and ML techniques into the evolution of 5G is critical for effectively handling this complexity and propelling 5G-advanced evolution towards 6G. 3GPP Release 18 provides details of AI/ML-enabled 5G NR air interface design to reduce complexity and enhance network performance for specific use cases such as enhanced channel state information (CSI) feedback, beam management, and positioning accuracy. It also specifies enhanced data collection

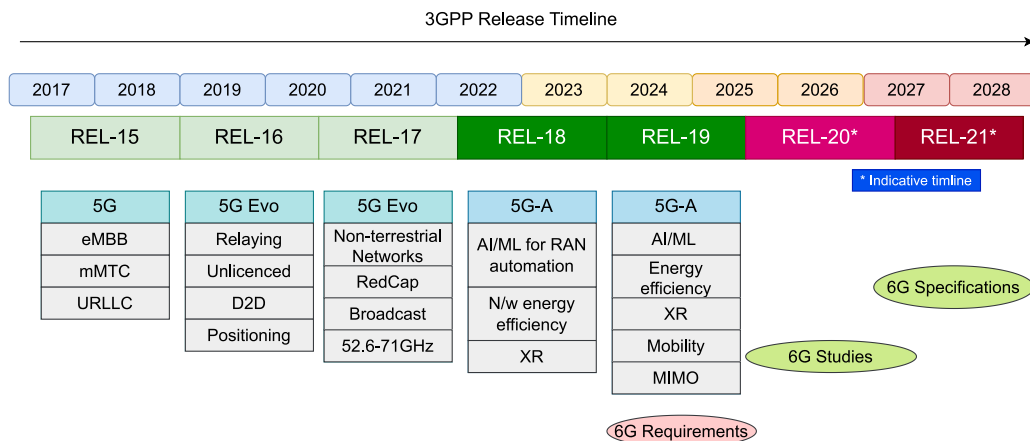


FIGURE 2. 3GPP timeline for 5G advance and 6G based on [4].

and signaling support for AI/ML-based load balancing, mobility optimization, and network energy-saving [6]. AI/ML-powered end-to-end (E2E) communication is a key enabler for removing the barriers toward 6G where E2E information and application of dynamic distributed computing infrastructure (e.g., cloud, edge, devices) for new capabilities will pave the way for enhanced performance of networks [7]. In line with this, extensive research is actively underway, led by dedicated researchers, to explore the application of AI/ML techniques to next-generation radio access networks (physical layer techniques, radio resource management/control, medium access control), system architecture aspects (management, orchestration, charging), and core network (network slicing, service function chain (SFC) management, virtual network function (VNF) placements, etc.) [8].

To review interference detection and mitigation in 5G and 6G networks, this survey focuses on AI/ML-enabled radio access networks (RAN). As 5GB networks are delineated to support various services and deployment scenarios, from autonomous vehicles and augmented reality to massive IoT deployments, interference issues become increasingly complex. Conventional interference mitigation techniques such as frequency reuse, spatial reuse, power control, channelization, equalization, interference cancellation, and adaptive beamforming that are commonly used in traditional wireless communication systems may not be suitable for handling unique challenges posed by emerging enabling technologies of 5GB networks. The distinctive attributes of 5GB networks, including massive MIMO, mmWave, THz frequencies, Hybrid beamforming, dynamic spectrum access, advanced modulation schemes, heterogeneous deployments, UAV-assisted communication V2X communication, and interference from non-cooperative sources, present challenges that conventional interference mitigation techniques may not effectively address. Hence, it is essential to design customized interference mitigation strategies that deal with the specific requirements of 5GB networks to optimize

performance and ensure URLLC communication. These problems encompass continuously changing interfering conditions, spectrum management, coexistence challenges, and dynamic resource allocation. AI/ML techniques, with their situational awareness, complexity handling, data-driven, predictive, adaptive, and automation proficiency, have been investigated to tackle these complex issues. AI/ML algorithms can dynamically allocate resources, adjust network parameters, and predict interference patterns in real-time for efficient and smarter interference management/mitigation, ensuring that wireless networks operate seamlessly in the presence of intentional or unintentional interference sources. Therefore, this survey covers various “Beyond 5G” technologies such as UAV-based networks, vehicular networks, heterogeneous networks, D2D networks, IoT networks, mmWave-MIMO networks, THz Communication networks, and Non-Orthogonal Multiple Access (NOMA)-based communication networks providing an exhaustive and detailed review of research works on AI/ML-enabled intentional/unintentional interference management techniques in these domains.

Before delving into the survey content, we provide a concise introduction to various ML techniques. ML, a major field of AI, enables network nodes and devices to perform tasks without explicit programming. ML algorithms can generally be categorized into three types [9]: Supervised Learning (SL), Unsupervised Learning, and Reinforcement Learning (RL) [10]. SL algorithms use labeled training data to build models for predicting outputs from input features. Examples include classification (e.g., KNN, Decision Tree, Random Forest, SVM) and regression (e.g., Linear, Polynomial, Exponential, Logistic). Unsupervised Learning algorithms detect patterns in unlabeled data, including clustering (e.g., K-means, DBSCAN, Gaussian Mixture) and dimensionality reduction. RL enables nodes to learn control strategies by interacting with the environment to maximize rewards. Examples include policy optimization (e.g., DDPG, TD3, PPO, SAC, A2C) and Q-learning (e.g., DQN, SARSA). Also,

TABLE 2. Comparison of our tutorial on AI/ML-enabled intentional and unintentional interference management with related existing survey and tutorial works (Not considered ○; partial discussion ◐; considered ●).

Ref.	Year	AI/ML-enabled	Intentional Jamming	UAV	IoV/V2X	HetNets	D2D	IoT	M-MIMO	THz comm.	NOMA	OFDM
[12]	2013	○	○	○	○	○	○	○	○	○	○	●
[13]	2013	○	○	○	○	○	○	○	○	○	○	●
[14]	2013	○	○	○	○	○	○	○	○	○	○	●
[15]	2015	○	○	○	○	●	○	○	○	○	○	○
[16]	2015	○	○	○	○	●	○	○	○	○	○	○
[17]	2016	○	○	○	○	●	●	○	●	○	○	○
[18]	2016	○	○	○	○	●	●	○	●	○	○	●
[19]	2017	○	○	○	○	○	○	○	●	○	●	○
[20]	2017	○	○	○	○	○	○	○	●	○	○	○
[21]	2018	○	○	○	○	○	●	○	○	○	○	○
[22]	2019	○	○	○	○	○	○	○	●	○	○	○
[23]	2019	●	○	●	○	○	○	○	○	○	○	○
[24]	2020	◐	●	●	○	○	○	○	○	○	○	○
[25]	2021	◐	○	○	○	●	●	●	○	○	○	○
[26]	2021	●	○	○	○	○	●		●	○	○	○
[27]	2021	◐	○	○	○	●	○	○	○	○	●	●
[28]	2021	●	○	○	●	○	○	○	○	○	○	○
[29]	2022	◐	○	●	○	●	●		●	○	○	○
This survey	2024	●	●	●	●	●	●	●	●	●	●	●

Deep Learning (DL) is a subset of ML that leverages multi-layer Artificial Neural Networks. These methods can be employed in SL, Unsupervised Learning, and RL scenarios. DL encompasses various architectural models, including Deep Neural Networks (DNN), Deep RL (DRL), Recurrent Neural Networks (RNN), Convolutional Neural Networks (CNN), Autoencoders, and Transformers [11].

B. PREVIOUS INTERFERENCE MANAGEMENT SURVEY WORKS

As reported in Table 2, some surveys and tutorials have been published on conventional interference management techniques such as [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]. In [12], cellular, ad-hoc, and infrastructure-based platforms are considered with single and multiple antenna OFDM systems. The study reviews successive interference cancellation (SIC) techniques and compares the effectiveness of various solutions in terms of bit error rate (BER) performances and computational complexity. In [13], a multi-cell OFDMA-based system is considered and the survey focuses on interference avoidance via inter-cell interference coordination (ICIC) such as fixed, adaptive, and real-time techniques with selective frequency reuse, selective power reuse, and selective invert power frequency reuse. Similarly, in [14], the authors survey the conventional frequency reuse, fractional frequency reuse, and cell coordination-based ICIC techniques in OFDMA-based cellular networks. Further, cell coordination is classified as centralized, semi-centralized, coordinated, distributed, and autonomously distributed. In [15], cognitive femtocells in

HCNs are considered, where different cognitive interference mitigation schemes are surveyed, such as power control, spectrum access, and antenna techniques with varying scenarios of deployment for femtocells such as dedicated channels, co-channel, open access, closed access, fixed downlink power, and adaptive downlink power. Similarly, in [16], to control the inter-tier interference among low power nodes (small cells) and high power nodes (macro cells) and intra-tier interference in low power nodes (small cells) in HCNs, different interference control schemes are surveyed such as spatial interference coordination at the transmitter and interference cancellation at the receiver.

In [17], to control the interference, interference alignment (IA) techniques are surveyed in different dimensions such as time, frequency, and space, and its application is discussed in different domains such as HetNets, D2D, cognitive radio, and cellular networks. However, the main challenges in the practical adoption of IA schemes are also discussed, such as the high overhead of CSI feedback and poor performance in low SNR scenarios. In [18], a survey of different radio interference and resource management techniques (spectrum- and energy-efficient) techniques is conducted. Specifically, interference management techniques focus on power control, coordinated multi-point communication, and MIMO-based interference mitigation techniques in the D2D and HetNets domains. In [19], the survey work is focused on a radio access technique, i.e., power-domain NOMA. NOMA seems to be a very promising access technique for future technologies. This survey paper provides detailed information on power allocation techniques, capacity analysis, and

user-pairing methods in the power-domain NOMA, where superposition coding is used at the transmitter, and SIC is utilized at the receiver. In [20], the multi-user MIMO (MU-MIMO) system is considered, and a detailed survey is provided for resource allocation techniques in the downlink transmission (TX) of the MU-MIMO system. Different precoding techniques and TX technologies exploiting spatial dimensions are surveyed precisely. Also, different methodologies are reviewed to optimize user selection jointly, the number of users and antennas, spatial signaling, and power allocation to maximize the performance and mitigate the interference in downlink MU-MIMO communication. In [21], a detailed overview of interference management and resource management in in-band and out-band D2D communication is provided. In [22], the authors thoroughly review interference management schemes in MIMO-based WLAN systems through the MAC and physical layers' mechanism design.

It is to be noted that all the above studies focus on conventional interference management techniques. However, in the current era of AI and ML, it is highly necessary to develop intelligent techniques for interference and network management to deal with the complex environment of next-generation networks. In [30] and [31], the authors provide a comprehensive review of methodologies applying deep learning to enhance the performance of wireless networks. In [31], the authors focus more on using artificial neural networks (mainly spiking, deep, and recurrent neural networks) in the domains such as UAV-based communication, virtual reality, IoT, and cache computing. However, these works are not focused on interference mitigation techniques. In [24], the authors provide a detailed review of intentional interference mitigation works utilizing ML techniques in the Global Navigation Satellite System (GNSS) for unmanned and manned aerial vehicles. There are some survey works [25], [27], [29], [32] that focus on ML techniques partially along with the conventional interference mitigation techniques with their scope to some fixed wireless applications such as IoT [25], D2D [25], [29], UAV [23], [29], HetNets [25], [27], [29], MIMO [29], and radar communication [32]. Further, we can find a detailed survey on AI-based technologies for resource allocation and interference management in the Internet of Vehicles (IoV) [28] and D2D communication with mmWaves [26]. In [33], the authors use DL techniques to focus on interference mitigation in radar communication. In comparison to existing surveys and tutorials, which mainly focus either on conventional interference mitigation techniques [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22] or on a few wireless applications with partial or full attention on AI/ML-enabled resource and interference management techniques [23], [24], [25], [26], [27], [28], [29], [32], [33], in this tutorial, to cover the literature gap we provide an in-depth literature review of AI/ML-enabled intentional and unintentional interference mitigation methodologies in vast domains of 5G and beyond

networks such as UAV-based communication, Vehicular networks (V2V and V2X), Hetnets, D2D, IoT, mmWave communication, MIMO, NOMA, and THz communication.

As summarized in Table 2, the only survey that reviews AI/ML-enabled solutions for intentional jamming is presented in [24]; however, the survey is limited to UAV communications. Thus, this work bridges the gap in the literature by not only covering AI/ML-enabled interference management solutions against intentional jamming but also reviewing the solutions by considering the Beyond-5G enabling technologies listed in the table, including UAVs, IoVs/V2X, HetNets, D2D, IoT, mm-MIMO, THz communications, NOMA and OFDM.

C. CONTRIBUTIONS

The main objective of this survey work is to provide a comprehensive review of the AI/ML-enabled techniques for interference mitigation in Beyond-5G networks. Here, we provide an in-depth tutorial on the recently emerging research directions and contributions in AI/ML-enabled interference management of next-generation wireless networks. It also addresses the major advantages and challenges of developing and deploying AI/ML-based interference mitigation techniques in different domains and technologies. *To the best of our knowledge, this is the first survey that assembles the recent state-of-the-art research advances in the use of AI/ML-based techniques for both intentional and unintentional interference management in different domains of Beyond-5G networks towards 6G (see Table 2).* The main contributions of this survey are summarized as follows:

- 1) A two-fold study of interference management based on the physical-layer perspective: intentional and unintentional interference. For intentional interference management, first, we provide the key characteristics of different types of jamming attacks and then provide a detailed review of a wide range of works that use AI-based techniques for jamming detection and mitigation in various wireless communication scenarios.
- 2) An in-depth investigation of the main features and types of unintentional interference affecting the performance of various schemes and enabling technologies of 5G and beyond network, followed by an extensive review of AI-based interference management works in different domains such as UAV-assisted networks, Vehicular Networks, Heterogeneous Networks, Device-to-Device (D2D) communication, Internet of Things (IoT), millimeter-wave (mm-wave) Massive Multiple-Input Multiple-Output (mMIMO), THz communications, and Non-Orthogonal Multiple Access (NOMA) networks. We provide a summary and comprehensive taxonomy of the network systems considered, objectives, and the ML-based approaches used to implement different wireless techniques to mitigate interference and achieve the set objectives. Further, we provide a qualitative comparison of the

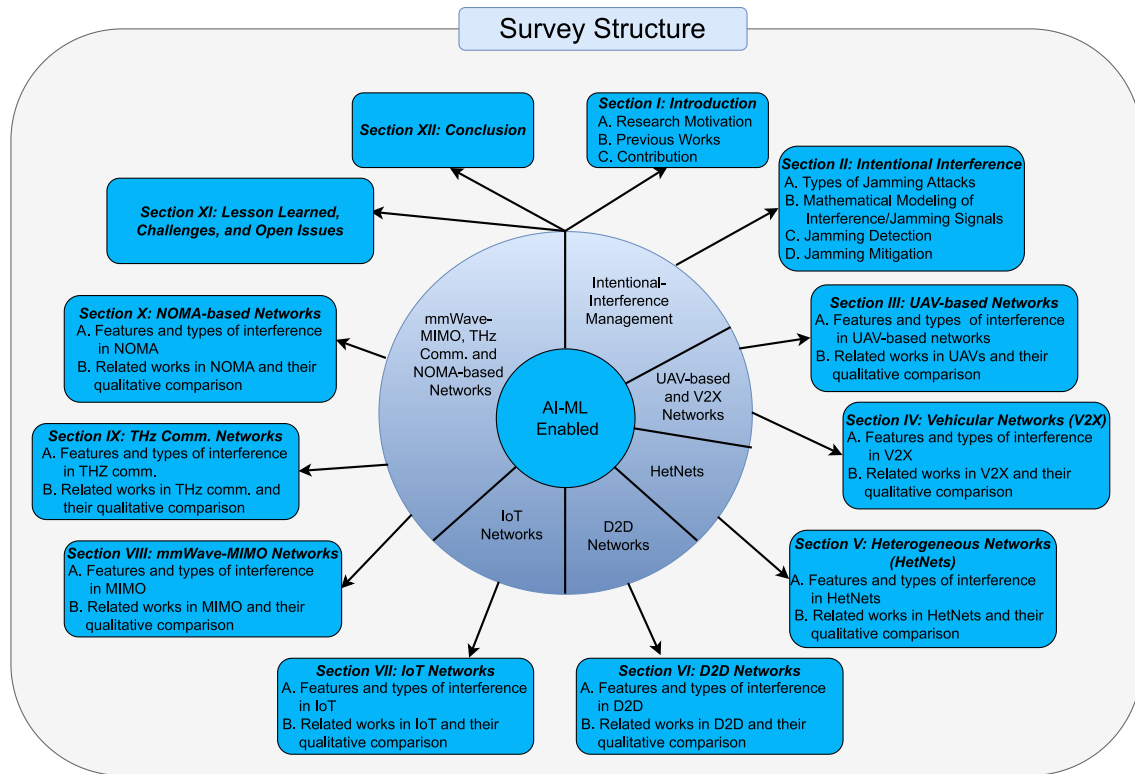


FIGURE 3. Structure of the survey.

different ML-based approaches regarding performance and complexity.

- 3) Unveiling the challenges associated with developing and deploying AI/ML-based interference mitigation techniques in different enabling technologies of 5G and beyond networks at the dawn of 6G. We also elucidate the potential future directions for using AI/ML-based techniques to enhance the performance of future-generation wireless networks.

The rest of this survey is organized as per the high-level structure presented in Fig. 3. Section II focuses on intentional interference, i.e., different types of jamming attacks from a physical layer perspective, and provides a detailed review of AI/ML-enabled jamming detection and jamming mitigation techniques. Section III onwards, this tutorial focuses on unintentional interference, providing the features, types of interference, and AI/ML-enabled techniques to mitigate that interference in different 5G and beyond enabling technologies. Section III begins with UAV-assisted wireless networks and reviews recently developed AI/ML-enabled interference mitigation techniques in this domain. Section IV covers the Vehicular networks, and Section V focuses on Heterogeneous Networks. Then, Sections VI and VII discuss the intelligent interference mitigation in D2D and IoT Networks, respectively. Further, Section VIII sheds light on AI/ML-enabled interference management in the mmWave-MIMO networks, and Section IX does the same for THz communication networks. Section X demonstrates

the reviews for NOMA-based networks. Then, Section XI describes the lessons learned, challenges, and future direction in utilizing AI/ML-enabled techniques for performance enhancement in next-generation wireless networks. Finally, Section XII provides the concluding remarks of the study.

II. AI-ENABLED INTENTIONAL INTERFERENCE MANAGEMENT IN 5G AND BEYOND NETWORKS

This section comprises three subsections. The first subsection characterizes the various categories of potential jamming attacks at the physical and medium access control (MAC) layer. The second subsection focuses on works centered on AI-enabled intentional interference detection, and the last subsection showcases the research works employing AI-enabled techniques for intentional interference mitigation across diverse domains of 5G and beyond networks.

A. DIFFERENT TYPES OF JAMMING ATTACKS AT PHYSICAL AND MAC LAYERS

A jamming attack specific to physical or MAC layers emits interference in radio signals as noise to disturb the communication between transmitter and receiver. Jamming deteriorates the SNR and makes it difficult to retrieve the information from the received signal [34]. Jamming is also called intentional interference created against a specific target by the jammer. In contrast, unintentional interference is generated due to the use of the same frequency bands by nearby transmitters. Jamming attacks can be classified into

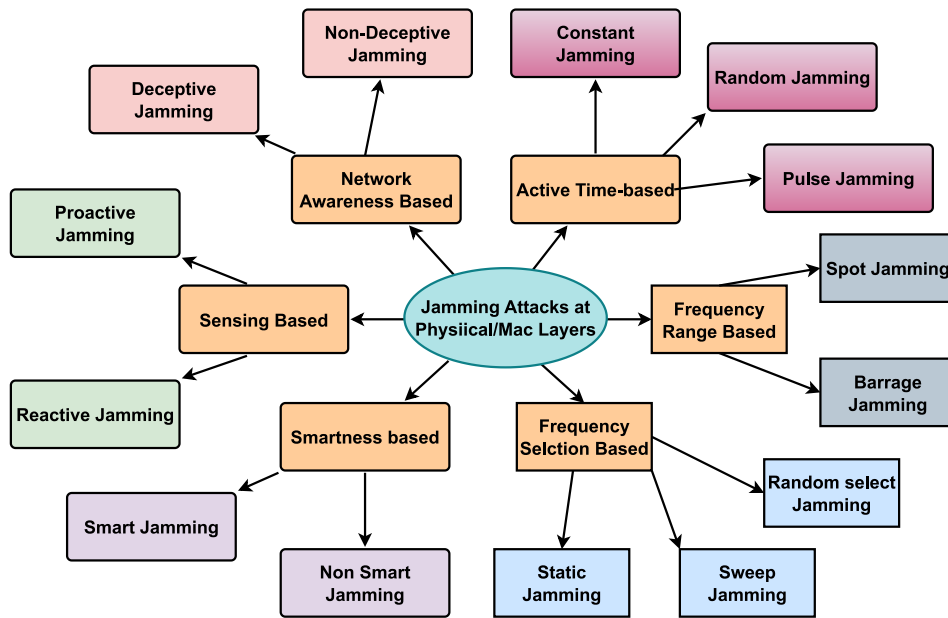


FIGURE 4. Types of jamming attacks at physical and MAC layers.

different categories, which are presented in Fig. 4 and are explained as follows:

- 1) *Sensing-based jamming attacks*: In this category, jamming attacks are divided into two types: reactive jamming and proactive jamming [35]. The reactive jammer first senses the activity on the channel and then activates in response to the detection of certain events or signals, such as initiating communication over the channel. Reactive jamming is normally used to avoid continuous TX and minimize the detection risk and energy consumption [36], [37]. In proactive jamming, the jammer transmits interfering signals irrespective of ongoing communication. Proactive jamming disrupts communication more generally and is effective against communication systems with predictable patterns and constant TX characteristics.
- 2) *Network-awareness jamming attacks*: Deceptive and non-deceptive are two classifications of jamming attacks in this category. In deceptive jamming, based on network awareness, the jammer transmits signals that copy the protocol, format, and characteristics of the legitimate communication signals [36]. A deceptive jammer aims to make the target system accept the jamming signal as authentic. On the other hand, a non-deceptive jammer interferes with communication signals to disrupt the communication without aiming to deceive the target system.
- 3) *Active-time-based jamming attacks*: In this category, jamming attacks are classified into three types: constant, random, and pulse jamming [35]. A constant jammer transmits continuous and steady interfering signals on the radio frequency the target system uses for communication. This type of jammer aims

to maintain constant interference on a frequency and disrupt devices' communication operating on that frequency [38]. It is worth noting that constant jamming is a type of proactive jamming. Similarly, random jamming can also be proactive because it does not listen to ongoing communication over the channel before transmitting the jamming signals for random intervals. A random jammer transmits interfering signals that randomly change the frequency and timing of TX. Random jamming makes it difficult for the target system to predict or detect the jamming signal and counteract the random interference [38]. In pulse jamming, the jammer transmits intermittent interference pulses instead of continuous TX. The jamming signal occurs in periodic bursts, creating confusion for the target to establish reliable communication. Pulse jamming avoids easy detection and is an energy-efficient jamming technique that transmits periodically, not constantly.

- 4) *Frequency Range-based jamming attacks*: Spot and barrage jamming come under this category. In spot jamming, the jammer transmits all its power on a single frequency at a time to disturb the target communication. It is a very powerful jamming technique but can be easily avoided by continuously changing communication frequency. In barrage jamming, a range of frequencies are jammed at once instead of a single frequency to decrease the SNR of the target system [34]. However, as the jammer divides its power in the range of the frequencies, the output power of the jamming is reduced, and jamming becomes less effective.
- 5) *Frequency selection-based jamming attacks*: Based on the frequency selection, jamming can be classified

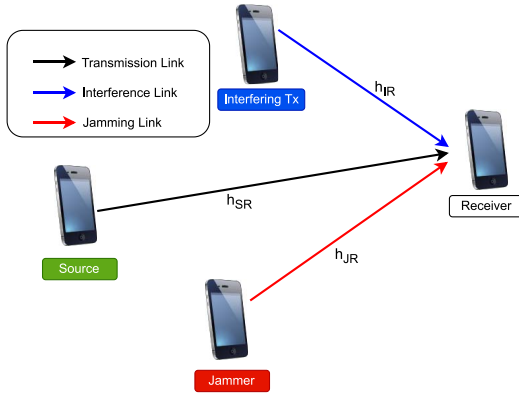


FIGURE 5. Simple Scenario to Model Jamming and Interfering Signals.

into three types: static, sweep, and random select jamming. Static jammer transmits interfering signals continuously on a single frequency or over a range of frequencies and effectively blocks communication. However, static jamming can be easily detected compared to dynamic jamming techniques. Sweep jamming rapidly shifts the frequency of the jamming signal over a range of frequencies and can jam multiple frequencies in succession [39]. Sweep jamming can overcome countermeasures that might be effective against static jamming. In random select jamming, a jammer transmits interfering signals randomly on different frequencies. It is challenging to counteract random select jamming because of its unpredictable nature.

- 6) *Smartness-based jamming attacks*: Smart and non-smart jamming, in this category, refer to a level of intelligence and adaptability a jammer applies in creating interference signals [40]. Smart jamming uses intelligence and advanced techniques to adapt and optimize the jamming process and minimizes the chances of detection and counteraction [35]. Non-smart jamming uses simple and static methods of interference without any intelligence and adaptability used for jamming.

B. MATHEMATICAL MODELLING OF INTENTIONAL/JAMMING AND UNINTENTIONAL INTERFERING SIGNALS

To present the mathematical modeling of jamming and interfering signals, a simple scenario is considered with a source and receiver pair, one jammer, and one interfering transmitter, as illustrated in Figure 5. It is worth noting that the following explanation is easily extensible to the scenario with multiple jammers and multiple interfering transmitters. The signal transmitted by source S in timeslot t can be denoted by X_S , while X_J and X_I represent the jamming signal emitted by jammer J and interfering signal emitted by transmitter T , respectively, in timeslot t . Let h_{SR} denote the channel coefficient of the channel between source S and

receiver R and h_{JR} represent the channel coefficient between jammer J and receiver R , while h_{IR} characterize the same for the channel between interfering T and receiver R . It is worth noting that the channel coefficient considers the channel attenuation, antenna characteristics, and distance-dependent path loss. The received signal Y_R at receiver R in timeslot t with respect to channel c can be mathematically expressed as:

$$Y_R = h_{SR}X_S + \lambda_J h_{JR}X_J + \lambda_I h_{IR}X_I + n_o \quad (1)$$

where $\lambda_J \in \{0, 1\}$ and $\lambda_I \in \{0, 1\}$, respectively, represent whether channel c is jammed by jammer J and whether transmitter T is emitting interfering signal using same channel c for its own TX, in timeslot t . n_o represents a zero-mean complex Gaussian random noise with variance σ_o^2 .

The transmit power of the source, jammer, and interfering transmitter can be defined as $P_S = E\{|X_S|^2\}$, $P_J = E\{|X_J|^2\}$, and $P_I = E\{|X_I|^2\}$, respectively. Following equation (1), the signal-to-interference plus noise ratio (SINR) at the receiver can be expressed as:

$$\text{SINR} = \frac{P_S |h_{SR}|^2}{\lambda_J P_J |h_{JR}|^2 + \lambda_I P_I |h_{IR}|^2 + \sigma_o^2} \quad (2)$$

Based on the operations of the jammer J and interfering transmitter T in time slot t , the SINR value at the receiver can be defined as follows:

- 1) Channel c is ideal because another transmitter T is not using this channel, i.e., $\lambda_I = 0$, and jammer is not active in channel c , i.e., $\lambda_J = 0$. In this case, both intentional and unintentional interference do not exist and the SINR value of the received signal at receiver R is presented as:

$$\text{SINR} = \frac{P_S |h_{SR}|^2}{\sigma_o^2} \quad (3)$$

- 2) Channel c is in use by another transmitter T , i.e., $\lambda_I = 1$, and jammer is not active in channel c , i.e., $\lambda_J = 0$. In this case, Jamming or intentional interference does not exist and the SINR value at receiver R is presented as:

$$\text{SINR} = \frac{P_S |h_{SR}|^2}{P_I |h_{IR}|^2 + \sigma_o^2} \quad (4)$$

- 3) Channel c is ideal, i.e., $\lambda_I = 0$, and jammer is active in channel c , i.e., $\lambda_J = 1$. In this case, unintentional interference does not exist but the jamming, i.e., intentional interference is present and the SINR value at receiver R is presented as:

$$\text{SINR} = \frac{P_S |h_{SR}|^2}{P_J |h_{JR}|^2 + \sigma_o^2} \quad (5)$$

- 4) Lastly, channel c is in use by another transmitter T , i.e., $\lambda_I = 1$, and jammer is also active in channel c , i.e., $\lambda_J = 1$. In this case, both unintentional and jamming, i.e., intentional interference exist. Hence, the SINR value of the received signal at receiver R is presented as:

$$\text{SINR} = \frac{P_S |h_{SR}|^2}{P_J |h_{JR}|^2 + P_I |h_{IR}|^2 + \sigma_o^2} \quad (6)$$

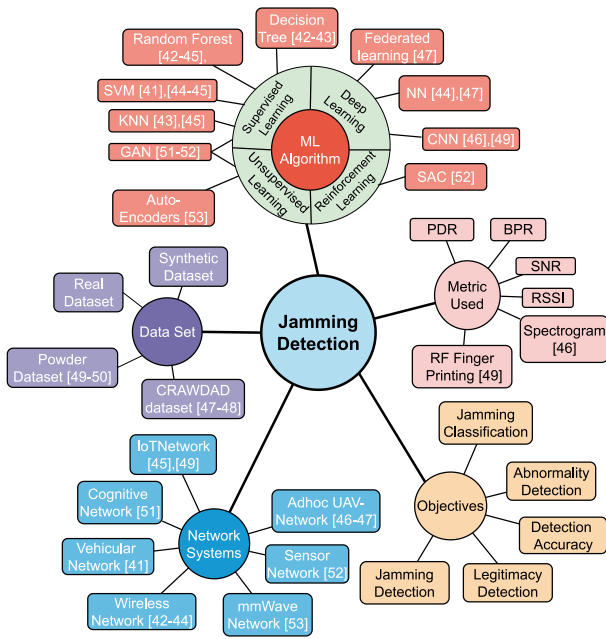


FIGURE 6. Summary of jamming detection related works.

A TX from source to receiver is successful if the receiver decodes the information without error, which requires the SINR value to exceed a threshold. Jamming signals intentionally degrade the received signal quality. While unintentional interference can be managed through proper resource allocation and power control of known transmitters, jammers are unknown and beyond the service provider's control. Therefore, developing effective jamming detection and mitigation techniques, alongside managing unintentional interference, is crucial for 5G and beyond networks.

C. AI/ML-BASED JAMMING DETECTION IN 5G AND BEYOND NETWORKS

This subsection provides state-of-the-art information related to AI/ML-based research, specifying network systems, parameters, techniques, and ML algorithms employed for jamming detection and classification purposes. A summary of all types of terrestrial and non-terrestrial network systems, objectives, metrics, data sets, and ML techniques used in jamming detection-related works is depicted in Fig. 6.

The work in [41] proposes a support vector machine (SVM)-based mechanism to detect the jamming attack in a vehicular network. In this work, joint sufficient statistics of packet drop probabilities are used to generate the training data. The simulation results present the effectiveness of the proposed technique in terms of jamming attack detection probability. Then, a novel ML-based hybrid jamming detection algorithm is proposed in [42], where the decision tree (DT) algorithm is used to classify known attacks, and an isolation forest (IF) is used to detect unknown attacks. Known attacks are those that are used for training purposes. To evaluate the proposed algorithm, a real dataset is generated from wireless modems transmitting in the

C-band with four types of different jamming attacks (spot, barrage, pulse, and sweep jamming). Numerical results show that the proposed algorithm outperforms the baseline algorithm (hierarchical combination of DT and one class-SVM) in terms of jamming classification and detection accuracy.

Furthermore, three ML-based algorithms are used to detect and classify constant, random, and reactive jamming attacks in [43] that are random forest, k-nearest neighbors, and decision tree. In this work, simulation data is generated using the NS-3 simulator to train the algorithm, and the metrics used for jamming detection are received signal strength indicator (RSSI) and packet delivery ratio (PDR). Numerical results show that the random forest algorithm is the best for detection accuracy. In the same line, the efficiencies of several ML-based algorithms such as SVM, random forest, and neural network (NN) are compared in detecting the jamming signals in [44]. In this work, a dataset is generated using features such as PDR, bad packet ratio (BPR), and RSSI that identify jamming signals and is used to train, test, and evaluate the ML-based models. The simulation results show that Random Forest has higher jamming detection probability and accuracy than SVM and NN-based techniques.

The authors in [45] consider the wireless IoT network where RSSI is collected via dedicated anchor nodes to detect the jamming attack. Then, different ML-based algorithms such as random forest, SVM, and KNN are used for jamming detection. Numerical results show that Random Forest performs the best among all other algorithms, and the detection accuracy of the Random Forest algorithm with the Real AdaBoost with five anchor nodes for real network data and simulated data are 89.7% and 98.6%, respectively. From the above works [41], [42], [43], [44], [45], it can be concluded that although there are multiple available ML techniques for jamming detection and classification such as SVM, Decision tree, KNN, NN, and Random Forest, however, Random Forest is the best choice for resulting the highest detection accuracy in different datasets of different application scenarios.

In [46], OFDM-based UAVs' network is considered where to detect and classify different jamming attacks two different techniques are proposed. First, an ML-based feature classification technique is proposed using numeric features such as SINR, energy threshold, and OFDM parameters. Second, using the spectrogram images collected through the experimental testing procedure, a spectrogram-based classification model is designed using the CNN-deep learning algorithm. Also, the performance of both techniques is evaluated in terms of accuracy and false alarm (false positive- FP) rates. This work claims that the spectrogram-based CNN model outperforms the feature-based ML techniques such as KNN, Decision Tree, Logistic Regression, Naive Bayes, and Random Forest in terms of jamming detection and classification accuracy and FP rates. Furthermore, for a flying Ad-hoc network, a

federated learning-based on-device jamming attack detection scheme is proposed in [47]. Also, leveraging the Dempster-Shafer theory, the client group prioritization technique is used to enhance the performance of the proposed three-layer NN-based algorithm. The proposed algorithm is evaluated using CRAWDDAD [48] and the ns-3 simulated datasets. Numerical results show that the proposed federated learning scheme outperforms the traditional distributed learning solution regarding jamming attack detection accuracy.

In the work [49], an efficient fine-grained augmentation-driven DL approach is proposed to identify legitimate transmitters via RF fingerprinting to ensure the security of industrial IoT-enabled cyber-physical systems. A fine-grained augmentation approach improves the learning performance of a DL model while tackling the varying propagation channels. Further, an enhanced classifier structure is also presented to enhance the accuracy of the proposed approach. The POWDER dataset [50] is used to experiment. It is claimed and demonstrated that the RF fingerprinting identification accuracy performance improves when training data are augmented in a waveform-specific fine-grained manner compared to the tapped delay line and clustered delay line-based augmentation approach. Also, a sensitivity analysis is presented for the fine-grained approach concerning different SNR, SIR, and SINR levels regarding accuracy performance.

The authors in [51] consider a cognitive radio communication environment where to detect the abnormalities inside the radio spectrum and to make the physical layer secure, two AI-based techniques, (Generative Adversarial Networks (GANs) and the dynamic Bayesian networks (DBNs)) are proposed. Conditional-GAN is applied for high dimensional data, while DBN is applied on low dimensional data. Simulation results demonstrate the effectiveness of C-GAN and DBN models in abnormality detection accuracy over the Cyclostationary Feature Detection (CFD) and Energy detector, respectively. Also, the work in [52] proposes a soft actor-critic (SAC) deep-RL framework for anomaly detection in a target process where remote sensors monitor the process. Further, to detect the adversarial jamming attacks on the communication link between sensors and the decision-making unit, a GAN-based framework is proposed. Simulation results show the effectiveness of the proposed techniques in terms of detection accuracy and time complexity over the conventional Actor-Critic algorithm. Thus, [51] and [52] state that using GAN-based approaches is quite useful in detecting the abnormalities in spectrum or adversarial jamming attacks.

An autoencoder-based jamming detection and mitigation approach is proposed against beam training jamming attacks in a manufacturing environment where an automated guided vehicle communicates with a base station (BS) via mmWave carrier [53]. The proposed auto-encoder-based jamming detection algorithm learns through RSSI and can capture more than 80% of attack events, while the proposed auto-encoder-based jamming mitigation algorithm improves the average SINR up to 15 dB over the case without using

any jamming mitigation strategy. However, the accuracy performance of the autoencoder-based jamming detection algorithm is not compared with any benchmark schemes. It can be observed from above stated works that all different ranges of ML techniques such as discriminative, autoencoder, and generative models have been investigated and proven effective for jamming detection and classification tasks.

D. AI/ML-BASED JAMMING MITIGATION TECHNIQUES IN 5G AND BEYOND NETWORKS

Building on the previous section, this subsection delves into AI/ML-based techniques for jamming mitigation. Figure 7 summarizes the related works, covering network systems, objectives, wireless techniques, and ML algorithms used for interference mitigation.

A novel framework using a deception mechanism is proposed to defeat a super-reactive jammer that makes the jammer continuously attack the channel [54], and the jamming signals are used to transmit data using ambient backscatter communication technology. Further, a DL-based detector with an LSTM network is proposed to detect the backscattered signals that adapt to noise and channel distributions and learn the received signals' dependencies. Simulation results demonstrate an interesting outcome that the transmitter can achieve better BER performance using the proposed method if the jammer attacks the channel with a higher power. In the case of a multi-user communication scenario, a Markov game framework is used to model the anti-jamming problem, and to solve this problem, a collaborative MARL-based anti-jamming algorithm is proposed [55]. The proposed solution mitigates mutual interference while tackling external sweep-jamming attacks. The impact of FP and false negative (FN) is considered in sensing error. Simulation results claim that the proposed collaborative MARL-based algorithm outperforms independent Q-learning and sensing-based methods regarding jamming mitigation and normalized rate.

To mitigate the jamming attacks in a flying ad-hoc network, an RL-based model-free Q-learning algorithm with an epsilon-greedy policy is proposed [56]. Also, an on-device federated learning-based jamming detection technique directs the jamming mitigation algorithm. The simulation results show that the proposed algorithm outperforms the baseline methods, such as random policy and Dijkstra's algorithm, to reduce the number of en-route jammer location hop counts from source to destination. A novel active inference framework is proposed for an anti-jamming resource allocation strategy in a cognitive-UAV to ground base station communication environment [57]. Specifically, an active generalized dynamic Bayesian network is proposed to represent the external environment and jointly encode the spectrum's dynamic physical signal and UAV-jammer interaction. Further, the action and planning are considered a Bayesian inference problem that can be solved by avoiding states under attack (abnormal state). Simulation results verify that the proposed technique outperforms the frequency

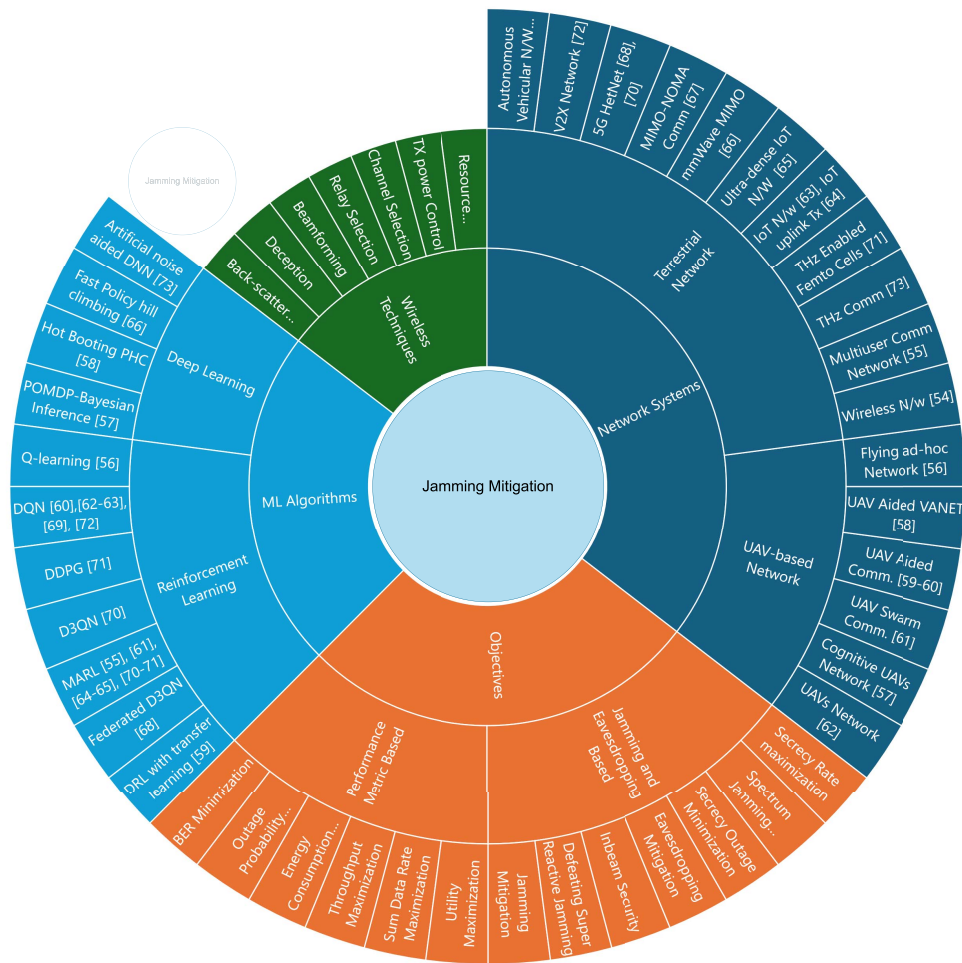


FIGURE 7. Summary of jamming mitigation related works.

hopping and Q-learning-based techniques regarding jamming mitigation and convergence speed.

A hot-booting Policy Hill Climbing (PHC)-based UAV relay strategy is proposed to enhance the performance of UAV-assisted vehicular networks against jamming attacks on roadside units (RSUs) [58]. Simulation results demonstrate that the proposed relay strategy outperforms the Q-learning-based scheme regarding BER and UAV utility performance. In work [59], a UAV-aided cellular framework is considered to resist smart jamming where a UAV selects a DRL-based relay policy to serve those mobile users whose associated BSs are under jamming attack. Specifically, the problem is formulated as a dynamic relay game and its NE solution is derived in terms of BER and UAV energy consumption. The NE solution is obtained by applying deep RL and transfer learning utilizing the observed current communication status and the previous anti-jamming relay experiences. Simulation results show that the proposed DRL-based scheme outperforms the Q-learning and PHC-based approaches regarding BER and UAV energy consumption performance. The authors in [60] utilize prospect theory to formulate a smart-attack game on UAV TXs. Further, a deep

Q-learning-based algorithm is proposed to optimally allocate UAV TX power on multiple radio channels to mitigate smart jamming, eavesdropping, and spoofing attacks. Numerical results show that the proposed algorithm enhances the utility and secrecy capacity of the UAV in comparison to Q-learning and Win or Fast Learn (WoFL)-PHC-based schemes.

To mitigate the jamming attacks and improve the BER and energy consumption performance in a UAV swarm communication network, a MARL-based scheme is proposed to control the Tx power and relay selection [61]. In this scheme, policy distribution is formulated to enhance policy exploration, and transfer learning is used to exploit the shared swarm experiences to fasten the initial learning. Further, a deep RL-based scheme is proposed to remove state quantization error for dynamic channel states due to high swarm mobility. Simulation results show that the proposed scheme outperforms the MAPPO and MADDPG-based approaches regarding BER, utility, and energy consumption. However, the computational complexity of the proposed scheme increases with the number of UAVs, batch size, quantized power levels, and the number of learning samples [61]. Furthermore, a distributed DRL-based energy-efficient and

low complexity framework is proposed to enhance the BER performance under jamming attacks in UAVs communication network [62]. Specifically, using this framework, each relay UAV selects its Tx power independently based on historical state-related information without knowing the location, trajectory, and battery level of other UAVs and jammers. Then, a power control game is formulated and solved using a deep RL-based anti-jamming relay approach to achieve faster and better performance. Simulation results show that the proposed schemes outperform the optimal power allocation scheme against jamming regarding BER and energy consumption performance.

For IoT devices to improve SINR and save their energy under jamming attacks, a DRL-based power control scheme is proposed [63]. Knowing the current IoT Tx status and the jamming strength, DQN determines the Tx power of the IoT devices. The authors implemented the proposed scheme on USRP to evaluate the performance of the anti-jamming technique. Experimental results show that the proposed scheme outperforms the benchmark Q-learning-based power control scheme against jamming. Especially for the uplink TX of IoT devices, a novel spectrum anti-jamming technique is proposed in [64]. Specifically, a value function approximation-based MARL technique is used to optimize the configuration policies for the IoT devices sequentially. Simulation results show that the proposed scheme outperforms various benchmark schemes regarding average normalized throughput. In [65], a model-free MARL-based algorithm is proposed to mitigate the jamming attack in ultra-dense IoT networks. In this paper, a quality of service (QoS) constrained anti-jamming problem is modeled as a Markov game, and to get its NE solution, a mean-field deep Mellow-based anti-jamming technique is developed. The simulation results show that the proposed technique simultaneously enables spectrum sharing and jamming attack mitigation.

In the downlink mmWave-MIMO communication scenario, an RL-based power control technique is proposed to control the jamming attack [66]. Specifically, a fast PHC-based power control algorithm is used to choose the BS Tx power over multiple antennas. Simulation results show that the proposed scheme outperforms the benchmark technique regarding sum data rate and average SINR. Furthermore, to mitigate the smart jamming attack in downlink NOMA TXs [67], an RL-based power control algorithm is proposed for BS power allocation equipped with multiple antennas without knowing the radio channel and jamming parameters. The power allocation problem is formulated as a zero-sum game and its Stackelberg equilibrium is derived. Further, the use of Dyna architecture and hot-booting techniques accelerate the learning speed of the proposed power allocation algorithm. Simulation results show that the proposed scheme outperforms the standard Q-learning-based anti-jamming techniques in terms of sum data rates of users.

A federated DRL-based beamforming and power control algorithm is proposed to enhance the achievable rate of

Femto-cell users in 5G HetNets in the presence of one jammer in each femtocell [68]. Specifically, the Markov decision process (MDP) is used to transform the non-convex optimization problem into a MARL problem, and a dueling DDQN (double deep Q-network) with federated learning is used to solve the problem. Simulation results show that the proposed scheme outperforms DDQN and DQN schemes regarding the achievable rate. In work [69], an autonomous vehicular network is considered, where to mitigate jamming attacks and to enhance the anti-eavesdropping secrecy capacity, a distributed Kalman Filtering algorithm is proposed to track jammers, and a hierarchical DQN-based technique is proposed to design anti-eavesdropping channel selection and power control policy. Simulation results show that the proposed technique outperforms available techniques regarding secrecy capacity and data rate under jamming attacks.

Along with jamming attacks, eavesdropping is also a serious problem from a secrecy point of view, where an illicit user tries to steal the information of ongoing communication. The following are related works utilizing AI/ML-based techniques to tackle the eavesdropping problem and enhance secrecy rate. A multi-agent cooperative DRL-based secrecy-aware joint channel, power, and beam-forming control algorithm is proposed to maximize the secrecy EE of pico-cell users in a 5G heterogeneous network [70]. Specifically, the problem is transformed into a MARL problem utilizing MDP, and dueling DDQN is used to find the optimal solution of MDP. Simulation results demonstrate that the proposed scheme outperforms MA-DQN and MARL schemes regarding the secrecy EE of pico-cell users. Furthermore, a deep RL-based power control algorithm is proposed to improve the secrecy rate of femtocell users in 6G HetNets [71]. Particularly, the non-convex optimization problem is transformed into a multi-agent deep RL problem using MDP, and a DDPG-based power control scheme is used to solve the problem. Simulation results show that the proposed scheme outperforms existing schemes regarding the average secrecy rate per femtocell user.

To secure V2X links in the cellular-based vehicle to everything (V2X) network [72], a deep Q-learning-based power allocation scheme is proposed to maximize the secrecy rate of vehicles while keeping the constraints on power and interference. Further, a reward function is developed for the deep Q-learning network to allocate power to vehicles efficiently. The simulation results demonstrate the effectiveness of the proposed algorithm. A THz communication network with randomly distributed eavesdroppers is considered in work [73]. In this paper, to increase the secrecy rate and to decrease the secrecy outage probability, an artificial noise (AN) aided DNN-based algorithm is proposed to control the Tx power, carrier frequency, and AN signal power. Simulation results show that the proposed scheme outperforms the conventional transmitter AN scheme regarding secrecy outage probability.

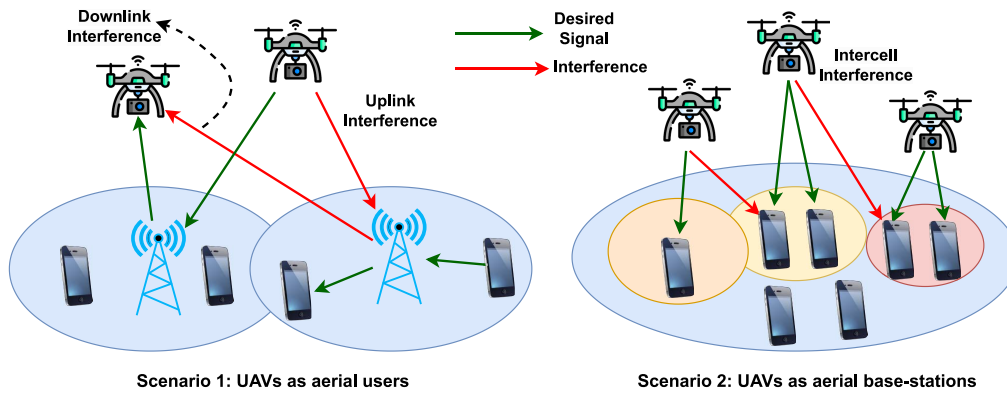


FIGURE 8. Illustration of UAV-based communication system with desired and interfering signals.

AI/ML-based techniques are indeed helpful in enhancing the performance of communication networks. However, in some situations, learning techniques can also be a reason for deteriorating network performance where a learning technique itself can be used to create jamming attacks [74]. The authors in [74], propose a DeepJam strategy for creating a jamming attack on the Zigbee traffic by exploiting the cross-technology interference (CTI) in the coexisting scenario of Wifi-Zigbee networks. DeepJam captures the temporal pattern of past traffic and predicts future traffic using a deep-learning technique. DeepJam attacks the Zigbee Tx that is not disrupted by CTI and reduces the victim's throughput significantly compared to conventional jamming strategies. Furthermore, to take advantage of learning techniques, securing the training and testing data is highly important. Otherwise, the performance can be negatively affected if the data is distorted. Some works represent the effect of distorting training and testing data on the system performance. For example, in [75], to reduce the spectrum sensing accuracy of ML-based spectrum sensing methods in a cognitive radio-based IoT environment, an embedded poisoning method (EPM) is proposed. The proposed EPM is a combination of embedded communication method waveforms that produce interference and poison data label hidden attacks on the training data of ML-based spectrum sensing techniques. Simulation results show the efficiency of the proposed method in reducing the sensing accuracy compared with traditional white-box attack methods. In work [76], the poisoning and evasion attacks using particle swarm optimization are exploited in training and testing datasets to reduce the classification accuracy of DL-based algorithms used in traffic sign recognition systems for autonomous vehicles. Simulation results demonstrate the effectiveness of the proposed attacks in terms of accuracy reduction. Hence, to take full advantage of AI/ML techniques, the data used for training and testing of the model should be highly secure.

From the next section onwards, this survey paper focuses on providing a detailed review of AI/ML-related works for unintentional interference mitigation in different domains of communication networks, such as UAV-based, vehicular

networks, HCNs, D2D, IoT, mmWave-MIMO, THz, and NOMA based networks. In this context, every section is dedicated to a particular domain and consists of two subsections. The first subsection, discusses the key features, characteristics, different challenges, and types of interference related to a specific domain. Then, in the second subsection, a detailed review and qualitative comparison of AI/ML-based research works are presented that focus on tackling these challenges and enhancing the performance of wireless communication networks in that particular domain.

III. INTERFERENCE MITIGATION IN UAV-BASED N/W

A. FEATURES AND TYPES OF INTERFERENCE IN UAV-BASED COMMUNICATION SYSTEMS

Nowadays, UAVs are widely utilized for monitoring, surveillance, rescue operations, military, and public safety operations [77]. In the above-stated scenarios, UAVs act as aerial users which communicate with the existing cellular ground BS for information sharing (uplink) and getting instructions (downlink) for further actions. However, due to high altitude and favorable line-of-sight (LoS) channel conditions, aerial UAV users can cause high interference to terrestrial ground users during uplink communication. On the other side, UAV aerial users can receive more interference from neighboring base stations during downlink communication [78]. Also, UAVs act as aerial BSs to provide effective solutions to serve temporary high-traffic demands during high-crowd events such as concerts, festivals, and stadium games [79]. In such scenarios, UAVs can face and create intercell interference with other UAVs aerial BSs, and cellular ground BSs utilizing the same spectrum bands. In both scenarios, the illustration of UAV-based communication systems having both desired and interfering signals is presented in Fig. 8. Since UAVs are power-constrained devices, in both scenarios, interference control via proper resource management is mandatory for better EE and throughput results. This section provides a detailed survey of AI/ML-related works utilizing intelligent techniques to avoid inter-cell or co-channel interference (CCI) in UAV-based communication systems. This section is mainly divided

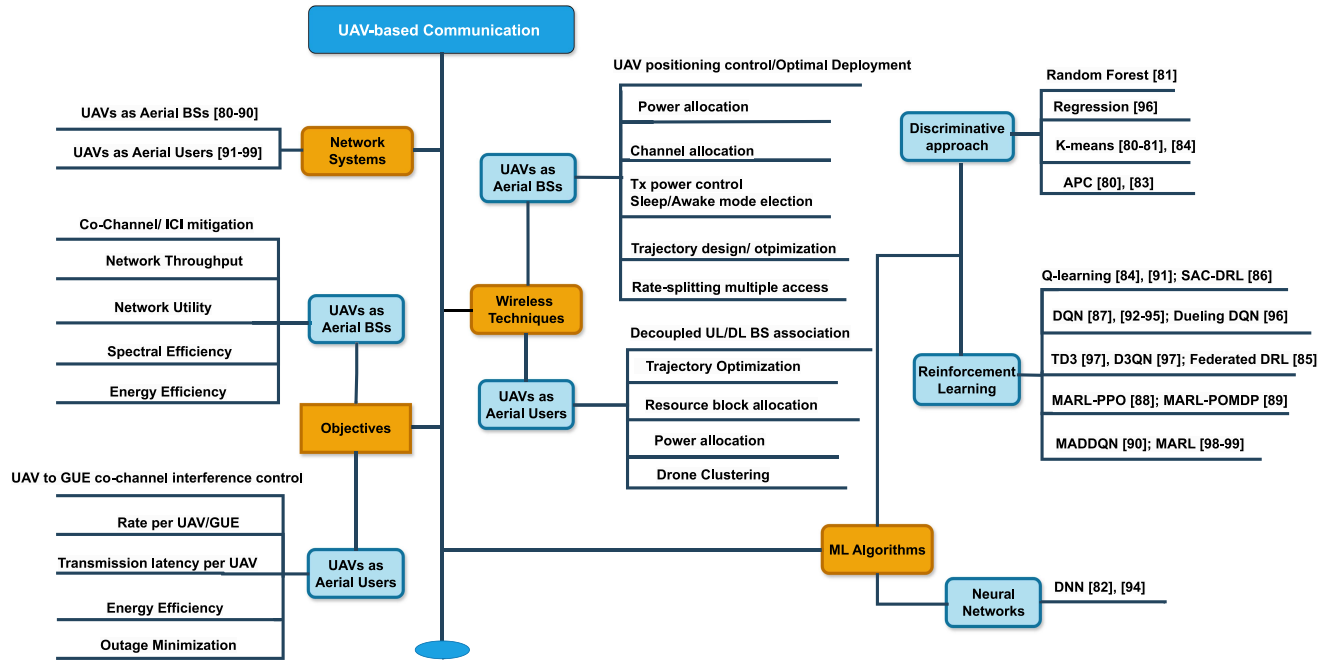


FIGURE 9. Summary of related works for interference mitigation in UAV-based networks.

into two parts: One for UAVs as aerial BSs and the other for UAVs as aerial users. The summary of all discussed related works comprising the network systems, objectives, wireless techniques, and ML algorithms used for interference mitigation in UAV-based networks is presented in Fig. 9.

B. RELATED WORKS IN UAV-BASED COMMUNICATION SYSTEMS AND THEIR QUALITATIVE COMPARISON

1) UAVS AS AERIAL BASE STATIONS (BSS)

The research in [80] considers a drone small cells (DSCs)-aided wireless communication scenario, in which to mitigate the inter-DSC (co-channel) interference and to enhance the EE and throughput, two key unsupervised learning techniques are proposed. The first is the APIM (affinity propagation interference management) scheme, designed to mitigate interference by putting the interfering DSC in sleep mode. The second scheme is KMPA (K-means position adjustment), which associates the uncovered GUE (ground users) with enhanced signal quality to the serving active DSC. Simulation results show that the proposed learning-based multiple drone management framework enhances the EE of DSCs by 25% and the SINR of GUEs by 56%, respectively, when compared to the baseline scheme where all drones transmit with maximum power. In the same line, a multiple DSCs scenario is considered to provide coverage to GUEs in [81]. In this paper, a prediction-based proactive drone management (P2DM) framework is designed to mitigate network inter-cell interference and improve the EE of DSCs. The proposed technique works in two phases. Firstly, a highly accurate mobility prediction model is built in the offline phase using Random forest as per the historical data. Then, the prediction model is launched in the online

phase to predict the user position, which helps proactively determine whether a DSC should be awake or asleep at the next timeslot. Further, in the online phase, the K-means technique is utilized to control the position and TX power of DSCs to improve their EE. Simulation results show that the proposed P2DM technique demonstrates better results in terms of execution time and EE compared to the existing method based on genetic or heuristic algorithms.

The researchers in [82] consider the UAV-assisted mission-critical services where UAVs provide extended network coverage to the affected area using spectrum-sharing cognitive radio networks. This work proposes an optimization scheme to maximize the throughput of both primary and secondary networks by optimizing the power allocation and number of UAVs deployed as relay nodes. The main constraint is that the interference on the primary users should not be beyond a tolerable limit. A DNN model is proposed to significantly reduce the execution time under real-time solutions of mixed-integer UAV deployment problems. It is claimed that low computation complexity and execution time are the main benefits of the proposed approach. The authors in [83] propose a hybrid unsupervised and RL-based technique to mitigate the inter-cell interference in multiple UAVs-mounted small-cell scenarios. In this, the unsupervised affinity propagation clustering (APC) scheme explores the interference structure of multiple UAVs and puts the most serious interfering UAVs into a sleep state. Further, RL is applied to adjust the TX power of active UAVs to optimize the total throughput. It is shown that the proposed scheme outperforms the pure Q-learning and genetic algorithm (GA) regarding execution time and throughput.

The multiple UAVs' deployments as aerial BSs are considered to provide emergency communication services to GUEs [84]. In this work, the combination of K-means and Q-learning algorithms is proposed to mitigate the CCI and maximize the sum capacity of the network by keeping minimum data rate requirement constraints. Specifically, the K-means algorithm is used to optimize the 2D horizontal placement of UAVs, and the Q-learning algorithm is used to optimize the power allocation to UAVs for varying heights. The proposed approach outperforms the equal power allocation and particle swarm optimization (PSO)-based power allocation schemes regarding sum capacity. To mitigate the CCI in UAVs network, an effective multi-agent collaborative environment learning algorithm is proposed by realizing the dynamic UAV positioning and resource allocation (in terms of power levels and sub-channels) to the GUEs [85]. In this work, the combination of DQN and federated learning (FL) is considered where each UAV learns and gets trained independently based on the DQN algorithm and regularly updates network training parameters through the sharing of non-private data to the FL cloud server to realize the collaboration of multiple UAVs. It is claimed through simulation results that the proposed method is better adaptable to a dynamic environment and outperforms the multi-agent DQN without collaboration in terms of network utility (throughput maximization and power consumption minimization) optimization.

Next, the research in [86] considers a system with multiple UAVs as aerial BSs to serve multiple GUEs using the same spectrum without knowing users' positions and channel parameters. In this paper, the authors propose a soft actor-critic (SAC), i.e., a DRL-based solution to control the CCI and maximize the total throughput and fairness among users by optimizing the trajectory and TX power of UAVs in a centralized way. To achieve the balance between exploitation and exploration, the reward function that combines sparse with non-sparse reward is carefully designed. Simulation results show that the proposed SAC outperforms the SAC without power control regarding throughput and fairness. The study in [87] considers an ultra-dense UAV network to serve the GUEs. Here, a discrete mean field game (MFG) framework is used to formulate the power control problem to improve EE and mitigate inter-cell interference. The MFG helps imitate the interactions among UAVs in ultra-dense networks. Then, MFG framework is transformed into an MDP and a DRL-based MFG (DRL-MFG) algorithm is proposed to obtain the equilibrium solution. Simulation results show that the proposed methodology outperforms the random power allocation and Q-learning-based techniques for power allocation regarding EE and interference mitigation.

In the study [88], multiple UAVs-assisted full-duplex cellular networks are considered. In this paper, to mitigate the interference and enhance the spectral efficiency, the downlink-uplink decoupled association technique is proposed utilizing the rate split multiple access (RSMA) method.

Furthermore, for optimizing the sum rate of users in the uplink and for the multicast group in the downlink having constraints on power budget and backhaul capacity of UAV, a multi-agent DRL (MADRL) technique is utilized in a distributed manner to optimize beam-forming and user association in the network jointly. Compared with standard coupled association techniques and multiple access techniques such as SDMA, TDMA, and NOMA, the proposed method performs better regarding the sum rate of users. The authors in [89] consider an uplink communication scenario in which multiple UAVs cooperatively fly over the GUEs as aerial BSs to collect data from them without any centralized controller. While considering the interference management through the SINR metric and fairness among users, an optimization problem is formulated to maximize the total throughput and EE. In this paper, the optimization problem is formulated as a decentralized partially observed MDP (DEC-POMDP), and to solve this problem, the trajectories of UAVs and the GUEs associations are jointly optimized using an approach that combines the coalition formation game (CFG) for user association and MADRL based technique for trajectory optimization and energy consumption minimization. The benefit of the proposed technique is that it is trained offline in a centralized manner and gets executed online in a decentralized manner. Simulation results demonstrate that the proposed algorithm outperforms greedy association, the best response-based association, and iterative algorithm-based techniques in terms of fair throughput and energy consumption in a distributed manner. The authors in [90] consider the multiple UAVs as aerial BSs that provide wireless coverage to static and mobile GUEs. In this paper, to maximize the EE of the network, a cooperative multi-agent decentralized DDQN (MAD-DDQN) approach is proposed to mitigate the inter-cell interference from nearby UAV cells and to jointly optimize each UAV's 3D trajectory and the number of connected users. The proposed technique outperforms the random policy and MADDPG (multi-agent deep deterministic policy gradient) approach regarding EE without degrading the outage performance.

2) UAVS AS AERIAL USERS

The authors in [91], propose a multimodal Q-learning framework to locate an unknown interference source in a dynamic environment. The proposed framework combines pattern recognition with Q-learning and can dynamically adjust the algorithm's parameters based on the changing environments. The simulation results confirm that the proposed approach can outperform various baselines in terms of both accuracy and distance detection. To mitigate the interference among cellular-enabled UAVs and the GUEs, the authors propose the influential decoupled base station association and the different spectrum resource allocations for uplink and downlink communications [92]. To avoid CCI among UAVs and GUEs uplink communications, UAVs are served using mmWaves, and UHF are considered for GUEs. Also, different spectrums are allocated to UAVs'

data links and control links. Since UAV communication is power-constrained, two power allocation algorithms based on Q-learning and DQN are proposed to optimize the EE of the network. The proposed scheme outperforms EE and sum rate per UAV/GUE compared to the conventional coupled benchmark schemes.

A DRL-based algorithm is proposed to maximize the downlink SINR of 5G cellular network-connected UAVs by mitigating the severe LoS link interference from the non-associated ground BSs (GBSs) [93]. The proposed algorithm controls the TX power of GBSs in a centralized way. Instead of CSI, the proposed algorithm requires only the UAVs' coordinates and their received SINR every millisecond. The proposed technique provides better results regarding UAVs' downlink SINR than Q-learning and the industry-standard fixed power allocation algorithm. The authors in [94] consider the joint trajectory design and GBS association optimization while managing the CCI antennas in cellular-connected UAVs networks. This paper aims to minimize the trajectory length with connectivity outage constraints. Here, the DNN is used to provide a better GBS association, and to train a DNN, the signal strength radio map of a given area is constructed. Also, a novel DRL approach is proposed to learn optimal trajectory, in which the classical DRL framework is supported with a self-imitation learning model to accelerate training by utilizing successful transition experiences. Simulation results show that the proposed approach outperforms the nearest BS association and the highest SINR association techniques regarding trajectory length and connectivity outage performance.

The uplink communication scenario for the cellular-connected UAVs is considered in [95]. In this paper, an echo state network (ESN) cells-based DRL framework is utilized online by UAVs in a distributed manner for optimizing their path planning, power allocation, and association vectors with ground BSs along the path toward their destination. The objective of utilizing this framework is to minimize the interference that UAVs cause on the GUEs and the UAVs' TX latency. Simulation results show that in comparison to the heuristic approach of following the shortest path toward the corresponding destinations, the proposed scheme performs better regarding the rate per GUE and TX latency per UAV. The study in [96] considers a single UAV connected to the cellular network, which flies over the urban area straight from a starting point to the destination point. To maintain the QoS and to reduce the handover frequency, a regression and dueling deep Q-learning-based solution is proposed for intelligent BS association of the UAV. Assuming that the UAV is aware of BS locations, received signal power, surrounding building topology, and interference, the proposed solution selects the best BS to be associated with to maximize the data throughput received by the UAV during its trip. Simulation results demonstrate that the proposed technique outperforms the heuristic association methods regarding data throughput.

In study [97], a cellular connected-UAVs scenario is considered where for harmonious coexistence of UAVs and GUEs, CCI is mitigated by joint optimization of time-frequency resource block (RB) allocation and beam-forming design to minimize the ergodic outage duration (EOD) of UAVs. This work considers a practical building distribution-based path loss model and small-scale fading. To solve the radio resource management problem, a DRL-aided technique is proposed, where deep double dueling Q network (D3QN) deals with RB coordination in the discrete action domain, and twin delayed deep deterministic policy gradient (TD3) handles beam-forming design in continuous action regime. To optimize the EOD performance, the hybrid D3QN-TD3 solution is trained via interacting with the MDP-mapped outer and inner environments in an online centralized manner. Then, the trained D3QN and TD3 agents are exploited offline in a decentralized manner to achieve independent EOD performance gain. Simulation results present the effectiveness of the proposed technique in terms of EOD performance compared to several baseline techniques, such as Maximum ratio beam-forming and bufferless DRL methods. In research [98], a wireless network of multiple UAVs is considered, controlled by ground control stations (GCS). This paper proposes a hierarchical MARL-based algorithm to mitigate inter-cell interference and efficiently utilize the battery power of UAVs. The proposed algorithm works in two steps to maximize UAVs' EE. Firstly, a distributed inner-loop RL algorithm is used to control the TX power of UAVs. Then, a centralized outer-loop RL algorithm is used to find the optimal frequency reuse factor to control the inter-cell interference. The proposed algorithm outperforms the random action algorithm with a fixed frequency reuse factor and a hierarchical multi-agent Q-learning algorithm with binary transmit power controls regarding EE.

The Internet-of-Drones (IoD) system served by cellular BSs is considered in [99]. To jointly mitigate the interference and enhance the physical layer security in the presence of an eavesdropping drone, a resource allocation scheme consisting of clustering the legitimate drones and TX power control of cellular BSs is proposed. Different clusters are allocated with orthogonal RB to mitigate the CCI, and in each cluster, the NOMA scheme is used for uplink communication to enhance spectral efficiency. The cellular BSs are assumed to have full-duplex communication capability and simultaneously transmit artificial noise signals to weaken the eavesdropper links. To solve the optimization problem with the objective of sum-secrecy-rate maximization, the problem is decomposed into drone clustering and power allocation sub-problems. To address the drone-clustering sub-problem, a MARL framework is devised, and the transmit and jamming power allocation sub-problem is solved by employing fractional programming, successive convex approximation, and alternating optimization techniques. The better performance of the proposed technique SIREN (security and interference management with RL and NOMA) over the benchmark

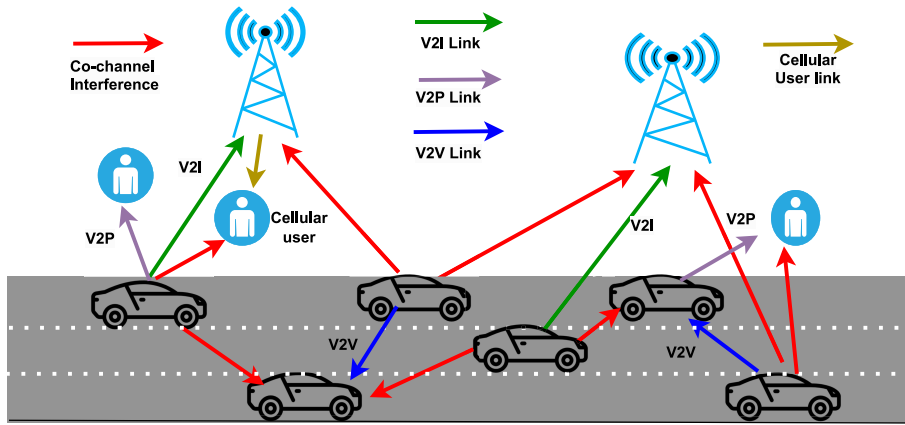


FIGURE 10. Illustration of V2X communication system with desired and interfering signals.

clustering and power control schemes in terms of secrecy capacity is verified via simulations.

IV. INTERFERENCE MITIGATION IN VEHICULAR COMM.

A. FEATURES AND TYPES OF INTERFERENCE IN V2X COMMUNICATION SYSTEMS

With the rapid evolution of autonomous driving and interconnected vehicles, the V2X communication technology is pivotal in facilitating data exchange among vehicles, RSUs, edge servers, and pedestrians. Two primary radio standards enabling V2X communication are Dedicated Short-Range Communication (DSRC) and Cellular V2X (C-V2X). DSRC is an IEEE 802.11p technology developed from standard Wi-Fi, specifically for use within vehicular communication [100]. Meanwhile, C-V2X is defined by 3GPP standards, where LTE-V2X is presented in Releases 14 and 15, and 5G-V2X is presented in Releases 16 and 17. Both DSRC and C-V2X standards use the 5.9GHz band to transmit the necessary information to deliver basic safety messages between vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), and Vehicle-to-pedestrians (V2P). In addition to using the 5.9GHz band, C-V2X also uses existing cellular networks, forming vehicle-to-network (V2N) communication alongside V2V, V2I, and V2P [101]. As C-V2X uses direct (PC5 interface/sidelink) and indirect (Uu interface/network) communication methods, fatal and time-critical safety messages can be delivered through direct communication. In contrast, less time-sensitive messages can be efficiently delivered through indirect communication. Therefore, this adaptability ensures that the most appropriate and efficient communication method is allocated based on the specific use cases. Thus, to provide high-quality communication for on-road vehicle safety, it becomes highly important to maintain QoS constraints such as ultra-high reliability and low latency of V2V link and V2I link communication. In cellular underlying V2V, the co-existence of V2V and V2I link, and cognitive IoV networks, CCI degrades the quality metrics. An illustration of the V2X communication system

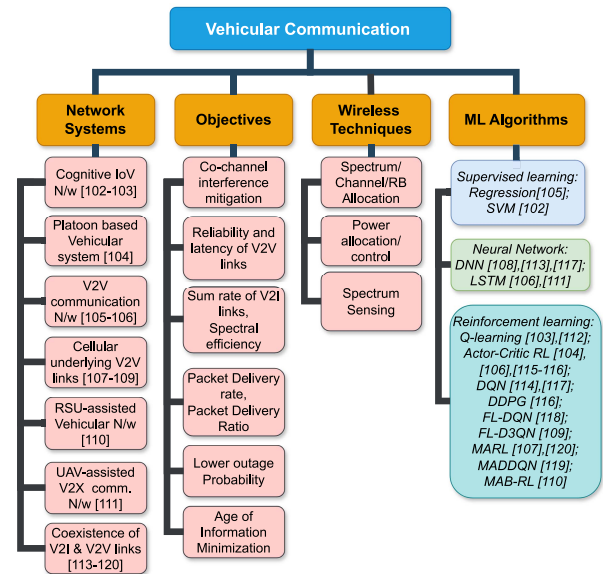


FIGURE 11. Summary of related works for interference mitigation in vehicular communication networks.

with interference effects is presented in Fig. 10. Hence, this section provides a detailed survey of intelligent AI/ML-based techniques that mitigate CCI via proper resource allocation and spectrum sensing. The summary of the network systems, objectives, wireless techniques, and ML algorithm utilized by state-of-art works in this section can be seen in Fig. 11.

B. RELATED WORKS IN V2X COMMUNICATION SYSTEMS AND THEIR QUALITATIVE COMPARISON

A cognitive radio-assisted IoV network is considered in [102]. Here, a novel congestion-aware hybrid learning-based spectrum sensing (SS) and resource allocation algorithm is proposed to avoid CCI between the primary cellular users and the secondary vehicle users and enhance spectrum efficiency. Specifically, the hybrid learning spectrum agents (HLSAs) perform SS based on DL to detect

the vacant spectrum band and then allocate those spectrum bands to the requesting vehicle using a trained SVM classifier. Simulation results demonstrate the efficacy of the proposed scheme regarding detection probability, sensing, and allocation accuracy compared to the existing state-of-the-art schemes. Similar to [102], in [103], a cognitive IoV network is considered that opportunistically accesses primary users' licensed spectrum without interrupting their normal communications. In this paper, to enhance spectral efficiency and avoid harmful interference to primary users, a joint collaborative big spectrum data sensing and Q-learning-based dynamic spectrum access scheme is proposed. Simulation results indicate that the collaborative spectrum access mode achieves a higher average throughput, lower interference power, and lower communication outage than the underlay and overlay spectrum access modes.

In the research in [104], a platoon-based vehicular cyber-physical system is considered, where the platoon vehicles are served by a RSU. This paper proposes a soft actor-critic-based two-timescale DRL algorithm to balance the tradeoff between traffic capacity and communication efficiency. The objective of this work is to minimize the inter-vehicle distance (IVD) and to maintain the QoS requirement of the V2I link by jointly optimizing the transmit power and the beamwidth of the RSU and IVD of the platoon, which in turn controls the inter-beam interference among different V2I links. Further, a decentralized technique using the Gaussian process regression (GPR) approach is proposed to allocate the power and resource blocks to vehicles so that their age of information (AoI) does not exceed a predefined threshold [105]. The proposed technique learns the network dynamics (such as wireless channels and interference), estimates the future AoI of vehicles, and allocates resources in a distributed manner. Simulation results show that the proposed technique outperforms the baseline techniques with random power allocation and GPR-based power allocation without exploration regarding AoI violation probability.

A 5G V2V network is considered in [106], where to minimize the CCI and maximize link capacity and packet reception ratio, a novel MADRL-based resource allocation algorithm is proposed. Here, each vehicle acts as an individual agent and use the actor-critic RL technique. Specifically, all agents share their observations over critic networks for centralized training using DDQN to tackle the non-stationarity in a multi-agent setting. Further, each agent uses DPG in the actor network to select the appropriate combination of RB and Tx power in a decentralized way. This paper also addresses the hidden node problem, half duplex problem, and merging collision issues. Simulation results demonstrate that the proposed method outperforms the DDQN-based schemes. In [107], a semi-persistent scheduling (SPS) scheme is considered for resource allocation and avoid CCI in cellular-C2X Mode 4 V2V communication to maximize the distance-weighted TX utility while enhancing the reliability and latency-based QoS performance. Specifically, the maximization problem

is formulated as a Markov game. A MARL approach is designed to solve it, which helps vehicles select radio resources independently based on their local observations. Further, a multi-head attention method is proposed to improve the training efficiency. Simulation results show that the proposed RL-SPS outperforms the baseline approach with random allocation.

V2V communication systems underlaid uplink cellular networks are considered in [108]. This paper proposes a DNN-based transmit power control algorithm to minimize the interference among cellular users and vehicles and maximize the sum rate of all users. Simulation results verify that the proposed algorithm outperforms fixed transmit power, weighted minimum mean square error (MMSE), and Dinkelbach's methods in terms of achievable sum rate. Further, a cellular-underlaying V2V communications system is considered in [109]. In this paper, to mitigate the CCI between V2V and cellular communication, a novel federated MADRL-D3QN approach is proposed for the decentralized joint optimization of channel selection and power control for V2V communication. The objective of the proposed approach is to maximize the sum rate of cellular users and the packet delivery rate of V2V communication. Also, to accelerate the multi-agent learning training process and address limited partial observation, federated learning is adopted to synchronize the DRL models of all agents and maintain a global model.

In [110], a dynamic multi-vehicle interference environment using mmWaves for BS communication is studied. A combinatorial multi-armed bandit (CMAB) RL framework with adaptive combinatorial Thompson sampling (CTS) maximizes network throughput by selecting beams for all vehicles in high-mobility settings. To reduce complexity, a sequential Thompson sampling (TS) algorithm performs individual beam selection. Simulations show both strategies approach the optimal rate of the genie-aided solution using the best beam each time slot.

In [111], a UAV-assisted cellular V2X communication system is explored, where UAVs and cellular BS provide V2I communication with multiple orthogonal spectrum resources, shared with V2V links. To avoid co-channel interference (CCI) and maximize bandwidth efficiency, a resource allocation scheme for air-slicing is proposed, meeting data rate and latency constraints. An LSTM-based algorithm optimizes resource allocation by forecasting vehicle and UAV mobility. Simulations show this scheme outperforms DQN and DDPG-based methods. Further, an uplink random multiple access (RMA) oriented sparse code multiple access (SCMA)-V2X network is considered in [112], where V2I users and V2V users randomly select codebooks from a codebook pool. The total available bandwidth is divided into two parts for V2I and V2V user's TX to avoid mutual interference. In this paper, to maximize the sum rate of V2I links while maintaining the reliability (outage probability) performance of V2V links, a decoupled Q-learning-based algorithm is proposed for codebook selection,

and a convex optimization method is used to find the optimal bandwidth allocation (OBA) to V2I users and V2V users. Simulation results show that the proposed RMA-SCMA with OBA outperforms the conventional schemes, such as fixed bandwidth allocation, regarding sum rate. A high-mobility V2X network is considered in [113]. This work aims to satisfy the diverse QoS requirements of different types of links, such as low delay for V2I connections and ultra-reliability in terms of outage performance for V2V connections while managing the CCI between V2I and V2V links. To achieve the objective, a parallel DNN framework is designed for TX power allocation and vehicle local computing ratio allocation. Further, the Hungarian method is employed to obtain the optimal spectrum allocation to V2V and V2I links. Simulation results show that the proposed scheme performs better than other general algorithms in terms of diverse QoS achievements.

In the V2X communication system, shared spectrum resources are allocated to V2I and V2V links along with their Tx power control to manage the CCI and to maximize the data rate of V2V links and successful TX probability of V2I links In [114]. Specifically, the DQN algorithm is used to solve the sub-band assignment and the DDPG algorithm is used for continuous power allocation. Also, a meta-based DRL algorithm using the off-policy model-agnostic meta-learning (MAML) is used to adapt trained policy quickly in the new dynamic environment. Numerical results verify that the proposed DRL-based approach outperforms the DQN-based algorithm that quantizes continuous power and easily adapts to the new dynamic environment. An IoV communication network is considered in [115]. This paper proposes a distributed model-free actor-critic RL framework for joint mode selection (reuse mode, dedicated mode, and cellular mode), RB assignment, and power control in D2D-V2V-based IoV networks. This work aims to maximize the sum capacity of V2I and V2V communication and minimize the outage probability to ensure the URLLC requirements of unicast V2V links. Moreover, the efficient transfer actor-critic (ETAC) learning approach is proposed to enhance the learning efficiency and convergence speed. Simulation results prove that the proposed ETAC RL approach outperforms the classical actor-critic learning, the Q-learning, and the random search approaches regarding sum capacity and outage probability.

In research [116], a cellular-V2X is considered where to mitigate the CCI between V2I and V2V links, and a hybrid centralized-distributed approach is proposed for radio resource management. Specifically, to maximize the system capacity while guaranteeing the reliability of V2V links, a graph-matching method is utilized for centralized channel allocation, and a distributed DDPG-based algorithm is used for power allocation through actor-critic learning. Numerical results show that the proposed method outperforms other DQN-based techniques regarding system capacity. Further, to avoid the CCI among V2I and V2V links in V2X network, a hybrid architecture consisting of centralized decision-making

and distributed resource sharing is proposed to maximize the long-term sum rate of all vehicles In [117]. Further, a DNN-based algorithm and quantization technique is used by each vehicle to compress the CSI information before sending the feedback to centralized decision-making BS so that the network signaling overhead can be reduced. To decide on resource allocation to all vehicles, DQN is utilized by centralized BS. Also, a distributed decision-making and spectrum-sharing architecture is proposed for each V2V link to reduce the execution time for resource allocation. Simulation results demonstrate that the proposed schemes are robust to feedback interval variations, input, and feedback noise and achieve near-optimal performance.

The authors in [118] present a DRL-based decentralized mode selection and resource allocation (channel and power) approach for cellular V2X communications to maximize the sum capacity of V2I users while ensuring the latency and reliability requirements of V2V pairs. A two-timescale federated DRL-based algorithm is further developed to overcome limited local training data at vehicles and obtain robust models. On a large time scale, clusters of nearby vehicles are formed using a graph-based clustering technique. Then on a small timescale, vehicles in the same cluster cooperate to train the robust global DRL model through federated learning. Further, the global DRL model can be directly downloaded to newly activated V2V pairs to avoid a time-consuming training process.

To mitigate the interference caused by unstable V2V links to the V2I links[119], the authors propose a multi-agent DDQN-based scheme to efficiently allocate the RBs to V2V links such that the sum capacity of V2I network can be maximized while satisfying the delay and reliability constraints for V2V links. Further, the proposed scheme adopts centralized training processes and distributed execution to ease the implementation. Also, a binarized weight algorithm is introduced to accelerate the DNN learning process and improve the computational complexity. A novel decentralized MARL-based spectrum access algorithm is proposed to maximize the throughput of V2I links while maintaining the successful PDR above a threshold for better reliability in V2I links in a cellular V2X network[120]. Each vehicle agent consists of an action selector module and a message generator module, and the agents learn emergent communication to enable collaboration among themselves. For inter-agent communication, an additional dedicated channel is used. Simulation results show that the proposed algorithm outperforms the baseline techniques without inter-agent communication regarding throughput and PDR.

V. INTERFERENCE MITIGATION IN HETEROGENEOUS CELLULAR NETWORKS (HCNS)

A. FEATURES AND TYPES OF INTERFERENCE IN HCNS COMMUNICATION SYSTEMS

HCNs represent a paradigm shift in wireless communication, offering a diverse and dynamic environment where

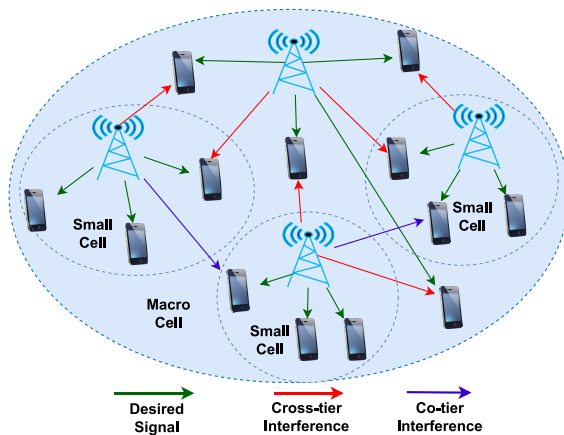


FIGURE 12. Illustration of HCNs communication system with co-tier and cross-tier interference.

macrocells coexist with small cells. This architectural innovation is significant in the context of next-generation mobile communication technology. This flexible and efficient approach enhances the overall network performance by increasing network capacity, improving spectral efficiency, meeting higher data rate demands, and improving user experience [121]. However, with all these advantages, the coexistence of macro and small cells introduces the challenge of co-tier (within the same tier) and cross-tier (among different tiers) interference management [122]. An illustration of a two-tier HCN with one macro cell and multiple small cells causing co-tier and cross-tier interference with each other is presented in Fig. 12. Consequently, effective interference mitigation strategies such as spectrum allocation, power control, cell range expansion (CRE) with biasing, and on-off switching of BSs are used to mitigate these challenges and ensure seamless connectivity. Spectral allocation, such as co-channel and orthogonal channel deployment to different tiers, is critical in controlling cross-tier interference. In the co-channel deployment scenario, almost blank subframe (ABS) [123] and fractional frequency reuse [124] are widely utilized techniques to manage cross-tier interference. With the orthogonal spectrum deployment, mainly co-tier interference needs to be tackled. This section provides an in-depth study of state-of-the-art works utilizing intelligent ML-based techniques to provide effective strategies to mitigate co-tier and cross-tier interference in the HCNs domain. Fig. 13 summarizes network systems, objectives, wireless techniques, and ML algorithms utilized by the works discussed in this section.

B. RELATED WORKS IN HCNs COMMUNICATION SYSTEMS AND THEIR QUALITATIVE COMPARISON

The authors in [125] consider a HetNet consisting of a macrocell overlaid with femtocells. In this work, an ML-based interference classification and offloading scheme is proposed to mitigate the co-tier interference of femtocells and maintain the QoS for video delivery in terms of

throughput, delay, and packet loss rate. Specifically, using the random forest and CNN, two different algorithms are designed to classify the users with high and low interference classes. Then, the high interference traffic is offloaded to the nearby small cell to maintain the QoS. Simulation results show that the proposed scheme performs better than the proportional fair scheduling and cognitive approach regarding delay, throughput, and packet loss ratio. The study in [126], proposes a contextual multi-arm bandit (MAB) approach based semi-distributed scheme having one global learning agent and multiple local controllers for joint interference coordination and energy saving in Hetnets. In research [127], the authors proposed a big data-driven framework and MAB algorithm-based online learning method to mitigate the inter-cell interference in HetNets consisting of macro and small cells sharing the same spectrum. Specifically, the sampled data from the network is used by three modules, i.e., action availability selector, MAB with limited action availability, and network change detector to effectively configure the eICIC parameters such as ABS ratio and CRE bias without any prior information of networks about traffic load, topology, and scheduling algorithm. Numerical results show that the proposed scheme provides better results in terms of throughput when compared to benchmark online learning techniques such as stochastic gradient ascent (SGA) and MAB algorithms.

In study [128], a joint problem of interference coordination and energy saving is investigated in HetNets consisting of macro and pico cells. In this paper, the joint problem is formulated as an MDP assuming periodic patterns in users' traffic demand so that efficient network configuration can be selected by predicting network load. Further, to solve MDP, an approximate dynamic programming (ADP) algorithm is proposed using the certainty equivalent control (CEC) principle and ML techniques (neural network and polynomial regressor) while satisfying QoS constraints. It is claimed that the proposed scheme for joint control of interference coordination and energy saving performs better than the disjoint control regarding both QoS and energy consumption reduction. Furthermore, for the joint control of interference coordination and energy saving in HetNets, an online learning framework based on a new variant of the contextual bandit approach is proposed while fulfilling the rate threshold-based QoS constraints of UEs [129]. Specifically, a two-level framework is proposed with a central controller and multiple distributed local controllers (one per macro cell). The algorithm is named Bayesian Response Estimation. Threshold Search is implemented on the central controller to learn each control action and its suitability for QoS and energy consumption as a function of traffic intensity. Further, The local controllers make local decisions based on control actions learned by the central controller. Numerical results demonstrate that the proposed algorithm outperforms the benchmark schemes regarding energy saving, rate threshold, and convergence speed.

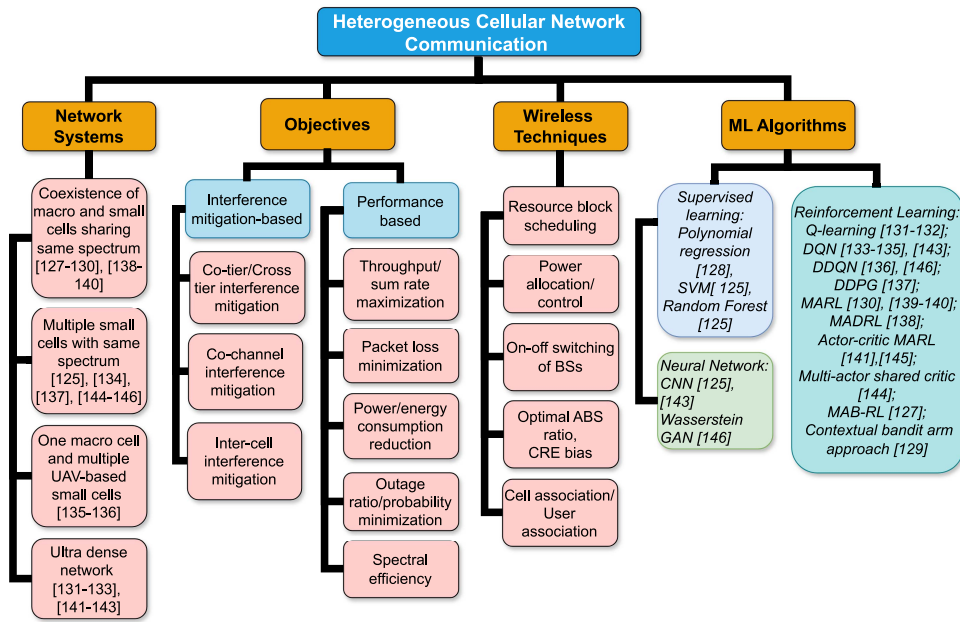


FIGURE 13. Summary of related works for interference mitigation in heterogeneous cellular networks communication.

In research [130], a power optimization problem is modeled using a multi-agent MDP-based distributed framework to mitigate interference in a dense HetNet consisting of a macro cell overlaid with small cells. To solve the problem, a reward function is designed based on an optimization problem, and a Q-learning-based distributed small cells' power allocation algorithm is proposed to maintain the QoS of macro UEs. It is claimed that the macro UE QoS can be maintained even in a high density of small cells by adequately selecting the distributed or cooperative learning approach. Then, in studies [131] and [132], the authors consider the small cells-based ultra-dense HetNet where Q-learning [131] and cooperative Q-learning [132] techniques are used to adaptively allocate the TX power to small cell base stations such that co-tier and cross-tier interference can be controlled while maintaining the minimum capacity requirements of macro cell UEs and small cell UEs. The numerical results show the improvement in the minimum capacity thresholds of macro and small cell UEs. Furthermore, both independent and cooperative Q-learning-based TX power management schemes are considered to control the co-tier and cross-tier interference in ultra-dense HetNet consisting of macro cells and small cells [133], where the ϵ -greedy policy is followed for optimal convergence. It is demonstrated that cooperative learning significantly impacts users' capacity and the sum capacity of cooperating small cells. Also, an optimal clustering algorithm is proposed to reduce the computational complexity, which decreases the cooperative Q-learning computation time by four times.

A 5G Hetnet consisting of one macro cell and multiple small cells with mobile users is considered in [134]. This paper proposes a DRL-based dynamic TDD configuration algorithm to allocate the uplink and downlink resources to

tackle inter-cell interference (ICI) and D2D link interference. The proposed Q-value iteration-based DRL algorithm reflects the complex states of HetNet and adaptively changes the TDD uplink and downlink ratio following evaluated rewards. Numerical results demonstrate that the proposed algorithm outperforms the conventional TDD resource allocation algorithm regarding packet loss rate and network throughput. Further, a HetNet with a macro cell and multiple UAV-based small cells is considered to offload the traffic of congested macro cells or to provide service at emergency sites [135]. In this scenario, sequential and DQN-based algorithms are proposed to optimize UAVs' positions and the ICI coordination parameters to control ICI and maximize spectral efficiency. Simulation results compare the two computationally less complex proposed algorithms with the brute force algorithm and show that the proposed techniques provide results very close to that achieved by the brute force method. Then, in research [136], a DDQN-based algorithm is proposed to solve a similar problem stated in [135] with better spectral efficiency performance.

An interference alignment scheme is adopted to mitigate the intra-cell interference and suppress the inter-cell interference in HetNets consisting of dense small cells [137]. A centralized power control scheme is adopted in macro BSs. Specifically, the power control problem to maximize the sum spectral efficiency is formulated as an MDP and is solved by the DDPG approach. Numerical results show that the proposed scheme outperforms the existing benchmark algorithms (random, Q-learning, DQN, etc.) regarding convergence, complexity, and spectral efficiency performance. The study in [138] considers a beam-forming framework, for HetNet with multiple APs such as Macro, micro, and femtocells, that utilizes the multi-agent DRL

to mitigate the inter-AP interference and to maximize the system downlink sum rate. Specifically, each AP is treated as an agent, which exploits an online policy DNN to generate an AP's beamforming vector and an online Q-function DNN to evaluate the appropriateness of this beamforming vector. Further, a centralized reward scheme is proposed to train Q-function DNNs and the policy DNNs of all the APs, which gets updated in a distributed manner. Numerical results show that the proposed method outperforms the benchmark beamforming schemes in terms of convergence speed and the downlink sum-rate performance of the system.

MARL-based decentralized cell association and interference management strategies are proposed by optimizing the downlink TX power and CRE bias parameter in a Hetnet consisting of macro and pico cells [139]. Simulation results show that the proposed scheme outperforms the static eICIC solutions, such as fixed CRE bias, fixed ABS, and resource partitioning, regarding per-user average throughput. Further, a joint power control and cell association-based ICIC approach is proposed in HetNet consisting of macro cells and pico cells [140]. Here, in the time domain, pico cells learn power allocation and CRE bias to fulfill the QoS requirements of its UEs and macro cell self-organizes its TX power to serve macro UEs while satisfying pico cells' interference constraints. In the frequency domain, single-flow CA (carrier aggregation) and multi-flow CA approaches are investigated in which users can be served simultaneously by one BS and two (or more) BSs. The proposed MARL-based solution is compared with several benchmark schemes, such as static CRE and resource partitioning. Numerical results state that the proposed scheme outperforms the benchmark schemes regarding average UE throughput in the time domain and cell-edge UE throughput in the frequency domain.

In research [141], the authors investigate the inter-cell interference mitigation problem for autonomous ultra-dense networks by proposing a fully distributed self-learning interference mitigation (SLIM) scheme using a model-free MARL framework. Precisely, the actor-critic algorithm controls power to mitigate inter-cell interference while guaranteeing QoS to users. Further, the Mean Field Theory avoids agents' interaction and addresses the dimensional complexity of joint action in MARL by approximating the action value function. Numerical results show that the proposed scheme outperforms the benchmark schemes, such as random power control, soft frequency reuse, and gradient-based distributed power control, regarding TX rate and power consumption reduction. Furthermore, a joint channel allocation and BS ON-OFF switching scheme is proposed to manage the CCI in densely deployed HetNets [142]. Specifically, for channel allocation, two algorithms are proposed. The first algorithm is implemented at the BSs level, where allocating the channel of a non-cooperative game dynamically is solved using a no-regret learning approach. The second algorithm is executed at the UE level, where each UE provides feedback about the minimum CCI channel to its

associated BSs. Then, BSs use the UE feedback to select their channels. Further, both channel allocation algorithms use an ON-OFF switching scheme to minimize energy consumption. Simulation results demonstrate that the second channel allocation algorithm combined with the ON-OFF switching scheme performs better in load balancing, users' rate, and energy consumption compared to an interference-aware dynamic channel allocation benchmark scheme.

In [143], the authors propose an RL-based TX power optimization scheme for ultra-dense small-cell systems to control downlink inter-cell interference without knowing neighboring cells' channel states and interference distribution. Additionally, a DRL-based framework reduces computational complexity and enhances communication efficiency. Numerical results show these schemes outperform the benchmark big-data self-organizing network (Bi-SON) in throughput, energy consumption, and interference mitigation.

In [144], a DRL-based multi-agent power control algorithm with centralized training and distributed execution is designed to mitigate interference in Hetnets with multiple APs sharing the same spectrum. The multiple-actor-shared-critic (MASC) method trains local DNNs of each AP agent separately, controlling TX power based on local observations. The proposed scheme outperforms conventional power control algorithms, such as fractional programming and weighted MMSE, in computational complexity and sum rate.

In research [145], the power control problem is solved by a single agent and MARL utilizing a soft actor-critic (SAC) framework to mitigate the multi-cell interference in the uplink communication scenario and enhance the EE. Numerical results show that the proposed scheme approaches the global optimum provided by the branch and bound algorithm with less computation complexity. The authors in [146] consider a HetNet with multiple small cells where an intelligent resource (sub-channels) scheduling scheme is proposed for ICIC. This paper's ICIC problem is formulated as a distributed inverse RL problem. Further, to solve this problem from both Q-value and policy perspectives, a self-imitating learning (SIL) algorithm incorporating Wasserstein GAN and DDQN algorithms is proposed. Numerical results demonstrate that the proposed scheme outperforms the benchmark schemes regarding overall network throughput.

VI. INTERFERENCE MITIGATION IN D2D COMMUNICATION NETWORKS

A. FEATURES AND TYPES OF INTERFERENCE IN D2D COMMUNICATION NETWORKS

Due to the significant increase in users and the growing demand for seamless connectivity everywhere, integrating innovative technologies and concepts has become imperative. Consequently, incorporating D2D communication technology is a pivotal consideration for 5G and beyond networks [147]. D2D communication has emerged as a transformative paradigm in wireless communication, allowing devices close to each other to communicate directly without central BS. D2D communication enhances

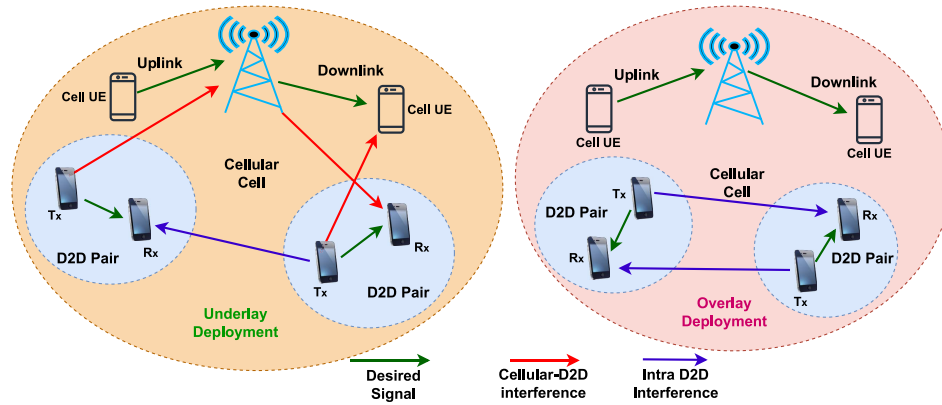


FIGURE 14. Illustration of interfering signals in D2D communication networks for both underlay and overlay deployment.

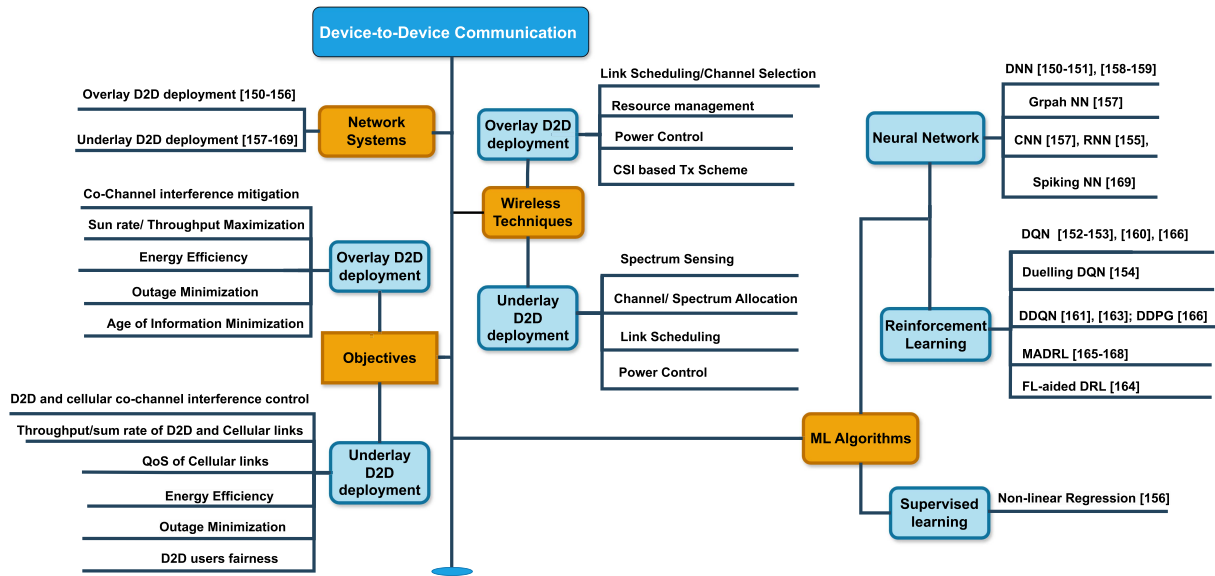


FIGURE 15. Summary of related works for interference mitigation in D2D communication domain.

the network capacity and performance by achieving high data rates, spectral efficiency, EE, and expanding network coverage with minimal latency [148]. However, integrating D2D technology into conventional networks presents several challenges, including power control, spectrum allocation, and interference management. Two deployment scenarios exist for the In-band D2D communication: underlay and overlay deployments with the existing cellular networks [149]. In underlay deployment, D2D pair communication relies on the cellular BS for coordination, control, and resource allocation. In this scenario, D2D pairs communication coexists with cellular communication, sharing the spectrum with existing cellular users. Therefore, devices engaged in D2D communication can create CCI with cellular users and other D2D users. In overlay deployment, D2D communication is independent of existing cellular infrastructure, and a separate/dedicated set of frequency bands is explicitly allocated for D2D communication. Hence, CCI management is required among different D2D pairs in overlay deployment.

An illustration of D2D communication networks consisting of desired and interfering signals is presented in Fig. 14 for underlay and overlay scenarios. This section provides a detailed survey of state-of-the-art works for interference management in both D2D deployment scenarios through proper TX power control and spectrum allocation schemes utilizing different ML techniques. The summary of network systems, objectives, wireless techniques, and ML algorithms utilized by works discussed in this section can be visualized in Fig. 15.

B. RELATED WORKS IN D2D COMMUNICATION NETWORKS AND THEIR QUALITATIVE COMPARISON

1) OVERLAY D2D COMMUNICATION NETWORKS

In research [150], the RF or VLC band selection problem is considered for D2D communication to maximize EE and minimize the outage probability. In this work, utilizing the limited channel knowledge in terms of received power and sum interference at D2D receivers, a DNN-based band

selection solution is proposed. To further improve the band selection decision, a heuristic algorithm, which utilizes the output of the DNN algorithm, is proposed. Numerical results show that the proposed scheme outperforms the heuristic algorithm without DNN regarding EE, outage ratio, and computation complexity. In [151], the authors investigate a link-scheduling scheme to jointly optimize the throughput and AoI for D2D communication networks having a last come, first serve-packet replacement (LCFS-PR) policy for packet TX. Specifically, a DNN structure maps the geographic location information to optimal scheduling parameters without CSI feedback. Numerical results show the trade-off between AoI and throughput, i.e., one metric can be improved at the cost of another metric. Also, as the number of scheduling links increases, DNN-based algorithms can approach near-optimal performance while offering lower computational complexity than iterative methods.

The authors in [152] consider frequency division duplex (FDD)-based overlay D2D communication network, mainly focusing on feedback overhead reduction and resource management for CCI. Specifically, the partial or binary feedback system is used to reduce the feedback overhead, and a DRL-based TX policy is used to maintain the average sum rate per the level with full feedback. Numerical results show that compared to the full feedback method, the proposed partial and binary feedback methods reduce the feedback overhead 49% and 92%, respectively. In work [153], two algorithms are proposed for solving the joint channel allocation and power control problem to mitigate the CCI among D2D communications and to maximize the weighted sum rate. The first algorithm is based on fractional programming (FrP), a centralized approach that requires global CSI, which is unsuitable for scalability and creates a high signal overhead. The second algorithm is based on DRL, which uses the D2D pairs to optimally select the channel and control the TX power in a distributed manner by utilizing local information. Numerical results indicate that the DRL-based algorithm performs near to the FrP-based algorithm without global CSI and with significantly less signaling overhead.

Further, for CCI mitigation and average sum rate maximization in a D2D communication network, a distributed TX scheme is proposed utilizing a dueling QDN algorithm that learns each state's value without learning the effect of each action [154]. Numerical results indicate that the proposed scheme outperforms no-control methods in average sum rate and matches the performance of centralized opportunistic and sub-optimal schemes at high SNR. However, the feedback and signaling overhead is significantly low in the proposed distributed scheme compared to the centralized schemes. The research in [155] proposes an interference feature extraction-aided RNN scheme to control the TX power of transmitters in a distributed manner with full frequency reuse for D2D pairs. Linear filters are designed to extract different local interference patterns from outdated channel information, and they enable the network to learn and predict the real-time interference pattern to

make power decisions. Further, an input reduction process is designed to reduce the computational time and system overhead. Simulation results show that the proposed scheme outperforms the benchmark schemes such as maximum power Tx and RNN-RL regarding the average sum rate.

2) UNDERLAY D2D COMMUNICATION NETWORKS

The study in [156] considers the coexistence of D2D and UAV communication networks is considered where UAV senses the spectrum used by D2D communication before utilizing it to avoid CCI. This work develops an ML-assisted stochastic geometry framework to characterize the aerial spectrum efficiency (ASE) and coverage probability of UAV and D2D communication by analyzing the spatial false alarm probability and the spatial missed detection probability of a typical UAV spatial spectrum sensing. Specifically, the Gaussian non-linear regression method determines the PDF (probability density function) parameters for the received signal strength of UAV networks. This helps in further analysis of ASE and coverage probability. The proposed method helps find the optimal radius for spatial spectrum sensing and UAV TX power to maximize the ASE of UAV networks while maintaining the minimum ASE requirement for D2D communication. The authors in [157] consider D2D communication-based IoT systems where to solve the resource allocation problem a Graph Neural Network (GNN)-based supervised framework is proposed. Specifically, a layered structure is investigated where CNN is used for feature aggregation, and DNN is used for feature combination to update the feature vector at each node iteratively. Further, two algorithms are proposed for resource management. First for link scheduling in D2D communication and second for joint power and channel allocation for D2D under-laid cellular network scenario. Numerical results demonstrate that the proposed methods outperform the unsupervised GNN and conventional DNN approaches regarding average sum rate, training sample efficiency, and computational complexity.

In research [158], an underlay D2D communication network is considered. In this work, the authors utilize a heuristic equally reduced power scheme together with DNN to control the TX power of D2D users such that the sum rate of D2D users can be maximized while fulfilling the rate constraints of D2D users and interference constraints to cellular users. Simulation results show that the proposed scheme achieves a near-optimal sum rate with low computation complexity. However, the channel allocation was not discussed in this work. Further, a DNN-aided resource allocation (power allocation and channel allocation) scheme is proposed in [159]. The proposed scheme can work in a centralized manner with full CSI and in a distributed manner with partial CSI such that the spectral efficiency of the D2D users is maximized while maintaining a minimum data rate for cellular users. Further, a novel hybrid training strategy combining supervised and unsupervised training is

proposed. It is reported that the proposed scheme can achieve near-optimal performance with low computation complexity.

A centralized DRL algorithm is proposed for power allocation to D2D transmitters in an under-laid D2D communication network having a time-varying environment to maximize the system capacity and user quality of experience [160]. A new reward function is adopted to improve the user experience quality. Numerical results show that the proposed method outperforms the Q-learning and random assignment schemes regarding system capacity and user experience quality. In [161], an under-laid D2D communication network is considered where to avoid CCI and to maximize the sum throughput while maintaining the communication quality of cell edge users; a centralized DDQN-based spectrum access technique is proposed for D2D pairs to access the spectrum shared with cellular UEs. Further, to maintain resource allocation fairness among D2D pairs, a fairness factor is included in the objective function. Numerical results indicate that the proposed algorithm achieves optimal performance and outperforms the scheme, which requires BS cooperation. However, the proposed schemes are centralized in nature and the investigation for joint power control and spectrum allocation is missing in [160] and [161].

The research in [162] proposes a distributed DL-based power allocation algorithm to maximize the spectral efficiency of the under-laid D2D communication network while controlling the CCI to cellular communication. Specifically, the mapping between transmit power and location information is memorized by D2D devices to achieve the near-optimal result. Numerical results show that the proposed scheme achieves similar spectral efficiency to the Iterative approximated distributed rate maximization approach without involving other nodes. In this line, the authors in [163] propose a distributed resource matching (wireless channel matching and power allocation) scheme based on DDQN to maximize the sum rate and EE of D2D links while maintaining the SINR threshold of cellular users. Simulation results demonstrate that the proposed scheme outperforms the DQN-based resource allocation scheme regarding the sum rate. Further, a DRL-based federated learning-aided decentralized resource allocation scheme is proposed in [164] to maximize the throughput and minimize the power consumption while guaranteeing the QoS requirement of both D2D and cellular users. The proposed scheme is evaluated through simulation for 5G mm-wave and 6G THz scenarios. It is shown that the proposed scheme outperforms pure DRL and random methods in terms of throughput and power consumption.

The authors in [165] propose an actor critic-based MARL distributed algorithm for spectrum allocation to mitigate the interference to cellular and other D2D links and to enhance the sum rate of D2D links. The training is performed in a centralized manner by sharing global policies, actions, and historical states. Further, neighbor users' historical information is used to reduce training complexity. Simulation

results demonstrate that the proposed algorithm outperforms the Q-learning and DQN-based algorithms regarding sum rate, the outage probability of cellular links, and convergence time. A DRL-based scheme for joint RB allocation and power control of D2D pairs is introduced to maximize the sum rate while ensuring fairness among all users, as described in [166]. Specifically, the MDP converts the optimization problem into the MADRL problem. To solve this problem, multi-DQN is used for RB allocation and distributed DDPG is used to convert stochastic policy to deterministic policy. Furthermore, a conventional optimization scheme with DDPG is used for power control of D2D pairs and cellular UEs. Simulation results show that the proposed scheme outperforms distributed DRL schemes regarding sum rate and fairness among all users. Further, in [167], a Stackelberg game (SG)-guided MADRL scheme is proposed for power control and channel allocation to D2D users in a distributed manner. Specifically, based on SG equilibrium, a Stackelberg Q-value is defined to guide the learning direction and reduce the convergence time. Simulation results validate that the proposed SG-guided MADRL outperforms normal MADRL regarding channel capacity, average utility, and training time.

In research [168], a MADRL-based power control and spectrum allocation scheme is investigated to mitigate the CCI and optimize the throughput of cellular and D2D links while satisfying the delay constraints of D2D links. Specifically, priority sampling (PS) is used to learn the dominant features to overcome the impact of large state space. The PS-based D3QN (PS-D3QN) algorithm is used in a distributed manner for resource management. Simulation results show the proposed algorithm outperforms the DRL baselines regarding sum throughput. Then, in [169], the authors consider the spiking neural network cooperated with DRL for power control and channel selection to optimize the resource allocation for many D2D users in an underlaid D2D communication network. Further, mean-field RL (MFRL) is utilized to approximate the D2D users' interactions. Specifically, two algorithms are developed by integrating MFRL with spiking actor-critic (AC) and spiking proximal policy optimization (PPO). It is asserted that the proposed algorithm is effective for many D2D users and outperforms the simple PPO and AC-based algorithms regarding overall throughput and convergence rate.

VII. INTERFERENCE MITIGATION IN IOT NETWORKS

A. FEATURES AND TYPES OF INTERFERENCE IN IOT COMMUNICATION NETWORKS

IoT is a communication environment where many devices communicate with each other, and 5G is considered an essential and emerging technique for supporting IoT communication. It is anticipated that, by the year 2030, approximately 80 billion devices will be interconnected within a network [170]. 5G networks address the major existing challenges in cellular networks, such as low latency, ultra-reliability, and efficient massive connectivity while supporting

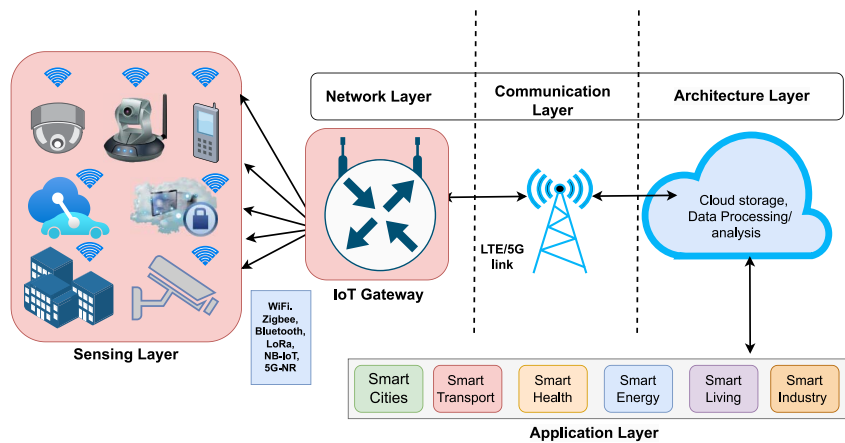


FIGURE 16. IoT network architecture.

mobility [171]. 5G-enabled IoT applications include services in smart cities, smart homes, self-driving cars, drones, virtual reality, healthcare, and industrial automation, which improve people's daily lives [172]. All the devices used in these applications communicate with each other or an IoT gateway (IoTG) or an access point through the low-power wide-area networks technologies, such as LoRa, WiFi, SigFox, ZigBee, and NB-IoT (Narrow-band IoT) [173]. Also, IoT devices and cellular users may transmit and receive data while sharing the same spectrum in 5G-enabled IoT. Usually, IoT devices communicate with an IoTG, and cellular users communicate with BS, using the same spectrum, causing interference at BS, cellular user receiver, and IoTG, which affects the IoT system's efficiency and reliability [174]. The layered architecture of an IoT network with sensing, network, communication, architecture, and application layer is presented in Fig. 16. In the literature, diverse solutions have been proposed to overcome the interference problem and enhance IoT networks' overall system throughput and EE via proper resource allocation, spectrum management, and TX power control. This section provides a detailed state-of-the-art survey of ML-based intelligent interference mitigation techniques in IoT networks. Fig. 17 illustrates the various network systems, objectives, wireless techniques, and machine learning-based algorithms examined across the studies discussed in this section.

B. RELATED WORKS IN IOT COMMUNICATION NETWORKS AND THEIR QUALITATIVE COMPARISON

An IoT network with Wi-Fi and Bluetooth low Energy (BLE) communications sharing the 2.4 GHz band is considered in [175]. This work presents an LSTM-based lightweight interference prediction model to predict the Wi-Fi interference on BLE communication for resource-constraint IoT devices. The proposed algorithm is compared with the Auto-Regressive Integrated Moving Average (ARIMA) model where the RMSE results on prediction error indicate the better performance of the LSTM-based

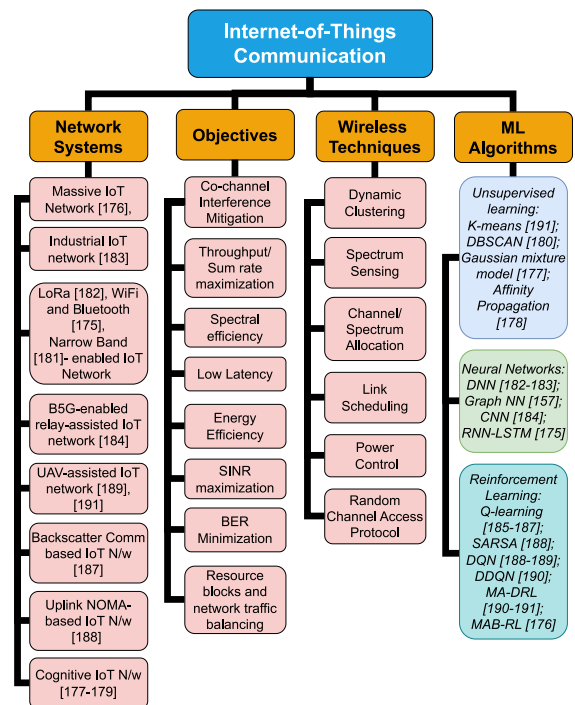


FIGURE 17. Summary of related works for interference mitigation in the IoT communications domain.

proposed approach than ARIMA model. Further, a massive IoT network is considered in [176], where to enhance the communication success rate, a Tug-of-War dynamics-based MAB algorithm is proposed for channel selection by each IoT device in a distributed manner. The proposed algorithm is implemented on an IoT device (Lazurite Pi Gateway) for experiment and on the ns-3 platform for simulation. Numerical results demonstrate that the proposed approach for channel selection outperforms the random frequency hopping and ϵ -greedy algorithm in terms of communication success rate in high-density IoT network environments.

The research in [177] considers a cognitive IoT network where a new receiver is designed for the secondary user (SU)

in a concurrent spectrum access (CSA) model to efficiently handle the primary user (PU) interference with minimal knowledge of PU system such as modulation type of PU interference. Specifically, an iterative framework consisting of a linear estimator, a demodulation and decoding module, and a clustering module is designed. The main focus of this work is on the realization of clustering module to cluster interference using Gaussian mixture model (GMM)-based expectation maximization (EM) algorithm. Further, extrinsic information technique and modulation constraints (MC) are used to enhance the performance of the conventional GMM-based EM algorithm, resulting in the Ext-MC-EM algorithm. It is shown that the proposed Ext-MC-EM algorithm outperforms the K-means algorithm and MC-EM and normal EM algorithm regarding interference mitigation and BER minimization. Further, in work [178], a similar system setup is considered as stated in [177] with a different objective: multiple PU-interference cancellation at the SU receiver. In this work, an affinity propagation (AP) clustering algorithm is used, which works well without cluster centroid initialization, which is not the case with the GMM-based EM clustering algorithm. Numerical results indicate that the proposed AP clustering algorithm outperforms the k-means and GMM-based EM algorithms to handle multi-interference. In the same line, the authors in [179] consider an asynchronous cognitive IoT (C-IoT) network with a system model similar to that considered in [177]. Since the PU and C-IoT devices are not synchronized at the symbol level, the interference at the C-IoT device receiver has the first-order Markov property over time. Considering this property, to improve the performance of the iterative receiver of SU, a hidden Markov Model-based AP (HMM-AP) clustering algorithm, which exploits the temporal correlation of the interference from PU, is proposed. It is claimed that the proposed HMM-AP algorithm outperforms the K-means, EM, and AP-based clustering algorithms in terms of BER minimization and interference mitigation.

In study [180], an ML framework is proposed to sense and model the interference in the IoT frequency band. To capture the short-time spectrum occupancy in the shared bands, which determines the interference, a software-defined radio (SDR) is used with an energy detection module. Also, to capture the frequency and temporal behavior of the spectrum, two models are used, namely the Poisson distribution arrival rate and semi-Markov chains. To improve energy detection performance, the use of an unsupervised ML algorithm, DBSCAN, is proposed that reduces the FP and FN rate in capturing the spectrum occupancy. The authors in [181] utilize the sparse cross-entropy minimization learning algorithm to recover the narrow-band (NB) interference of the NB-IoT network to the broadband LTE systems. Further, regularization is introduced to the loss function to enhance the recovery accuracy and convergence rate. Also, the proposed framework is extended to the MIMO system by exploiting the spatial correlation of the NB interference at multi-antennas. The proposed approach is claimed to be

more efficient than conventional methods in reducing NB interference in IoT networks within the LTE system and improving spectral efficiency.

A long-range (LoRA)-enabled static and dynamic IoT network is considered in [182], where to tackle the resource allocation issue an AI-empowered resource allocation (AI-ERA) approach is proposed. In the proposed approach, the DNN model is first trained offline using a synthetic dataset generated by ns-3. Then the pre-trained DNN model is utilized online to assign a suitable spreading factor to the end device before each uplink TX. Numerical results demonstrate that the proposed scheme outperforms the conventional adaptive data rate approaches and SVM in terms of packet success ratio and convergence period. Then, a DNN-based directional MAC (DMAC) protocol is proposed to tackle the deafness problem for industrial IoT networks [183] that utilize mmWave band for communication among IoT devices through direction antennas. The proposed DNN model first identifies the reason for communication failure, which can be due to interference or deafness, and then avoids deafness by rescheduling the TX through another beam direction. It is demonstrated that the proposed ML-based DMAC outperforms the conventional DMAC protocols regarding deafness duration minimization and throughput maximization.

In research [157], a wireless IoT network is modeled with a directed graph, in which nodes represent the desired links and edges represent the interfering links. Further, a GNN-based supervised learning framework is used to solve resource allocation problems involving integer parameters with less execution time. Specifically, a constrained cross-entropy algorithm generates the samples to reduce the computational complexity. Considering D2D communication as a promising technology for IoT networks, the proposed scheme is verified for D2D link scheduling and joint channel and power allocation for D2D underlaid cellular networks. Numerical results demonstrate that the proposed scheme outperforms the unsupervised GNN, the graph embedding method, and conventional DNN regarding average sum rate, stability, and computation complexity. The authors in [184] consider the multiband relay nodes assisted 5G-enabled massive IoT network where relay nodes are used for traffic offloading of cellular base stations. This work proposes a lightweight DL technique based on the CNN model to select the best channel for the relay node to transmit and receive data based on SINR metric quality. Also, the scheduling algorithm at the relay node utilizes channel quality measures to determine the coding and modulation rates to transmit data to the next node. It is claimed that the Shallow CNN model performs better than Deep CNN concerning throughput, processing delay, and memory consumption. However, the CSI feedback requirement for the proposed approach creates an extra overhead on the network.

A massive-scale IoT network of machine-type communication devices (MTCs) is considered in [185], where to reduce congestion and overload problems, MTCs are clustered by the K-means algorithm based on their spatial

distribution. Then, a centralized approach is presented for slot allocation utilizing a graph coloring-based algorithm. Furthermore, to mitigate inter-tier interference, a distributed block-based Q-learning algorithm with various reward schemes is proposed for block of slots scheduling in MTCDs constituting clustered IoT networks. It is shown that the proposed distributed slot allocation scheme outperforms the centralized approach in terms of average SIR and convergence capabilities. Then, a spectrum sensors-aided IoT network is considered In [186], where to enhance the spatial spectrum re-usability, an RL-based dynamic spectrum access (DSA) algorithm is proposed which learns the spectrum access probability for the slotted ALOHA protocol utilized by IoT devices. The proposed RL-based DSA approach is claimed to outperform the distributed Carrier Sensing Multiple Access with Collision Avoidance (CSMA/CA) protocol regarding area spectral efficiency.

The study in [187] proposes an RL-based power allocation scheme in a backscatter communications-based clustered IoT network. First, based on the considered radio environment, states, actions, and rewards of the agent are presented, and then, the Q-learning-based power allocation scheme is proposed to minimize interference and to improve the received SINR of the ambient backscatter IoT network. Numerical results demonstrate the proposed method performs better than the equal power allocation technique in terms of interference control and received signal strength improvement. The authors in [188] consider the uplink TX in a power domain NOMA-based IoT Network. This work proposes two RL-based optimal clustering and resource allocation strategies for IoT users in multi-cell networks to control inter-cell interference and enhance the long-term average sum rate. The first strategy is based on SARSA (State-Action-Reward-state-Action)-learning is suitable for light traffic, and the other strategy utilizing DRL can handle heavy traffic. Numerical results demonstrate that the proposed schemes outperform the conventional OMA (Orthogonal multiple access) based IoT network in terms of mean sum rate. A wireless IoT network is considered with multiple solar-powered UAVs as relay nodes that leverage NOMA to serve the IoT devices [189]. Specifically, a p-persistent slotted ALOHA protocol is used by IoT devices to access the shared channel, and UAVs utilize SIC to decode multiple pieces of information received from IoT devices. This work proposes a joint problem of dynamic multi-UAV altitude control and random channel access control as constrained MDP. Further, to solve this problem, an online model-free constrained DRL algorithm is proposed to learn an optimal control policy. It is asserted that the proposed approach outperforms baseline solutions such as Vanilla policy gradient for total network capacity and energy sustainability of solar-powered UAVs.

To ensure low latency and ultra-reliability in D2D communication for the Internet of Controllable Things, a multi-agent DRL algorithm addresses joint subcarrier and power allocation for D2D pairs [190]. Each D2D pair uses

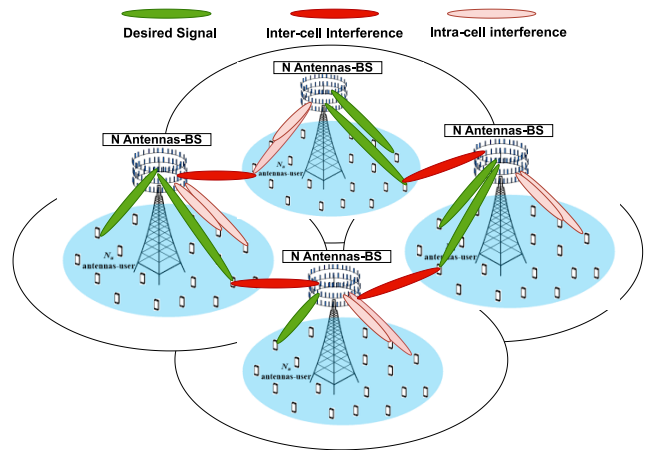


FIGURE 18. Illustration of a dense mmWave-MIMO Network.

multiple subcarriers, and subcarriers can be shared among pairs. The problem is framed as an MDP, and a DDQN-based algorithm learns the optimal policy without CSI knowledge. Each D2D pair, as a learning agent, iteratively adjusts its resource allocation strategy through trial and error, outperforming random subcarrier and power assignment schemes in spectrum efficiency and interference control. In a related study, multiple UAVs collect data from IoT devices [191]. The multi-UAV trajectory design problem, also formulated as an MDP, seeks to minimize mission completion time while avoiding collisions and interference. A multi-agent DRL algorithm optimizes UAV trajectories both distributively and centrally. Numerical results show the proposed approach outperforms IAQL and fully centralized methods in minimizing mission time.

VIII. INTERFERENCE MITIGATION IN MMWAVE-MIMO COMMUNICATION NETWORKS

A. FEATURES AND TYPES OF INTERFERENCE IN MMWAVE-MIMO COMMUNICATION NETWORKS

Millimeter wave (mmWave)-massive multiple input multiple output (MIMO) technology is one of the dominant enabling technologies that are being explored towards achieving the set goals of network performance for the next-generation wireless networks. The use of the mmWave-MIMO system brings together the advantages of huge bandwidth availability in the mmWave spectrum band, and the spatial diversity and multiplexing gains of a massive MIMO system [192]. In the mmWave-MIMO system, precoding (via Zero forcing (ZF), Maximum ratio TX (MRT), MMSE, and Dirty paper coding), beamforming (via analog, digital, and hybrid beamforming), and channel estimation with proper feedback are of vital importance to take advantage of spatial diversity and multiplexing gain [193]. These techniques also help mitigate intra-cell and inter-cell interference to enhance spectral efficiency, EE, and achievable rate performance in ultra-dense networks [194]. An example of a dense mmWave-MIMO communication network with inter-cell and intra-cell interference can be visualized in Fig. 18. Nowadays, with the advancements in ML algorithms, it

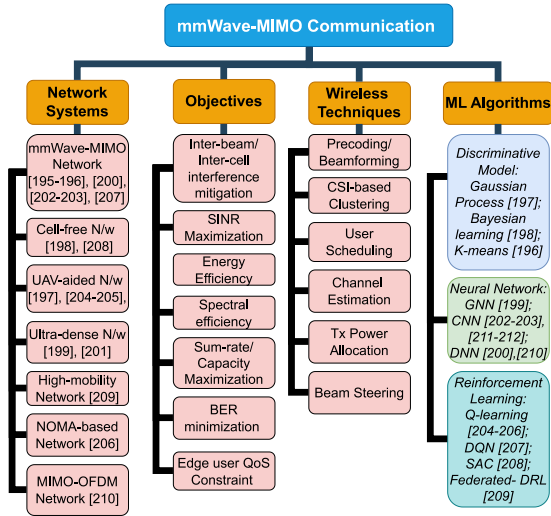


FIGURE 19. Summary of related works for interference mitigation in mmWave-MIMO communication networks.

is being widely explored by researchers to assist channel prediction, beamforming optimization, dynamic resource allocation, and interference management with intelligent ML algorithms. This section provides an in-depth study of ML-based state-of-the-art works for interference management and resource allocation in mmWave-MIMO systems. Fig. 19 presents the summary of different mmWave-MIMO network systems, different objectives, wireless techniques, and ML algorithms considered in the works covered in this section.

B. RELATED WORKS IN MMWAVE-MIMO COMMUNICATION NETWORKS AND THEIR QUALITATIVE COMPARISON

The study in [195] considers the initial access problem for mmWave communication to overcome the signal attenuation and interference problem with optimal beamforming techniques. In this work, a single-path channel model is considered where to get the optimal beamforming vector, an adaptive and sequential beam search algorithm named hierarchical posterior matching algorithm is proposed. The performance of the proposed algorithm is characterized by the error probability and the resolution of actively learning the Angle of Arrival (AoA), which indicates the link quality between users and BS. Numerical results demonstrate that the proposed algorithm can reach an acceptable resolution and lower probability of error of estimated AoA even in a low SNR regime and provide better spectral efficiency than bisection and random-search algorithms. In research [196], an unsupervised learning-based user scheduling scheme is proposed for a large-scale multi-antenna system to improve spatial diversity in uplink MAC and downlink broadcast scenarios. In this work, the Grassmannian K-Means algorithm is first used for clustering the users with high CSI similarity, and then, scheduling is done into MIMO access groups with low CCI. Numerical results show that the proposed scheme outperforms direct greedy scheduling in terms of CCI suppression and spectrum efficiency with lower computation complexity. A

UAV-aided mmWave communication scheme is proposed to improve the QoS of edge users in [197], where LoS links from primary and coordinated BSs to edge users are blocked, and therefore, UAV acts as a relay to communicate to edge users. In this work, it is assumed that due to resource constraints, UAV feeds back CSI only to the primary BS, and then the primary BS sends this information to the coordinated BS through the backbone network. To tackle the CSI feedback delay, a Gaussian process-based ML technique is proposed at coordinated BS to estimate the channel used to perform hybrid beamforming towards the UAV. Furthermore, a Max-SINR-based beamforming compensation scheme is proposed for the primary BS and UAV to eliminate the inter-BS interference. Simulation results demonstrate that the proposed method outperforms the schemes without compensation in terms of BER and sum rate, and performs close to perfect CSI feedback scenarios in terms of sum rate in the low and medium range of SNR. The authors in [198] design optimal hybrid beamformers for unicast, multicast, and broadcast cell-free multi-user mmWave-MIMO systems for capacity maximization and inter-user and inter-group interference mitigation with low complexity. In this work, the optimal beamforming solutions are first derived for the broadcast scenario while considering constraints of total power and per-AP power. Then, optimal successive minimum variance hybrid beamforming schemes are designed for unicast and multicast scenarios to mitigate inter-user and inter-group interference. Next, a Bayesian learning (BL)-based joint hybrid transceiver (with RF and baseband transmit precoding and receive combiners (RC) is designed that does not require AoAs, AoDs, or channel gain information. Lastly, a successive cooperative uplink hybrid beamforming scheme is proposed to maximize the SINR and data rate for the uplink scenario. Numerical results demonstrate that the proposed schemes outperform the no-coordination scenario in terms of interference cancellation and spectral efficiency and perform close to fully digital design.

The research in [199] considers an ultra-dense D2D mmWave network and formulates the joint beam selection and link activation problem for interference control between communication pairs and sum-rate maximization. To solve this non-convex and NP-hard integer programming problem, a GNN-based DL framework is proposed by considering the network's topology. Furthermore, a Lagrangian dual learning framework is proposed to train the parameters of the GNN model. Simulation results demonstrate that the proposed scheme outperforms the greedy approach in terms of weighted sum rate and reaches a near-optimal solution by comparing it with an exhaustive search solution in small-scale D2D mmWave networks. Further, a DNN-based framework is presented for fast and reliable initial access for mmWave networks in [200]. The proposed scheme utilizes only a subset of the available beams by mapping RSSs obtained from a subset of beams to the beam that is best oriented to the receiver and reduces the initial access time compared to the conventional continuous beam sweeping technique. Simulation results show that the

proposed scheme enhances the beam prediction accuracy in LoS and NLoS conditions compared to the conventional initial access scheme. The authors in [201], consider a dense mmWave network where a DL-based centralized solution is proposed for beam management and interference coordination to reduce the computational complexity of radio resource management. Specifically, an optimization problem is formulated for sum rate maximization in a dense mmWave network, and to solve this problem, a beamforming training-aided DNN-based algorithm is used to find the optimal beam directions, beam widths and transmit power allocation to each beam. Numerical results demonstrate that the proposed approach achieves a comparable sum rate to the conventional algorithm with much less computation complexity.

An interference-free achievable rate is introduced for wideband multiuser mmWave massive MIMO systems [202]. To avoid beam conflict and to maximize the sum rate, the beam allocation problem is investigated by decoupling the analog and digital precoders. Then, the beam allocation problem is treated as a multi-label classification problem, and to solve this problem, a CNN-based beam allocation scheme is proposed. A simulated environment is used to generate synthetic data to train CNN offline, and then the trained CNN is used online to predict the beam allocation for all the users. Numerical results show that the proposed scheme reduces the computational complexity to a large extent with marginal sacrifice in sum-rate performance compared to fully digital and Hungarian schemes. The authors in [203] investigate self-interference cancellation and hybrid beamforming (HBF) to enhance the spectral efficiency for full duplex mmWave relay systems. First, an alternating direction method of multipliers-based algorithm is proposed to eliminate residual self-interference and for fully digital beamformers optimization. Then, a majorization-minimization-based algorithm is proposed to optimize the Tx and Rx hybrid beamforming jointly. Furthermore, two learning frameworks (ELM-Extreme learning machine-based and CNN-based) are proposed for HBF design, which estimates the precoders and combiners of full duplex mmWave systems. Numerical results demonstrate that the proposed schemes perform better in terms of spectral efficiency and computation complexity compared to the orthogonal matching pursuit algorithm.

The study in [204] considers mmWave multi-UAV communication network consisting of one leader UAV and multiple follower UAVs. In this work, a Q-learning-based algorithm is proposed to deal with the high dynamics of UAVs for analog beam tracking, which uses coupling coefficients and past and current observations to solve the prediction problem. Further, the desired signal and interference powers are estimated using the received pilot signal for the selected analog beams. This helps decide the digital weights for optimal digital beamforming with the objective of SINR maximization. It is noted that the received power-based selected analog beams may not result in maximum SINR, hence, additional analog beams are

reserved as candidates during the beam tracking. Thus, combining analog beams with their digital weights, and achieving the maximum SINR provides the optimal hybrid beamforming solution. Numerical results demonstrate that the proposed scheme significantly increases beam switching and data TX efficiency. Further, in [205], the authors investigate the beamforming and beam-steering problem in a multi-UAV and multi-antenna mmWave network. In this work, cross-entropy estimation using a robust algorithm is proposed for optimal hybrid beamforming, and a mean field game (MFG) framework is proposed for realizing beam-steering to find the optimal mmWave channel. An RL-based learning approach is used to obtain the mean field equilibrium. Finally, a joint beamforming and beam-steering algorithm is presented to maximize the sum rate of the considered network. Numerical results demonstrate that the proposed scheme outperforms the classical algorithms in terms of sum rate, spectral efficiency, and EE. The authors in [206] consider a NOMA-based mmWave network where to mitigate the inter-beam inter-cell interference and to enhance the sum-rate capacity of the network, an RL-based algorithm is proposed to optimize the user-cell association and inter-beam power allocation jointly. Optimal user-cell association balances users' attachments across cells, improving the SIC performance. Numerical results demonstrate that the proposed scheme outperforms the equal power allocation scheme in terms of network sum rate.

In research [207], the authors consider the downlink communication scenario in a multi-access OFDM cellular network with multi-antenna BSs and single-antenna receivers. This work proposes a centralized DRL-based algorithm for joint beamforming, power control, and interference coordination to maximize the downlink SINR and sum-rate capacity. This algorithm does not require CSI and channel estimation, while it needs users' locations and SINR information for every millisecond as feedback. Numerical results show that the proposed scheme outperforms tabular Q-learning and fixed power allocation algorithms in terms of SINR and sum-rate capacity for voice bearers in the sub-6 GHz band, and achieves maximum sum rate with less time complexity for data bearers in the mmWave band. Further, the authors in [208] consider a mmWave MIMO cell-free network in which solving the beamforming problem in a centralized manner is highly complex. Therefore, to solve this problem in a distributed manner, dynamic clustering of the network into a group of cell-free subnetworks consisting of an independent set of APs and UEs is proposed based on the DRL algorithm. To mitigate inter-subnetwork and intra-UE interference while maximizing TX rate per UE, a hybrid beamforming scheme is proposed. Specifically, a DRL model continuous-action subspace is proposed for analog beam-steering, and within the environment of the DRL agent for analog beam-steering, digital beamforming is solved as a convex optimization problem. Numerical results demonstrate that the proposed scheme outperforms the conventional all-digital beamforming scheme regarding TX rate. The

study in [209] proposes a DRL-based algorithm for fast and efficient mmWaves beamforming in MIMO system to mitigate the signal attenuation and interference problem for serving highly mobile users (for eg. V2I and high-speed train scenario). In this work, to solve the EE maximization problem while fulfilling the QoS constraint of each user, access points (APs) association per user is considered, and hybrid beamforming matrices are obtained using the proposed DRL-based algorithm. Furthermore, to reduce the communication overhead a federated DRL-based distributed scheme is also proposed. Numerical results demonstrate that the proposed approach outperforms the existing pilot-based beamforming schemes in terms of achievable rate and EE.

Along with working on the transmitter side, the ML techniques have been proven very effective in improving the desired signal detection performance at the receiver side of the MIMO systems. The following are some works utilizing neural networks to improve computation complexity, BER, and interference mitigation performance at the receiver side of the MIMO systems. The authors in [210] present a DL-based signal detector model for the MIMO-OFDM communication system. The proposed scheme considers the iterative orthogonal approximate message passing (OAMP) algorithm which suppresses interference to a large extent and replaces its matrix inversion process per iteration with a conjugate gradient method to reduce its computational complexity. Then, inspired by an iterative soft-thresholding algorithm (ISTA), the revised OAMP detector is unfolded into a neural network (NN) through which the detection performance of the detector could be enhanced by tuning some key trainable parameters using DL techniques. Furthermore, a cyclic-prefix-free MIMO-OFDM receiver is proposed to improve the spectral efficiency. It is demonstrated that the proposed scheme reduces the detector's computational complexity and significantly improves detection performance in terms of BER. Further, through experiment, the robustness of the proposed scheme is verified in realistic over-the-air (OTA) scenarios. The research in [211] investigates the detection problem at the receiver for MIMO networks with multiple antennas while considering the interfering signals correlating the time and frequency domain. In this work, the joint use of an MLD (Maximum likelihood Detector) and a deep CNN is proposed to tackle the highly complex joint distribution estimation of the interfering signals. Specifically, MLD produces an initial detection result, and then deep CNN suppresses the interference and improves the detection by exploiting the local correlation among interference signals. Furthermore, an improved version of deep CNN is presented utilizing the cross-entropy of detection for defining the loss function of deep CNN which makes the residual interference follow the Gaussian distribution. Numerical results verify that the proposed detector with Deep CNN outperforms the conventional MLD regarding BER and MSE minimization. In [212], the authors investigate the DL-based linear detector for MIMO systems over correlated noise environment.

Specifically, the iterative framework of linear detectors and deep CNN is employed where the linear detector provides an initial estimate of the transmitted signals and then deep CNN is used to estimate the transmitted signals more accurately by capturing the local correlation among noise. The linear detectors used in this work are ZF (zero-forcing), MMSE, ZF-SIC (ZF with SIC), and MMSE-SIC. Numerical results demonstrate that the proposed scheme outperforms the conventional linear detectors in terms of BER.

IX. INTERFERENCE MITIGATION IN THZ COMMUNICATION NETWORKS

A. FEATURES AND TYPES OF CHALLENGES/INTERFERENCE IN THZ COMMUNICATION NETWORKS

THz-band communication is envisioned as a fundamental enabler of the 6G wireless networks. The THz frequency band, typically ranging from 0.1 to 10 terahertz (THz), holds promise to address the growing demands for ultra-high data rates (peak data rate: 1 Tbit/s and experience data rate: 10-100 Gbps), low latency (< 0.1 msec), high EE (100X) compared to 5G, ultra-reliability (99.9999%), high sensing and positioning accuracy, and massive and seamless connectivity that are expected in the 6G era [213]. The THz band has capability of multiplexing thousands of users simultaneously and can establish ultra-high-BW links. Beyond just addressing spectrum congestion, the THz band is poised to revolutionize society by empowering transformative applications [214] such as real-time virtual reality, augmented reality, holographic telepresence, e-healthcare, autonomous vehicles, imaging, localization, positioning, and tracking of drones etc. While THz channels offer distinct opportunities, they also pose unique challenges that are not observed in traditional wireless environments which are discussed as follows:

- 1) *Atmospheric and molecular absorption*: One of the significant challenges is the substantial impact of molecular absorption on path loss within THz band. This absorption loss is contingent on various factors tied to time and distance, such as the concentration and specific molecular composition encountered during propagation, notably water vapor. Consequently, THz channels exhibit extreme frequency selectivity, with bandwidth significantly contracting or expanding over time. These variations are dependent on factors like the distance between the Tx and Rx (also called distance selectivity) or the ambient humidity [215]. As a result, addressing these bandwidth and absorption loss variations necessitates substantial adaptability, especially concerning the physical layer (PHY) waveforms utilized for TX.
- 2) *Propagation loss*: The increase in carrier frequencies and extremely short wavelength of THz waves lead to higher spreading and intensified NLoS link propagation losses. These losses are primarily a result of

diffuse scattering, reflection, diffraction, and shadowing [216].

- 3) *Device and infrastructure development*: THz communication hardware is still in the development / experimental stage, and challenges exist in building compact, power-efficient devices and infrastructure [217].
- 4) *Phase Noise*: Phase noise is a critical problem in THz band that causes inter-carrier interference and severely impacts demodulation performance at Rx [218].
- 5) *Beam squint*: In THz massive-MIMO systems, wide-band TXs with large array apertures can introduce spatial-frequency wideband (SFW) effects. In this scenario, the channel response becomes frequency-selective not just due to the delay spread of the multi-path channel but also because of extensive propagation delay across the array aperture [219]. Consequently, the response of the BS array can be frequency-dependent, even in a LoS scenario. When using OFDM to combat intersymbol interference (ISI), the SFW effect causes the Direction of Arrival (DoA) and Direction of Departure (DoD) of signals to vary across subcarriers. This phenomenon, termed beam squint, requires frequency-dependent beam-forming/combining. However, typical hybrid array architectures of THz massive MIMO lack this capability. Hence, compensating for beam squint is crucial for THz massive MIMO-OFDM systems [220].
- 6) *Limited coverage range*: To tackle the restricted coverage range of THz networks due to high penetration and propagation loss, strategies involving ultra-dense deployments and highly directional antennas are envisaged in THz networks. However, such deployments substantially elevate CCI levels, limiting network density and presenting challenges regarding beam alignment and tracking. Hence, there is a need for innovative solutions that can effectively address these issues and enable efficient deployment of THz networks [221].

AI/ML can effectively address THz band challenges (e.g., atmospheric loss, co-channel, and inter-symbol interference) in THz communication systems. Techniques such as intelligent dynamic hybrid beamforming, beam steering, channel estimation and prediction, adaptive modulation and coding, OTFS modulation, orbital angular momentum, user clustering and association, and optimal resource allocation can mitigate these issues. We will discuss related works that propose AI/ML-based solutions for THz communication challenges. A summary of these works is shown in Fig. 20.

B. RELATED WORKS IN THZ COMMUNICATION NETWORKS AND THEIR QUALITATIVE COMPARISON

The authors in [222] investigate EE maximization problem in the THz-MIMO-NOMA cache-enabled network. Specifically, the EE maximization problem is divided into

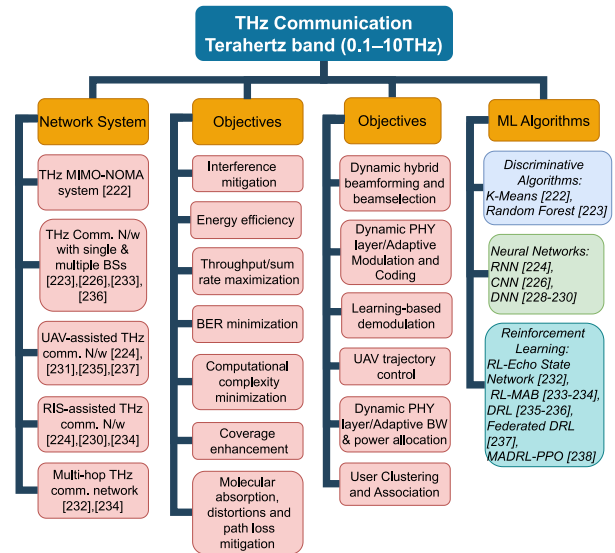


FIGURE 20. THz communications network system, objectives, and solution techniques.

three sub-problems: channel correlation-based user clustering, hybrid precoding, and optimal power allocation to users in clusters to mitigate intra-cluster and multi-cluster interference. For user clustering, an enhanced K-means algorithm is proposed for fast convergence. Then, a hybrid precoding scheme is adopted based on the sub-connection structure (where each RF chain is connected to only part of the antennas through phase shifters, not to all antennas in the array) to reduce power consumption and implementation complexity. Lastly, a distributed ADMM (Alternating Direction Method of Multipliers) algorithm is designed for power allocation to the users in clusters with imperfect SIC. It is demonstrated that the proposed scheme achieves better clustering convergence rate and EE performance than benchmark schemes such as conventional k-means-based clustering and equal/random power allocation. The study in [223] investigates the beam-selection problem in THz communication system as a multi-class classification problem. To solve this problem, the authors propose the use of a random forest ML algorithm that is trained by using the simulated data. The proposed scheme provides a better tradeoff between the computational complexity of hybrid beamforming and sum-rate maximization in comparison to classical beam selection schemes such as exhaustive search method and SVM-based beam selection.

A reconfigurable intelligent surface (RIS)-assisted drone THz-communication network is considered in [224], where a mobile drone establishes and maintains a reliable communication link with the BS through either a direct or a RIS-assisted link. To ensure a seamless communication link between the BS and the drone while considering the drone mobility, the authors propose a gated recurrent unit (GRU)-based RNN DL algorithm to predict the most optimal communication link (be it direct or RIS-assisted) and to

forecast the most suitable beamforming vector for that specific link. Furthermore, the deep MIMO data-generation framework [225] is used to build the scenario and to develop a dataset to evaluate the proposed scheme. Numerical results show that the proposed approach achieves high accuracy in determining the communication link (handoff) and serving beam selection. In study [226], to enhance the security against smart interference and to tackle the communication challenges in THz-band such as molecular absorption loss, frequency-selectivity, and distance-dependent bandwidth change, a DL-based bandwidth (BW) and modulation (MD) classification technique is presented to make the physical layer very flexible where Tx and Rx can change their PHY parameters such as BW and MD without coordination. Here, the Tx selects its BW and MD parameters using CSI reports Rx sends periodically. At the Rx side, the processed and down-converted baseband signal is sent to CNN to infer BW and MD of the incoming THz signal. Then CNN inference is used by a reconfigurable DSP logic to demodulate and recover the application data. This work extensively trains and tests CNN classifiers using the experimental data collected through TeraNova testbed [227]. The experimental dataset consists of In-phase/Quadrature (I/Q) samples collected at the 120-GHz RF frequency with five MD schemes (BPSK, QPSK, 8PSK, 16QAM, and 64QAM), three BWs (5, 10, and 20 GHz), and two SNR levels (low, high). Numerical results prove the effectiveness of the proposed DL-based modulation and BW classification scheme at THz frequencies with 78% and 90% accuracy in low and high SNR conditions.

The authors in [228] focus on tackling the hybrid distortions in THz communication systems, such as phase noise, phase shift, I/Q imbalance, and non-linearity of power amplifiers. These distortions cause a mathematical-model-deficit scenario that limits the performance and application of the conventional demodulation techniques in THz communication systems. Consequently, in this work, a DL-assisted demodulation technique is proposed to enhance the BER performance of the Rx system. Specifically, a multi-output deep feed-forward neural network is proposed for nonlinear mapping between the Rx symbols and the demodulation likelihood information vector. Furthermore, a training data generation method is presented in which prior knowledge of likelihood information is not required. Simulation results show that the proposed demodulation methodology outperforms the MMSE equalization and SVM-based schemes in terms of BER performance for THz communication systems under severe hybrid distortions.

In research [229], a multiuser THz communication system is considered with rapid variation in molecular absorption coefficient (MBC) within the spectrum of interest. In this work, to minimize the variation of molecular absorption loss among users and to maximize the aggregated multiuser data rate, a DNN-based adaptive sub-band bandwidth allocation and power control algorithm is proposed. The proposed DNN algorithm is trained offline by using a Lagrangian-based Loss

function. Then, the trained DNN is used to find the near-optimal solutions to the optimization problem. Simulation outcomes demonstrate that the aggregated multiuser data rate achieved using the proposed DNN-based algorithm surpasses the results obtained by the existing convex optimization approach, particularly in scenarios where the MBC exhibits highly non-linear variations within the spectrum of interest. The study in [230] tackles the coexistence challenge of eMBB and URLLC services in RIS-assisted THz communications and proposes a resource management framework optimizing parameters such as BS transmit power, RIS reflection coefficients, and resource block allocation. Specifically, an alternative decomposition method-based optimization policy is proposed that decomposes the resource management problem in two parts: one for resource block allocation and the second for reflection coefficient and power allocation and addresses the joint problem iteratively. Next, an LSTM network is trained using optimized parameters for real-time prediction and its learning rate hyper-parameter is tuned via the Bayesian optimizer. Finally, multiple LSTMs are integrated into an ensemble model to enhance the prediction accuracy. The numerical results show substantial eMBB spectral efficiency gains with RIS of size 11×11 , reaching up to 49% with the proposed optimization approach, 34% with the LSTM DL method, and 48% with the ensemble learning method, compared to a DRL-based scheme.

The authors in [231] consider a multi-drone-based network in which different spectrum bands such as microwave, mmWave, and THz bands provide data streaming services to users on the ground in blockage-rich mobile environments. This work investigates the sum rate maximization problem by jointly controlling the drones' flight, user association, and spectrum band selection for serving the users. It is assumed that a user can be served one spectrum band at a time, while a drone can serve multiple users using different spectrum bands based on channel conditions. To solve the formulated MINLP (Mixed Integer nonlinear programming) throughput maximization problem, an echo state learning-based utility prediction and flight optimization framework is proposed while combining it with an RL-based flight control distributed algorithm. Simulation results through a newly developed event-driven simulator, SimBAG (Broadband Air-to-Ground network simulator), demonstrate that to fulfill the demands of comprehensive coverage and high data rate, using the combination of different spectrum bands is a desirable solution, and THz-band has excellent potential to enhance the throughput in a blockage-rich environment by taking advantage of the mobility of flying drone BSs.

The study in [232] focuses on a multi-hop THz communication system and introduces a novel RL-based beam training approach. This approach aims to minimize the time spent on beam searching and enhance network spectral efficiency. Specifically, an MAB-based hierarchical beam training algorithm is proposed to optimize the number of beam training levels at each hop, facilitating rapid convergence and reducing beam searching time overhead.

Ultimately, this leads to improved spectral efficiency in the multi-hop THz communication system. Numerical results substantiate that the proposed scheme surpasses existing beam training schemes having a fixed and random number of training levels in terms of spectral efficiency. The research in [233] focuses on a dense THz communication system and introduces a time-domain interference mitigation technique using an RL-based algorithm at the Rx to improve the BER performance of the communication system. The proposed technique utilizes an MAB-based strategy to determine the best threshold value from multiple threshold clipping levels for a threshold-based nonlinear preprocessor at the Rx. Simulation results indicate that this approach converges rapidly and surpasses traditional interference mitigation methods like blanking and single-threshold clipping schemes in terms of BER performance.

The authors in [234] consider the multi-hop RIS-empowered THz communication system to tackle the propagation attenuation and atmospheric absorption challenges and enhance the coverage range in THz-band. Specifically, to increase the TX range between BS and users, multiple controllable RISs are deployed, and the joint design of analog beamforming at RISs and digital beamforming at BS is investigated using the DRL-based algorithm. Furthermore, two alternating optimization-based methods are proposed for initializing the beamforming metrics, which enhance the convergence rate of the proposed DRL-based algorithm. Numerical results show that the proposed DRL-based scheme outperforms the benchmarks regarding the THz communications system's total throughput and coverage range. The research in [235] explores an architecture utilizing UAVs for offloading computations of wireless users' devices in the THz band. The goal is to increase system capacity while meeting the real-time latency needs of wireless users having limited computational capacity. This work proposes an optimization problem to minimize the overall time delay, considering the resource and QoS constraints and swiftly changing environment. To optimize UAV placement, computation offloading, and resource allocation within this architecture, the study introduces the DDQN and DDPG algorithms. The simulation outcomes demonstrate the algorithm's efficient convergence and lower latency than full local computing and complete offloading methods.

In [236], a THz-communication network with multiple BSs coordinated by an edge server is studied. To minimize inter-cell interference (ICI) and maximize network throughput, a federated DRL algorithm for THz-beam searching is proposed, using DDPG-based learning with limited CSI. BSs update their DDPG models with partially estimated CSI through Federated Edge Learning (FEL) to mitigate ICI. Numerical results show the federated DRL-based beamforming scheme outperforms conventional ZF and MMSE beamforming in computation complexity and throughput.

In [237], a mobile edge computing (MEC)-assisted UAV swarm network is explored for task offloading of

ground users using THz-band communication. The goal is to maximize network utility by optimizing UAVs' resource allocation and trajectories. The NP-hard utility maximization problem is divided into two subproblems: local and offloaded computation resource allocation, solved by convex optimization, and optimizing MEC-UAV trajectories, data offloading, and communication resources, solved by a multi-agent PPO DRL algorithm with an Embedded Multi-Head Attention (MHA) mechanism. Numerical results show this scheme outperforms fairness-based resource block and power allocation and general MAPPO in network utility.

C. FEATURES AND TYPES OF INTERFERENCE IN NOMA-BASED COMMUNICATION NETWORKS

NOMA is a promising multiple-access technique to enhance spectral efficiency, user fairness, massive connectivity, and cell edge throughput in future wireless technologies [238]. Different multiple access techniques were utilized in previous wireless technologies such as FDMA in 1G, TDMA in 2G, CDMA in 3G, and OFDMA in 4G and 5G, where frequency, time, and code resources are divided among users to avoid interference for simultaneous information TX. In contrast with previous orthogonal multiple access (OMA) schemes, NOMA allows simultaneous TX of multiple users using the same frequency-time resources using power domain (PD-NOMA) or code-domain (CD-NOMA) multiplexing [19]. In PD-NOMA, multiple users' signals are superimposed using different power levels, while in CD-NOMA it is performed by using different signature codes [239]. Also, NOMA can be integrated with advanced technologies such as mmWave, massive MIMO, beamforming [238], and machine-type communication [240] to fulfill the ever-increasing demands of next-generation wireless networks. NOMA indeed enables the efficient utilization of resources at the Tx, however, it leads to a non-orthogonal mixture of signals interfering with each other at the Rx side [241]. To tackle this interference, more efficient protocols and algorithms are required for channel estimation, resource allocation, dynamic clustering, and user scheduling at Tx side, and more advanced signal processing techniques are required for SIC, and proper decoding of information at Rx [242]. An illustration of Uplink and downlink NOMA-based communication with SIC is presented in Fig. 21. Recently, ML techniques have been widely used to assist the NOMA technology by optimal user clustering, power allocation, channel allocation, optimal order of SIC decoding, and interference management based on learned patterns from historical data to enhance the network performance. This section provides a detailed study of ML-based works in the NOMA domain with different network systems architectures focusing on downlink and uplink communications, with different objectives, wireless techniques, and ML algorithms used to achieve set objectives. A summary of these details can be visualized in Fig. 22.

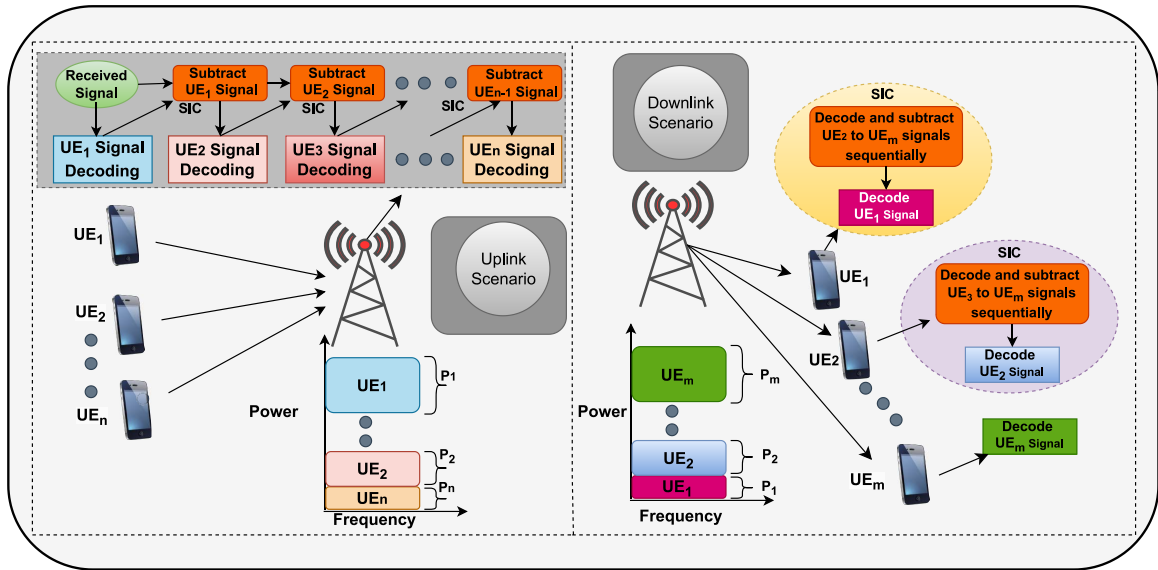


FIGURE 21. Illustration of uplink and downlink NOMA-based communication with Successive Interference Cancellation (SIC).

X. INTERFERENCE MITIGATION IN NOMA-BASED COMMUNICATION NETWORKS

A. RELATED WORKS IN NOMA-BASED COMMUNICATION NETWORKS AND THEIR QUALITATIVE COMPARISON

1) NOMA-BASED DOWNLINK COMMUNICATION SCENARIOS

Initially, we delve into the works centered on NOMA-based downlink communication scenarios leveraging ML techniques to augment the performance of various network metrics. The NOMA-based communication system is considered in [243], where assistance information such as modulation order of interfering users is necessary for canceling the inter-superposition-layer interference at the Rx using the SIC technique. In this work, a feature-based blind detection method is employed to derive assistance information from received constellation points. Specifically, the feature extraction process is based on the Anderson–Darling test, followed by a logistic regression-based classification to determine the modulation order of the interfering user. Numerical results demonstrate the superior performance of the proposed approach over the max-log algorithm in terms of detection rate and throughput. In research [244], a cell-free massive MIMO-NOMA network is considered to enhance the sum spectral efficiency (SSE). In this work, an advanced K-means algorithm is used for user clustering which defines the initialization of centroids of the clusters, and ZF precoding is used to mitigate inter-cluster interference. Then, to solve SSE optimization problem, optimal power is allocated to users for proper SIC decoding while considering the power constraints at APs and QoS constraints of user equipment. An efficient iterative algorithm is proposed based on the inner approximation method to obtain the optimal power allocation. Simulation results show that the proposed user clustering scheme outperforms the

random and conventional K-means-based clustering schemes regarding SSE.

Then, the authors in [245] propose a novel cluster-free NOMA framework in a multi-cell network system. In this work, an auto-learning GNN is used for distributed multi-cell coordinated beamforming and cluster-free SIC decoding. The objectives of this work are sum rate maximization and inter-cell and intra-cell interference mitigation while fulfilling the users' minimum rate constraints. Moreover, a bi-level learning algorithm is proposed to train the GNN architecture parameters and model weights to reduce the complexity and information overhead for distributed coordinated beamforming and cluster-free SIC optimization. Simulation results show that the proposed cluster-free NOMA framework outperforms the conventional cluster-based NOMA in terms of sum rate under multi-cell scenarios and reduces the information and computation overhead in comparison to fixed GNN and distributed ADMM (Alternating Direction Method for Multipliers) techniques. A NOMA-based heterogeneous IoT system is considered in In [246], wherein based on QoS targets, mobile users and IoT users are classified as PUs (Primary users) and SUs (Secondary users). In this work, to enhance the EE of the NOMA system with imperfect SIC, a peeling-based subchannel assignment scheme is proposed for PUs, and reweighted message-passing algorithm-based resource allocation is done for SUs. Also, a deep RNN-based algorithm is proposed to solve the problem of resource allocation for SUs. Furthermore, a rate-demands and priorities-based user scheduling method is put in use to synchronize the access of heterogeneous users with limited radio resources. Numerical results demonstrate that the proposed scheme outperforms the OFDMA system in terms of sum rate but at the cost of high power consumption. A NOMA framework is considered for cognitive radio networks [247], wherein a DL-based technique is utilized for

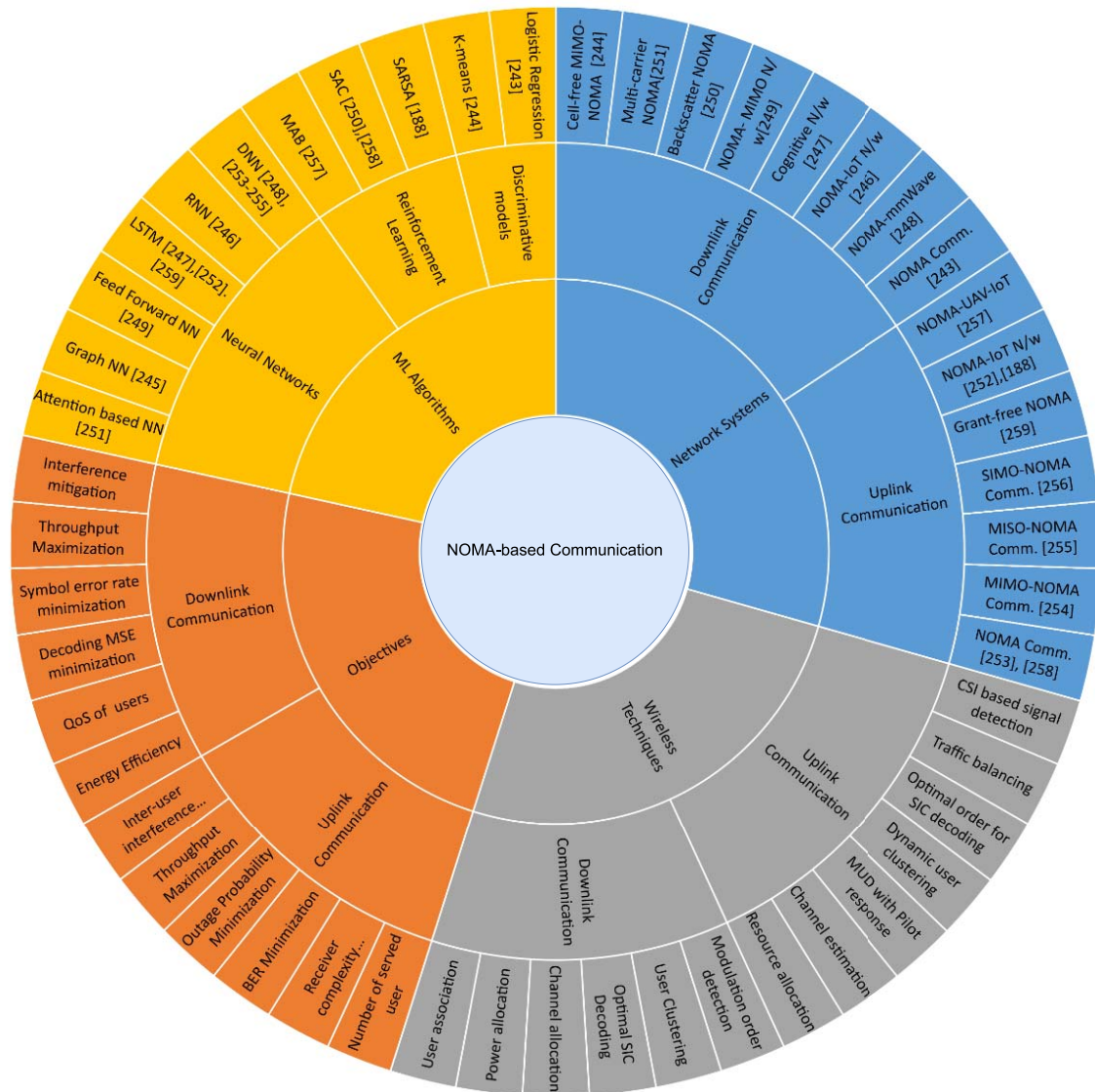


FIGURE 22. Summary of related works for interference mitigation in NOMA-based communication networks.

joint signal detection and fair power allocation for PUs and SUs accessing the same frequency bands. Specifically, an LSTM-based trained algorithm is used in online mode using which multi-user detection (MUD) is performed in one shot. The DNN algorithm's training is done offline using the pilot sequences. The proposed framework is implemented through an experimental setup using SDR with one PU and two SUs. Analytical and experimental results demonstrate the superiority of the proposed technique over the conventional SIC techniques in terms of Symbol error rate and system throughput.

A NOMA-based mmWave heterogeneous network is considered in [248] where it is assumed that up to two users can utilize a subchannel through SIC. This work aims to maximize the network's EE while fulfilling the users' QoS requirements and limiting inter-tier interference and power consumption. To achieve the set target, the Lagrange dual

decomposition method is used for user association, a semi-supervised learning-based regression algorithm is utilized for channel allocation and DNN is employed for power allocation. An iterative algorithm is used for generating the sample data. Simulation results verify the effectiveness of the proposed schemes over the conventional methods for power and subchannel allocation. The authors in [249] consider the MIMO-NOMA system wherein to tackle the issue of imperfect SIC, a FNNs (Feed-forward Neural networks)-based DL technique is proposed for joint optimal precoding and SIC decoding of multiple users. This work aims to minimize the mean square error (MSE) between the desired and decoded signals of the users. The numerical results demonstrate that the proposed scheme outperforms the conventional SIC schemes regarding MSE performance.

In research [250], a backscatter-NOMA network is considered that consist of one full-duplex BS, multiple downlink

users, and multiple uplink backscatter devices. This work aims to maximize the sum rate of uplink backscatter communication while fulfilling the QoS requirement of the downlink users. To achieve this objective, the reflection coefficient of backscatter devices and the NOMA-based resource allocation, i.e., Tx power of downlink users are jointly optimized using a soft actor-critic (SAC)-based RL algorithm. In this algorithm, the SAC agent explores and exploits the dynamic backscatter-NOMA network efficiently by entropy regularization. Simulation results demonstrate that the proposed scheme outperforms backscatter-OMA and random power allocation techniques regarding the average sum rate of uplink backscatter devices. The authors in [251] consider the downlink multi-carrier NOMA system wherein to maximize the sum rate and minimal rate of the network, a DRL-based algorithm is proposed for the optimal channel assignment to users. Specifically, the attention-based neural network having an encoder-decoder structure is used for policy formulation of the DRL framework. It is assumed that only two users can be served per channel using the power-domain NOMA technique. The optimal power allocation is done for both users for the given channel assignment so proper SIC decoding can be performed at the user end. Simulation results show that the proposed scheme outperforms the conventional joint resource allocation technique regarding sum rate and minimal rate performance.

2) NOMA-BASED UPLINK COMMUNICATION SCENARIOS

We now explore how ML techniques improve network performance in NOMA-based uplink communication scenarios. Researchers in [252] focus on grant-free NOMA IoT networks, proposing a DL-based algorithm for MUD. An LSTM-based network is trained offline using pilot signals, performing joint channel estimation and MUD. Numerical results show it outperforms conventional SIC in terms of BER and network capacity, especially with more IoT devices. The research in [253] proposes a DNN-based SIC technique for intelligent detection sequencing in uplink NOMA scenarios. The algorithm uses received signal information, channel characteristics, and Tx power to optimize the detection sequence. Numerical results demonstrate better BER and detection performance than conventional CSI-based SIC schemes.

Research focused on improving uplink communication in NOMA systems with machine learning techniques has shown promising results. Considered in one study, an uplink MIMO-NOMA system employs a DL-based successive interference cancellation (SIC) scheme [254]. Each step of the SIC process utilizes DNNs at the BS to decode users' signals. The DNN takes combined received signals and previously decoded symbols as inputs, outputting transmitted bit sequences. Pilot symbols transmitted at the beginning of each frame train the DNNs, demonstrating improved BER and computational complexity performance compared to conventional SIC methods. In another scenario, a MISO-NOMA system utilizes DNNs to construct transmitted signal

symbols [255]. The softmax function enhances detector reliability instead of relying on hard decisions. Soft information from detected signals is used in the SIC process and as input for subsequent DNNs, leading to superior BER performance over conventional methods. Furthermore, in an SISO-NOMA-based network, a pilot-assisted model-driven DL algorithm enhances receiver SIC performance [256]. This approach introduces new learnable parameters—noise cancellation and interference cancellation—to improve channel estimation and SIC-based signal detection accuracy. Numerical results highlight its superiority over conventional MMSE and DNN approaches in terms of both BER performance and computational complexity.

A NOMA-based uplink communication scenario is considered in a UAV-enabled IoT network in [257], where an adaptive RL-based resource allocation algorithm is proposed to maximize the sum rate of the network. Specifically, the Mamdani fuzzy inference system (MFIS) is utilized to supervise the MAB-based policy of the RL algorithm for adaptive exploration and exploitation. Both Joint maximum likelihood detection (JMLD) and SIC are used for information decoding at the receiver. It is claimed that the proposed scheme outperforms the random exploration and ϵ -greedy policy-based non-adaptive RL counterpart regarding sum rate, outage probability, and BER performance. Also, it is demonstrated that JMLD performs better than the SIC-based detection in terms of BER but at the cost of increased complexity. Further, A multi-cell uplink NOMA-IoT network is considered in [188], where multiple IoT users use NOMA technique to share the same RBs. This work aims to maximize the sum rate of the network with constraints on transmit power, channel gains, and the number of users. To solve the constrained sum-rate maximization problem, SARSA and DRL algorithms are proposed for low traffic and high traffic density, respectively. These algorithms provide dynamic clustering solutions for connecting IoT users to BSs, and sub-channel allocation based on the received power levels of IoT users. It is asserted that compared to OMA, the proposed NOMA-based solution provides better sum rate performance, and both SARSA and DRL exhibit lower complexity than the optimal benchmark scheme.

In research [258], a NOMA-based uplink communication system is considered with an imperfect SIC. In this work, an actor-critic learning-based DRL algorithm is proposed to optimize the dynamic user access and optimal decoding order for the SIC process to maximize the number of successfully served users. Simulation results show that the proposed algorithm outperforms the conventional access and decoding schemes in terms of the number of successfully communicating users with heterogeneous QoS demands and imperfect SIC. The authors in [259] consider an uplink grant-free NOMA system to reduce signaling overhead and latency for massive access. To reduce the collisions of uncoordinated access and to improve the system throughput, first, subchannels and user devices are divided into different clusters. Then, to improve the long-term throughput of

the clusters, a DRL-based algorithm is used for channel selection and power allocation. Here, discrete power control is adopted to reduce the computational complexity of DRL algorithm. Numerical results show that the proposed DRL-based algorithm for grant-free NOMA outperforms slotted ALOHA NOMA with 156% gain on system throughput when the number of devices is five times more than the number of subchannels.

XI. LESSONS LEARNED, CHALLENGES AND OPEN ISSUES

In this section, the intentional interference detection and mitigation works are reviewed for insightful outcomes, followed by a thorough review of unintentional interference management across different domains of 5G and beyond networks. For each domain, the discussion is structured into two main parts. Initially, we analyze insights and lessons learned from the state-of-the-art works featured in that domain to provide cognizance for researchers on *do's and don'ts*. Subsequently, we unveil the opportunities by outlining the challenges of utilizing the different ML-based techniques and delving into the open issues/future research directions concerning interference handling for that specific enabling technology within next-generation wireless communication networks.

A. INTENTIONAL-INTERFERENCE/JAMMING DETECTION AND MITIGATION

1) LESSONS LEARNED

For the jamming detection and classification: Supervised learning is the most widely used ML algorithm category for jamming detection and classification. Most often used algorithms include random forest [43], [44], [45], Decision Tree [42], [43], Support Vector Machine [41], [42], [44], [45], and k-nearest neighbor [43], [45]. The main features of the signals used for jamming detection and classification are PDR [41], [43], [44], BPR [44], and RSSI [43], [44], [45], [53]. To train these models data is generated either by simulations [41], [43], [44], [45] or by real-time experiments [45]. The performance of different algorithms is compared by metrics such as detection and classification accuracy. As reported by several works, the performance of random forest is superior to that of other supervised learning algorithms in terms of jamming detection accuracy [43], [44], [45]. Furthermore, researchers can utilize the GAN model to detect the anomalies inside the radio spectrum [51] and to detect adversarial jamming attacks [52]. Also, deep learning algorithms such as CNN can help detect jamming attacks using spectrogram images [46] and ensure the transmitter's legitimacy using RF fingerprinting [49]. Federated learning is also a viable candidate for detecting jamming attacks in a distributed manner [47]. Publicly available datasets such as CRAWDAD [48] and POWDER [50] are available for researchers to train and test their jamming detection algorithms.

Regarding jamming mitigation: For all different types of communication scenarios such as UAV-assisted

communication [56], [57], [58], [59], [60], [61], [62], IoT communication [63], [64], [65], mmWave-MIMO communication [66], [67], vehicular network communication [69], 5G HetNet communication [68], and THz communication [73], the main focus is on TX power control to mitigate the jamming attacks [57], [60], [61], [62], [63], [66], [67], [69]. Specifically for UAV-assisted communication, relay selection [58], [59], [61] is also crucial for jamming mitigation along with TX power control. The widely accepted approach to address power control and relay selection is the utilization of RL techniques. The performance metrics to evaluate jamming mitigation are presented in terms of BER [54], [58], [59], [61], [62], throughput [64], [66], [68], [69], and energy consumption [59], [62], [63]. For eavesdropping mitigation, channel allocation [70], [73] is also considered along with power control. Mainly, RL-based algorithms such as DQN [72], DDPG [71], D3QN [70] can be used for resource (power and channel) allocation such that secrecy rate [70], [71], [72] can be maximized.

2) CHALLENGES AND OPEN ISSUES

Designing and implementing ML-based jamming detection and mitigation techniques are acquainted with several challenges and open issues. Following are some challenges and future research directions related to ML-based jamming detection and mitigation.

First, collecting a diverse dataset with multiple features and labeled examples of jamming instances (simulated and injected into real data) via real-time setups for the training of ML models can be challenging. Data with limited features can obstruct the competence of ML models to get accepted as a general solution for various jamming scenarios. Since the jamming types or techniques may advance with time, adaptive ML models should be designed to detect innovative jamming attacks. Effective feature selection and jammed signal representation are crucial for jamming detection since sparse and imprecise feature representation can cause inaccurate results for jamming detection. Second, ML-based jamming detection and mitigation models should be of low computational complexity to tackle the problem in real-time processing for time-critical applications. Along with lower computational complexity, the EE of ML models is crucial for deploying those models on energy-constrained or battery-operated devices.

Third, the practical applicability of the proposed models is a phenomenal challenge. This is because a model trained in one environment may not be suitable for another environment. For example, the assumptions made in a simulated or lab environment may differ from the practical field. Designing a generalized model applicable to different operational environments is critical for practical implementation. At last, the proposed ML-model should be robust enough to handle adversarial attacks that can elude the model and hinder its deployment in real-world applications. Furthermore, a jamming detection model can access communication signals for analysis, which may

raise privacy concerns. Therefore, considering privacy is essential while designing an effective jamming detection and mitigation model. To address these challenges, it is crucial to consider advanced ML algorithms, refine data collection strategies, leverage domain adaptation techniques, and encourage collaboration among researchers and domain experts.

B. UAV-BASED COMMUNICATION NETWORKS

1) LESSONS LEARNED

We summarize the lessons learned from AI-enabled interference detection and mitigation in two steps: 1) lessons learned for scenarios with multiple UAVs serving as aerial base stations, 2) lessons learned for scenarios with multiple UAVs acting as aerial users.

Scenarios with multiple UAVs serving as aerial base stations: To mitigate co-channel interference and to maximize the throughput/sum rate [80], [82], [83], [84], [85], [86], [88], [89] and EE [80], [81], [85], [87], [89], [90] of UAVs-assisted networks, the main focus remains on UAV positioning [80], [81], [82], [84], [85], trajectory control [86], [89], [90], Tx power control [81], [82], [83], [84], [85], [86], [87], and switching the highly interfering drone small cells (DSCs) into sleep mode [80], [81], [83]. Further, to solve the UAV positioning and power control problem, researchers can use both distributed approaches such as multi-agent DRL [85], [89], [90] as well as centralized approaches such as Q-learning [83], [84], DRL [87], SAC DRL [86] and unsupervised learning algorithms [80], [81], [84]. It should be noted that a coordinated distributed approach using federated learning performs better than an uncoordinated distributed approach in terms of network throughput and EE [85]. To switch the DSCs into sleep mode, an unsupervised learning approach named affinity propagation can be used [80], [83], and for the UAVs positioning control, the k-means algorithm can be used as suggested by the related works [80], [81], [84]. MADRL algorithms such as MA-DQN [89] and MA-DDQN [90] can be utilized for UAV trajectory control. Researchers can use equal power allocation [84], random power allocation [87], pure Q-learning [83], [87], genetic [81], [83], and heuristic approach [81] as baselines to compare their solutions against.

Scenarios with multiple UAVs acting as aerial users: The main objectives are to reduce the co-channel interference from other ground base stations (GBSs) to UAVs in downlink communication [92], [93], [94], [96], [97], [98], [99] and to mitigate UAV-to-GUE interference in uplink communication [92], [95]. Various strategies are available to mitigate the interference. These strategies include BS association [92], [94], [95], [96], power control of GBSs [92], [93], [95], [98], [99], and UAV path planning/ trajectory design [94], [95] so that the rate per UAV/GUE [92], [93], [95], [96], EE [92], [98] can be maximized, and Tx latency per UAV [95] connectivity outage of UAV [94], [97] can be minimized. The most commonly used ML techniques include RL such as DQN for Tx power control [93] and trajectory

design [94], ESN (Echo state network)-DRL [95] for path planning, BS-association, and power allocation, and Dueling DQN [96] for BS-association.

2) CHALLENGES AND OPEN ISSUES

Employing ML-based interference mitigation techniques in UAV-assisted communication networks instigates many challenges and open issues. UAVs' flexibility in placement, trajectory designing, and air-to-ground channel modeling bring unique characteristics and operational challenges that call for proficient ML-based techniques to tackle interference mitigation problems in this highly dynamic environment. Below are some key challenges and open issues related to AI/ML-enabled interference management in the UAV-assisted communication networks domain.

First, in multiple UAV scenarios, since UAVs can have three-dimensional mobility with varying altitudes and operate in dynamic environments, interference estimation, prediction, and modeling become more complicated. Therefore, considering the temporal and spatial characteristics of UAV-assisted networks is important for ML models for interference control. For example, RNNs can be leveraged to deal with time-dependent data and predict the optimal trajectory of UAVs. Since UAV-assisted networks are highly dynamic with varying interference characteristics, ML models must be adaptive to real-time interference variations due to the mobility of UAVs. Also, a realistic path-loss and signal propagation model (Air-to-ground) should be considered for the design of interference mitigation techniques to be practically deployable in real-time scenarios.

As UAVs are energy and computational resource-constrained devices, the proposed ML models must be of lower computational complexity, higher EE, and capable of real-time interference mitigation with low latency in highly dynamic UAVs-assisted communication environments. In multi-UAV scenarios, the cooperative-distributed learning approaches must be investigated for effective interference management via Tx power control, resource block allocation, switching UAVs into sleep mode, trajectory control, and user association. While using discriminative ML approaches shows promising performance for UAV positioning and switching the drone's small cells into sleep mode, limited data may hamper the generalization of ML models to diverse and dynamic UAV-assisted communication scenarios. Lastly, ML models are expected to consider integrating UAV-assisted networks with existing cellular infrastructure and advanced signaling techniques such as mmWave-MIMO, NOMA, and THz Communication. Furthermore, the proposed ML models must be robust enough to handle the privacy and security concerns of UAV-assisted communication networks.

C. VEHICLE TO EVERYTHING (V2X) COMMUNICATION NETWORKS

1) LESSONS LEARNED

In this domain, researchers mainly focus on minimizing the CCI among V2I links, V2V links, and cellular users

sharing the same spectrum bands. Below, we list key lessons learned from the state-of-the-art works concerning AI-based interference detection/mitigation in V2X communication networks: To avoid co-channel interference, resource allocation in terms of resource blocks (RBs) and Tx power to V2V links and V2I links is a widely used approach. Some works focus on only resource block allocation [107], [112], [117], [119], [120], some on Tx power allocation [104], [108], and others on both channel and Tx power allocation [105], [106], [109], [111], [113], [114], [115], [116], [118]. The most commonly used objective function is to maximize the sum capacity of V2I links with latency and reliability constraints on V2V links [111], [113], [115], [116], [118], [119], [120]. In the cognitive Internet-of-vehicles network systems main objective remains to maximize the spectral efficiency while minimizing the interference to primary cellular users [102], [103]. It is achieved with efficient spectrum sensing and intelligent RB allocation (eg. via SVM classifier [102] and Q-learning [103]). Some other commonly used objectives are to maximize the sum rate of cellular users [108], packet delivery rate of V2V links [109], sum rate of all vehicles [117], network throughput of V2I links [110], and data rate of V2V links and success Tx probability of V2I links [114].

To solve the problem of resource allocation to V2V and V2I links, multi-agent RL can be utilized with a distributed approach (eg. MA-DDQN [119]), a collaborative distributed approach (e.g., using Federated learning with DQN [113], [118], and D3QN [109]), or with a centralized training and distributed execution approach [106], [107], [120]. It is possible to pursue a complete centralized approach for resource allocation using various ML techniques such as DNN [108], soft actor-critic [115], LSTM [111], as well as DQN and DDPG [114]. While utilizing mmWave to serve vehicles via V2I links in a highly mobile environment, optimal beam selection is critical to make seamless connections of vehicles with infrastructure. For example, a combinatorial MAB approach can be useful for optimal beam selection to maximize the overall network throughput [110]. Age of information is also a critical metric in a vehicular environment when working with AI/ML models in real-time. This problem can be addressed via decentralized resource allocation for vehicles using the Gaussian process regression technique [105].

2) CHALLENGES AND OPEN ISSUES

The use of machine learning techniques for interference mitigation in cellular-assisted V2X communication networks demonstrates several open issues and challenges due to the high mobility of vehicles and the continuously changing topology of vehicular networks. Cellular-assisted vehicular networks consist of simultaneous V2V, V2I, V2P, and cellular BS to cellular users that communicate by utilizing the same spectrum band, creating CCI with each other. Below we list some key challenges and open issues in the V2X communication networks domain.

Since vehicular networks experience high mobility and frequent topology changes, ML models should be able to adapt to the dynamic vehicular environment in real-time scenarios with acceptable time and space complexity. The ML models must be time-efficient to tackle the safety-critical/time-critical applications with low latency in real-time implementations. There are solutions with distributed and centralized approaches for interference management in vehicular networks. Proper investigation of the time and space complexity of distributed, cooperatively distributed, and fully centralized approaches is critical for practical implementation because ultra-reliability and low latency are essential in 5G-enabled vehicular networks. Considering the decentralized nature of V2V, V2I, V2P, and V2N communications, leveraging edge computing and investigating advanced distributed ML approaches can be effective for interference management in vehicular networks.

ML models should provide efficient solutions for resource allocation within limited bandwidth resources while considering heterogeneous data demands for V2V, V2I, V2P, V2N, and cellular communication links and provide consistent reliable performance. In the model considering ML or DL approaches, the collected data must contain sufficient diverse instances to generalize the model. Since vehicle/traffic density can vary depending on time and place, which may change the interference pattern, an ML model demonstrating competent performance in one environment may not be suitable in another. Therefore, to implement the ML model in real-time scenarios, the model is expected to adapt to such variations. Lastly, the ML models should be able to integrate advanced techniques such as mmWave-MIMO, NOMA, IRS, backscatter, and THZ communication with vehicular communication technologies. Furthermore, ML models must be resilient to potential threats, ensuring the privacy and security of vehicular communication networks while implementing interference management techniques.

D. HETEROGENEOUS CELLULAR NETWORKS (HCNS)

1) LESSONS LEARNED

In HCNs, ML techniques have been proven to be quite effective in mitigating inter-tier and intra-tier interference and enhancing the overall network performance. The types of network architecture considered in this domain are HetNets with the coexistence of macro cells and small cells [125], [126], [127], [128], [129], [130], [133], [134], [135], [136], [137], [138], [139], [140], HetNets with dense small cells [131], [132], [141], [142], [143], and multiple small-cell scenarios [144], [145], [146]. Following are some highlights for lessons learned from the state-of-art works in this domain: Commonly used network metrics to evaluate the performance of various interference mitigation approaches in HCNs are network throughput [125], [127], [133], [134], [138], [139], [140], [143], [144], packet loss ratio [125], [134], EE [126], [128], [129], [142], [143], [145], spectral efficiency [135], [136], [137], and QoS constraints of UEs [125], [128], [129], [130], [131],

[132], [140], [141]. Tx power control and cell association using CRE bias are widely used parameters to control the interference in HCNs. To obtain the optimal values of these parameters a decentralized MARL approach [130], [139], [140], [141], [144], [145], distributed Q-learning [131], [133], cooperative Q-learning [132], [133], or a centralized power control approach such as DDPG [137] and DRL [143] can be used.

Other than Tx power control, a few more techniques can be considered to mitigate interference in HCNs such as traffic offloading [125], [135], [136], ABS ratio [127], user classification/clustering [125], channel allocation [142], [146], BS ON-OFF switching [142], Dynamic TDD configuration [134], network load prediction [128], and beamforming [138]. To control the ABS ratio and CRE bias factor, the Multi Arm bandit [127] technique can be used, and for high and low interference users' classification, the Random Forest algorithm [125] is a feasible approach to be used. If RL is empowered to address interference mitigation, DQN [135], [143], DDQN [136], [146], SAC [141], [145], Mutli Actors Shared Critic (MASC) [144] have been shown efficient under different settings.

2) CHALLENGES AND OPEN ISSUES

HCNs consist of multiple tiers of diverse elements such as macro, small, pico, and femtocells. In HCNs and ultra-dense N/w, using ML techniques for inter-tier and intra-tier interference mitigation introduces challenges and open issues. Following are some key considerations in this domain:

Tx power control, cell association control using CRE bias, ABS ratio, network load prediction, traffic offloading, BSs ON-OFF switching, and user classification/clustering are some important factors/parameters to control inter-tier and intra-tier interference in HCNs. In case of dynamic network reconfiguration such as the addition, modification, or removal of cells, ML models must be adaptable to adjust these network parameters to minimize the impact of interference dynamically. Design of new ML models for dynamic spectrum access mechanisms along with Tx power control among different cells with users' QoS constraints is an open research direction, which can be tackled by the researchers in this field. Further, ML-based techniques may be investigated for optimal location prediction of small cells while considering user density, traffic, and interference patterns to enhance network throughput and spectral efficiency.

To handle user mobility and inter-tier handover, ML models should be designed with the capability of minimizing interference during the transition by predicting user mobility and optimizing handover decisions. A multi-connectivity approach can also be considered to handle user mobility and fulfill their QoS requirements. To mitigate the interference in HCNs, energy-efficient ML-based techniques with low computational complexity are required. In centralized approaches, feedback overhead, and high time and space complexity may be serious concerns. Therefore,

advanced distributed approaches can be investigated for interference management and performance enhancement of HCNs. Lastly, the ML model should be flexible to integrate new technologies into HCNs and must be robust enough to maintain the security and privacy of HCNs.

E. DEVICE-TO-DEVICE (D2D) COMMUNICATION NETWORKS

1) LESSONS LEARNED

We present lessons learned for AI/ML-based D2D interference management in two parts: 1) lessons learned from the scenario with overlay deployment of D2D communication with existing cellular networks, 2) lessons learned from the scenario with underlay D2D communication with the existing cellular networks.

Scenarios with overlay deployment of D2D communication with existing cellular networks: To control the co-channel interference among D2D pairs optimal TX policy [152], [154] with efficient link scheduling [151]/channel allocation [153] and Tx power control [153], [155] are of vital importance. Both centralized [150], [151], [152] and distributed [153], [154], [155] approaches can be used for defining optimal TX schemes. However, the main issue with centralized schemes is large feedback overhead. To reduce the feedback overhead, partial and binary feedback can be adopted if a DRL-based TX policy is chosen [152]. DNN can be effectively used in a centralized manner to map geographical location of D2D pairs to optimal link scheduling parameters [151], and for RF or VLC band selection to mitigate CCI [150].

RL-based schemes such as Duelling DQN is used for distributed TX schemes [154], and distributed DRL are useful instruments for joint channel and power allocation [153]. Also, RNN is beneficial for interference feature extraction to aid the distributed TX scheme [155]. It should be noted that the distributed approach in this scenario underperforms in comparison to a centralized approach [154]. On the other hand, decentralized schemes can still be preferred in the design if the priority is to mitigate the feedback overhead. To evaluate the performance of the proposed approach, the following metrics should be used: sum rate/throughput [151], [152], [153], [154], [155], EE [150], and outage probability [150].

Scenarios with underlay deployment of D2D communication with existing cellular networks: Under this scenario, with D2D communications, a crucial design constraint is that the QoS of cellular users should not degrade due to co-channel interference generated by D2D communication sharing the same spectrum [158], [159], [160], [165]. Other common performance metrics to evaluate co-channel interference mitigation techniques are sum rate [157], [158], [160], [161], [163], [165], [166], spectral efficiency [159], [162], throughput [162], [164], [168], [169], fairness among D2D users [161], [166], EE [163], [164], and channel capacity [167]. Co-channel interference mitigation techniques primarily center on Tx power control and channel

allocation. Some methods exclusively address Tx power control [158], [160], [162], while others solely focus on channel allocation [161], [165]. Additionally, it is also viable to take into account both Tx power control and channel allocation simultaneously [157], [159], [163], [164], [166], [167], [168], [169]. Centralized AI/ML-based approaches for power control include Centralized DQN [159], and DNN-based centralized approach [158]. For spectrum access/channel allocation, a centralized approach can employ DDQN [161]. Furthermore, for joint resource allocation (Tx power control and channel allocation), centralized GNN and CNN [157], and centralized DNN [159] can be employed.

In distributed approaches, to map the location and Tx power, distributed DL can be utilized [162]. For channel allocation, the Actor-Critic-based MADRL technique is a feasible candidate for centralized training and distributed execution process [165]. For joint channel and power allocation design, a distributed DDQN [163] can be adopted. For joint channel and power allocation, the MADRL technique is widely implemented with different algorithms such as DQN for Channel allocation and DDPG for Power allocation [166], federated learning-aided DQN [164], and spiking neural network with co-operated DRL (actor-critic and PPO-based) algorithms [169]. It should be noted that a co-operated distributed approach will result in better performance when compared to a fully distributed approach regarding throughput, power consumption, and convergence rate [164], [169].

2) CHALLENGES AND OPEN ISSUES

In D2D communication networks, direct communication between devices offloads traffic from the cellular BSs and enhances the network's spectral efficiency, EE, and coverage capacity. Achieving all these benefits needs efficient interference management techniques via proper scheduling, spectrum allocation, and Tx power control. Interference mitigation in in-band overlay and underlay D2D communication networks using AI/ML models presents several challenges and open issues as highlighted below.

ML models for interference detection and mitigation in D2D communication networks should be adaptive to real-time variation in user density, traffic load, and network conditions by predicting/learning the interference pattern in changing environments and dynamically reconsidering the interference mitigation strategies with low latency. While considering the centralized approach, sufficient data should be collected to train the model for its generalized applicability to diverse D2D communication environments. Furthermore, feedback overhead is critical in applying the centralized approach in D2D communication networks considering the energy-constrained end devices.

For a decentralized approach, EE and computational complexity are important parameters to measure its efficacy for energy-constrained devices. The state-of-the-art calls for ML models that consider QoS requirements, such as throughput latency and outage probability, and that dynamically

adapt to meet contextual requirements. ML models are expected to consider the coexistence of cellular networks and D2D communication while handling directional interference, optimal spectrum usage, and the scalability concerns of continuously increasing user density and demands. Lastly, the design of robust ML models against privacy and security threats by considering authentication and privacy-conserving techniques along with interference mitigation approaches in self-governing as well as cellular under-laid D2D communication N/w remains an open issue.

F. INTERNET-OF-THINGS (IOT) COMMUNICATION NETWORKS

1) LESSONS LEARNED

In the IoT communication domain, a vast variety of network systems are considered along with corresponding state-of-the-art interference mitigation solutions. Key points regarding different types of network scenarios are as follows: In a cognitive IoT network, the main aim is to mitigate the primary user (PU) interference and minimize BER at the secondary user's receiver [177], [178], [179]. In such a setup, to support the SU receiver, PU interference is clustered using GMM for single PU [177], Affinity Propagation (AP) for multi-PU interference [178], and HMM-AP for asynchronous cognitive-IoT case [179]. For an IoT network utilizing D2D communications, resource allocation in terms of Tx power and channel allocation is done either in a centralized (eg. using GNN and CCN [157]) or distributed manner (eg. using MADRL-DDQN [190]). Specifically, for LoRa-enabled IoT communication networks, a DNN-based algorithm can be used to assign suitable spreading factors to IoT devices for uplink communication to mitigate the effect of interference and enhance the packet success ratio [182]. Random channel access probability for Slotted-ALOHA protocol can be controlled with RL [186] or DRL [189] techniques for the IoT devices to access the shared channel for uplink communication.

Tx power allocation to IoT devices is also an important factor in controlling the interference, which can be achieved by RL or DRL [188] techniques both in centralized [188] and distributed manner. The utilization of interference modeling [180] and interference prediction [175] can play a crucial role in interference mitigation. This can be effectively accomplished through the application of the DBSCAN approach [180] for high interference detection accuracy and an LSTM-based approach [175] for minimizing prediction errors. To mitigate the NB-IoT interference on LTE, sparse learning is effective for sparse cross-entropy minimization [181]. Distributed channel selection by each IoT device using the MAB-RL technique is also a possible solution to mitigate the interference and increase the communication success probability [176]. Some other commonly used objective functions to evaluate the network performance in IoT communication domain are throughput/sum rate [157], [183], [184], [188], [189], spectral efficiency [181], [186], [190], and low latency [190].

2) CHALLENGES AND OPEN ISSUES

Employing ML techniques for interference mitigation via Tx power control, channel selection, random channel access control, and interference modeling/prediction in IoT communication networks unveils various challenges and open issues owing to the unique characteristics of IoT environments. Following are some open issues and opportunities for the researchers in this domain. IoT N/w have applications in various fields such as smart cities, smart health, smart transport, smart energy, and smart industry consisting of devices with diverse characteristics in terms of energy and computational capability constraints and different communication technologies such as LoRa, WiFi, SigFox, ZigBee, NB-IoT, and 5G-NR and so on. Hence, for effective interference mitigation, ML models must consider the diverse characteristics of IoT devices and communication technologies.

Since IoT devices have limited energy and computation resources, designing low-complexity ML algorithms and utilizing edge computing with distributed approaches for interference mitigation and resource allocation need exploration. It is worth noting that ML models must be capable of providing interference mitigation solutions in real time to fulfill the latency constraints of time-critical IoT applications and practical implementation of these algorithms in real-world scenarios. Not to mention the adaptability of the designed models to the diverse network topologies and varying traffic loads with mobile devices in IoT communication networks while considering the interference mitigation strategies.

Scalability is a unique challenge in IoT networks where a large number of devices can be connected within one network. Therefore, ML models designed for interference detection and mitigation must handle scalability challenges in IoT while being robust to preserve the security and privacy of IoT communication networks. Lastly, ML models should consider integrating IoT communication with cutting-edge technologies such as mmWave-MIMO, IRS, and THz communication, implementing corresponding interference mitigation strategies.

G. MMWAVE-MIMO COMMUNICATION NETWORKS

1) LESSONS LEARNED

In mmWave-MIMO-based communication networks, the optimal design of beamforming vectors is critical to avoid inter-beam/intra-cell and inter-cell interference and improve the network capacity performance. Various ML-based algorithms can solve beamforming and beam selection problems with low computation complexity. Following are the important lessons learned from ML-based state-of-the-art works in the mmWave-MIMO-based communication domain: In the mmWave-MIMO communication network, the beam allocation problem to multiple users can be treated as a multi-label classification problem which can be solved by a CNN-based algorithm with very low computational complexity [202]. Also, a DNN-based framework

can facilitate RSS-based mapping for beam selection for multiple users to reduce the initial access time [200]. In a mmWave-MIMO communication network, for beamforming design [207], beam width and beam direction control [201], Tx power control [201], [207], and interference coordination, DNN [201] and DRL-based [207] algorithms can be effective for sum-rate maximization. For a high-mobility user scenario, fast-beamforming and multiple AP association can be solved by using a centralized DRL-based approach and distributed federated-DRL-based approach to maintain the QoS of mobile users and enhance the EE of the network [209].

For the D2D-mmWave-MIMO communication network, a GNN-based algorithm can be competent at joint beam selection and link activation to control the interference between communication pairs and maximize the sum rate of the network [199]. In UAV-assisted MIMO-based communication networks, the Gaussian process-based ML technique can be used for channel estimation [197] and Q-learning can be used for hybrid beamforming [204] and beam steering [205] to improve the sum rate of the network. For a cell-free environment, DRL-based dynamic clustering of users and APs and hybrid beamforming design is proposed to mitigate the inter-UE and inter-cluster interference in a distributed manner [208]. To improve the BER as a signal detection performance metric at the receiver side of the MIMO communication system, a DL-aided signal detector for MIMO-OFDM is a suitable solution [210]. Also, the joint use of an MLD and a deep CNN for the MIMO system having correlated interfering signals [211], and the joint use of a linear detector and a deep CNN for the MIMO system over correlating noise environment [212] have been proved very effective in comparison to conventional detectors.

2) CHALLENGES AND OPEN ISSUES

Utilizing ML techniques for efficient beamforming, beam selection, channel estimation, and Tx power control for interference coordination has been proven effective in enhancing the sum rate performance of mmWave-MIMO communication networks. However, the use of ML techniques in this domain poses several challenges and open issues, as well as opportunities, which are listed as follows.

First, mmWave-MIMO communication possesses complex channel characteristics with high path loss and directional beams and sparse channel knowledge. Therefore, accurate channel estimation, beamforming design, and beam allocation to multiple users make the design of ML-based techniques complex. A diverse and large amount of data collection is required for a classification-based ML model to effectively solve the beamforming and beam selection/allocation problem to mitigate the interference and enhance the network capacity. However, transfer learning and synthetic data generation can be exploited to deal with data scarcity issues. The next challenge is to ensure the compatibility of the developed ML models with many beams and beamforming configurations while optimizing

the interference mitigation strategies to deal with scalability challenges. Besides, ML models must be designed adaptively to varying network conditions such as channel conditions, traffic load, and QoS requirements by dynamically optimizing the beam management, users' clustering, and interference mitigation strategies. In a mmWave-MIMO network, high energy consumption is a serious issue. Hence, the development of ML models to jointly address EE and interference mitigation strategies in mmWave-MIMO networks remains an open issue for researchers in this field.

To deal with correlated noise and interfering signals during signal detection at the receiver side in a mmWave-MIMO communication system, some advanced ML-based techniques should be explored to enhance the BER performance of the receiver. With this in mind, the state-of-the-art is seeking ML models that are capable of achieving efficient interference alignment in mmWave environments through which interference from multiple transmitters is deliberately aligned in an orthogonal subspace to the desired signal, which results in improved signal detection quality at the receiver. Last but not least, designing ML models that meet the latency requirements of real-time applications and are robust enough to preserve the security and privacy of mmWave-MIMO communication networks is another research direction in this field.

H. TERA-HERTZ (THZ) COMMUNICATION NETWORKS

1) LESSONS LEARNED

In THz communication networks, along with the adaptive and dynamic design of beamforming and beam selection, making the physical layer flexible in choosing PHY parameters such as bandwidth and modulation schemes is of critical importance to avoid inter-symbol/inter-carrier interference and tackle THz band challenges such as molecular absorption loss, frequency-selectivity, and distance-dependent bandwidth change to enhance the network capacity performance. Various ML-based algorithms are capable of solving these challenges of THz communication systems. Below are the important lessons learned from ML-based state-of-the-art works in the THz communication domain:

A Federated DRL-based framework with a distributed DDGP model at different BSs can be employed for THz beam selection having limited CSI to mitigate inter-cell interference and maximize the throughput in multiple BS network scenarios. This approach can be adopted instead of the conventional ZF and MMSE-based beamforming techniques when improved computational complexity and network throughput are aimed at [236]. The beam selection problem in the THz communication system can be solved as a multi-class classification problem. Without seeking sophisticated models, it is viable to go with Random Forest ML algorithm for a viable tradeoff between the computational complexity of hybrid beam forming and sum-rate maximization [223]. Combined use of RF, mmWave, and THz spectrum bands can serve the data streaming request of users in a multi-drone environment. A specific spectrum

band can be selected to serve a particular user based on its channel condition. A combined echo-state learning and RL-based framework would be an effective approach to maximize the throughput via proper flight control of drones, user association, and spectrum band selection [231].

Making the physical layer flexible in terms of selecting the Tx power [229], bandwidth [226], [229], and modulation scheme [229] for Tx signals is an effective solution to tackle the communication challenges in THz-band such as molecular absorption loss, frequency-selectivity, and distance-dependent bandwidth change. A CNN-based bandwidth and modulation classification technique can introduce such flexibility to the physical layer where Tx and Rx can exchange their PHY parameters (bandwidth and modulation) without coordinating with each other [226]. Further, a DNN-based adaptive sub-band bandwidth allocation and power control can achieve flexible TX and enhanced sum rate of the THz communication network [229]. In addition, Re-configurable Intelligent Surfaces (IRS) can be effective in tackling the propagation attenuation and atmospheric absorption losses and enhancing the coverage range in THz-band [224], [230], [234]. Specifically, the joint design of analog beamforming at RISs and digital beamforming at BS bridged with a DRL-based approach can improve the THz communication network's total throughput and coverage range [234].

On the receiver side, learning-aided demodulation can effectively handle the phase noise, phase shift, and I/Q imbalance-related challenges in the THz communication system. In this direction, a multi-output deep feed-forward NN for nonlinear mapping between the Rx symbols and the demodulation likelihood information vector can be preferred instead of MMSE equalization or SVM-based schemes to aim for low BER performance for THz communication systems under severe hybrid distortions [228]. The use of the THz communication band for task offloading is also encouraged [235], [237]. Specifically for UAV-assisted communication systems, to optimize UAV placement/trajectory, computation offloading, and resource allocation, DDQN [235], DDGP [235], and PPO-based [237] DRL algorithms can provide better system capacity [235] and network utility [237] performance. Multi-arm bandit-based RL algorithms are also effective tools for beam training at the Tx side to enhance spectral efficiency [232] and for time-domain interference mitigation at the Rx side to enhance BER performance [233] in THz communication systems.

2) CHALLENGES AND OPEN ISSUES

The utilization of ML techniques for adaptive hybrid beamforming, beam selection, and physical layer flexibility at Tx and Rx have been proven to be effective in dealing with THz band challenges and interference management in THz communication networks. However, the development of ML-based algorithms for interference mitigation in the THz communication domain consists of several unique challenges and open issues which are introduced as follows.

First of all, considering the beam selection problem as a multi-class classification problem needs sufficient diverse labeled training data for THz communication environments so that the designed ML model can adapt to the real-time variation in channel conditions due to distance selectivity and frequency selectivity of path loss. Accurate channel modeling, estimation, or prediction is quite complex in THz communication scenarios due to high atmospheric absorption, diffraction, reflection, scattering, and shadowing effects. Thus, designing ML models that overcome this challenge in THz communication networks remains an open issue. The ML models must be dynamic in changing the physical layer parameters such as modulation scheme, Tx power, and BW allocation to adapt to the time and distance-dependent variations in BW and propagation characteristics in real-world applications with ultra-low latency.

Next, the ML model should efficiently handle the inter-carrier interference due to phase noise, beam squint, and spatial frequency wideband problems via adaptive frequency-dependent beamforming and beam-combining at the transmitter side. On the receiver side, employing time-domain interference mitigation and learning-aided adaptive received signal mapping with demodulation schemes is crucial to deal with phase noise and beam-squint issues in the THz communication domain. Sixth, the use of the THz band in mobile environments such as autonomous vehicles is feasible with frequent handover and optimal beam steering and beamforming designs. The use of ML techniques in such scenarios may be promising to deal with these issues.

Integrating RIS with the THz communication system is crucial for increasing the coverage range, where the ML model should be efficient in beamforming while taking care of the unique channel characteristics of THz communication. While ML models must be energy-efficient to ascertain the sustainability of THz communication networks in resource-constrained scenarios, it is also worth noting that the complexity of the ML model should be compatible with the hardware constraints of the THz communication system along with being robust to maintain the security and privacy of the THz communication networks.

I. NOMA-BASED COMMUNICATION NETWORKS

1) LESSONS LEARNED

In NOMA-based communication systems, the optimal design for the SIC process at the Rx side and efficient resource allocation (channel and power allocation) at the Tx side are pivotal to mitigate the intra-cell, inter-cell, and inter-user interference and improve the network performance. The important lessons learned and key takeaways from the state-of-art contributions of ML-based techniques in these aspects of NOMA-based downlink and uplink communication scenarios are expressed as follows:

NOMA-based downlink communication scenario: The detection of modulation order of interfering signal can be quite useful to support the SIC process at the Rx side which in turn enhances the network throughput of the

NOMA-based communication system. A feature extraction method followed by a logistic regression-based classification algorithm can achieve modulation order detection of interfering signals [243]. To fulfill the QoS demand of users, mitigate the interference, and enhance the overall network performance, users association [248], users clustering [244], channel allocation [246], [248], and power allocation [246], [248] are critical factors.

For channel allocation, regression-based [248], Deep RNN [246], attention-based DRL [251] algorithms can be employed to achieve high EE [246], [248] and sum-rate maximization [246], [251]. For power allocation to user groups in NOMA-based communication, DNN [248], Deep RNN [246], [247], SAC-RL [250]-based algorithms can be adopted as their pros and cons have been discussed earlier. To tackle the imperfect SIC problem and to minimize the MSE between the desired and decoded signals of the users, a Feed-forward NN-based [249] technique is proposed for the SIC decoding of multiple users. In a multi-cell environment to enhance the sum rate performance, a cluster-free-NOMA framework is a viable solution for inter-cell and intra-cell interference mitigation by utilizing a distributed and coordinated auto-learning GNN-based approach for beamforming and cluster-free SIC decoding in different cells [245].

NOMA-based uplink communication scenario: To improve the BER performance with lower computational complexity, a DNN-based algorithm can be employed to detect the optimal sequence for SIC detection [253]. Usually, received signal information and Tx power information are used for training the proposed DNN. Also, a DNN can be used at each SIC step to decode each user's signal [254], [255]. To maximize the sum rate of NOMA-based uplink communication, efficient resource allocation, in terms of channel [188], [257] and power allocation [257] is required. ML-based techniques such as MAB-RL [257], SARSA [188], and DRL [188] can be exploited to obtain better performance than conventional resource allocation approaches.

To maximize the number of successfully served users having different QoS requirements, an actor-critic learning-based DRL algorithm can be exploited to optimize the dynamic user access and optimal decoding order for the imperfect SIC process [258]. For an uplink grant-free NOMA system, to reduce the signaling overhead, latency, and collisions of uncoordinated access and to improve the system throughput, first, subchannels and users are divided into different clusters. Then, a DRL-based algorithm is used for channel selection to clusters and power allocation to users [259]. Besides, for a grant-free NOMA system, LSTM-based network [252] and model-driven DL [256] algorithms can be trained by using the pilot signals so that they enable multi-user signal detection. Adopting these models for channel estimation and multi-user signal detection can lead to improved BER performance in comparison to the conventional SIC process [252], [256].

2) CHALLENGES AND OPEN ISSUES

ML techniques are widely being investigated for user clustering, user scheduling, power allocation, and SIC decoding sequence detection for interference mitigation and enhancing the sum capacity in NOMA-based N/w. However, there are some open issues and challenges in the development of ML-based algorithms for interference mitigation in the NOMA-based communication domain which are presented below.

ML-based dynamic user clustering, user scheduling, channel allocation, and power allocation are required to adapt to changing channel conditions with low latency for a successful implementation in real-time scenarios. Since correct channel estimation is crucial for optimal power allocation and user scheduling in a NOMA-based communication system, the ML models must be competent in estimating the channel condition with high accuracy in a dynamic wireless environment. Besides, there is a need to resolve the issue of high communication overhead for channel estimation for the practical deployment of the ML model. Along with the objective of sum rate maximization, ML algorithms should focus on user fairness while allocating power resources to different users. In NOMA systems, a user with good channel conditions may cause high interference to users with poor channel conditions while communicating on same spectrum resources. Hence, optimal power allocation is vital in NOMA to mitigate intra-tier and inter-user interference.

Given that SIC decoding is an essential part of the NOMA receiver system, diverse and sufficient data should be collected to train ML models for detecting the sequence for SIC decoding and for being adopted in real-time implementations. As NOMA is being integrated into various technologies such as mmWave-MIMO, IoT, and massive machine-type communication where BSs and UEs have different demands and characteristics, it is crucial to develop ML models for resource allocation and interference management for NOMA-based communication in diverse network architectures by considering specific characteristics of different technologies. Lastly, ML algorithms to realize NOMA techniques should be energy efficient and less computationally complex for real-time practical implementation. However, an ML model must be robust enough to maintain the privacy and security of NOMA-based networks.

J. LLMS FOR INTERFERENCE DETECTION AND MITIGATION

The evolution of wireless N/w towards 5G and 6G paradigms brings unprecedented advancements in speed, connectivity, and data handling capabilities. However, these benefits also introduce complex interference challenges. Large Language Models (LLMs) like GPT-4 offer innovative solutions for detecting and mitigating interference in these advanced networks. LLMs can analyze vast amounts of diverse data in 5G and 6G environments, where network architectures are more dynamic and heterogeneous, from enhanced Mobile Broadband (eMBB) to massive Machine

Type Communications (mMTC). By training on specialized datasets, including network traffic, sensor logs, and real-time user equipment data, LLMs can develop nuanced understandings of interference patterns unique to 5G/6G networks [260], [261].

The application of LLMs in 5G/6G networks for interference detection and mitigation represents a forward-thinking approach to addressing the complexities of modern wireless communications. Their ability to process vast and varied datasets, adapt to evolving network scenarios, and provide actionable insights makes them invaluable. However, leveraging these benefits fully demands addressing challenges related to reliability, computational efficiency, and seamless integration with 5G/6G technologies. Achieving the ultra-reliability and low latency requirements of 5G/6G is critical, particularly for LLMs handling real-time interference detection and mitigation [262]. The scalability of LLMs in vast 5G/6G networks is a concern, requiring efficient algorithms and distributed processing approaches. Therefore, integrating LLMs with other 5G/6G technologies like Massive MIMO, mmWave, and edge computing requires careful design to ensure seamless operation. With the increased number of connected devices, ensuring data privacy while utilizing LLMs becomes more challenging and essential. The dynamic nature of 5G/6G networks necessitates continuous learning and adaptation capabilities in LLMs to stay effective [263].

XII. CONCLUSION

To achieve the goals of next-generation wireless networks such as global wireless coverage, sustainable energy-efficient networking, massive connectivity, ultra-high reliability, and low latency, various enabling technologies exist and complement each other such as UAV-assisted networks, vehicular networks, HCNs, IoT, D2D, mmWave-MIMO, NOMA, and THz communications. However, every enabling technology has its own specific characteristics to be taken care of for efficient deployment of that technology. In that line, interference management is a critical issue to tackle for every technology for proficient utilization of scarce bandwidth and power resources and to improve the overall network performance and QoE of users. Integration of AI/ML techniques in 5G networks is inevitable and being widely investigated to deal with interference management in complex and dynamic environments. This survey paper has provided a detailed study of AI/ML-based works for intentional jamming and unintentional interference management focusing on the physical layer perspective. For tackling intentional interference, the key characteristics of different types of jamming attacks have been discussed, and then a detailed review of AI-enabled techniques has been provided for jamming detection and mitigation in various wireless communication scenarios. For intentional interference, every enabling technology of 5G networks towards 6G is considered individually discussing their unique features, characteristics, and interference effects followed by an in-depth review of AI-enabled interference

management works along with the qualitative comparison of the different ML-based approaches in terms of performance and complexity. Furthermore, the challenges associated with developing and deploying AI/ML-based interference mitigation techniques in different enabling technologies towards 6G networks are presented exhibiting the potential future directions of using AI/ML-enabled techniques to enhance the performance of future-generation wireless networks.

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POONAM LOHAN (Member, IEEE) received the B.Tech. degree in electronics and communication engineering (ECE) from the Guru Jambheshwar University of Science and Technology, Hisar, India, in 2009, the M.Tech. degree (Gold Medalist) in ECE from the National Institute of Technology Kurukshetra, India, in 2011, and the Ph.D. degree in Electrical Engineering from the Indian Institute Of Technology Delhi, New Delhi, India, in November 2019. She was a Postgraduate Trainee with Magneti Marelli UM Electronics Ltd., Gurugram, India, from April 2011 to April 2012. She was an Assistant Professor with the Punjab Engineering College (Deemed to be University), Chandigarh, India, from Jan 2020 to Jan 2023. She is currently associated with the Smart Connected Vehicles Innovation Centre, University of Ottawa as a Postdoctoral Researcher. Her research interest pertains to AI/ML-enabled NextG wireless networks, heterogeneous cellular networks, vehicular networks, and unmanned aerial vehicle (UAV)-assisted networks in terms of coverage probability, capacity, and secrecy data rate analysis, wireless channel modeling, interference management, resource-allocation/scheduling for utility maximization of the network, VNF placement, and SFC provisioning. She received the Best Faculty Award from the Punjab Engineering College (Deemed to be University) in September 2022. She is an Active Reviewer of the IEEE INTERNET OF THINGS JOURNAL, IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, IEEE COMMUNICATION LETTERS, IEEE GLOBECOM, IEEE ICC, and IEEE VTC conferences.



BURAK KANTARCI (Senior Member, IEEE) received the Ph.D. degree in computer engineering. He is a Full Professor and the Founding Director of the Smart Connected Vehicles Innovation Centre and the Next Generation Communications and Computing Networks Research Lab, University of Ottawa, Ottawa. He is the author/co-author of 250+ publications in established journals and conferences, and 15 book chapters. Continuously, he listed among the top-cited scientists in telecommunications and networking based on the data reported by Stanford University since 2020, and since 2021, based on data collected from Microsoft Academic Graph, research.com has listed him among Canada's top computer scientists. He holds an Exemplary Editor Award from IEEE Communications Surveys and Tutorials in 2021, and multiple best paper awards from various conferences, most recently from IEEE Globecom2021, Wireless World Research Forum 2022, and IEEE ICC2023 and IEEE VCC2023. He is a recipient of the Minister's Award of Excellence from the Ontario Ministry of Colleges and Universities in 2021. He is the recipient of 2023 Technical Achievement Award of IEEE ComSoc Communications Software Technical Committee. He was a Distinguished Speaker of the Association of Computing Machinery from 2019 to 2021. He is currently a Distinguished Lecturer of the IEEE Communications Society and IEEE Systems Council. He has been a keynote/invited speaker or a panelist in 40 events. From 2019 to 2020, he chaired the Communications Systems Integration and Modeling Technical Committee of the Institute of Electrical and Electronics Engineers. He has been the general chair, the program chair or the track chair in 30+ international conferences. He is an Editor of the IEEE COMMUNICATIONS SURVEYS AND TUTORIALS and IEEE INTERNET OF THINGS JOURNAL, and an Associate Editor for IEEE NETWORKING LETTERS and for *Vehicular Communications* (Elsevier).



MOHAMED AMINE FERRAG (Senior Member, IEEE) received the bachelor's, master's, Ph.D., and Habilitation degrees in computer science from Badji Mokhtar - Annaba University, Annaba, Algeria, in 2008, 2010, 2014, and 2019, respectively. He served as an Associate Professor with the Department of Computer Science, Guelma University, Algeria, from 2014 until 2022. Concurrently, he held the position of Senior Researcher with the NAU-Lincoln Joint Research Center of Intelligent Engineering,

Nanjing Agricultural University, China, from 2019 to 2022. As of 2022, he is a Lead Researcher with the Artificial Intelligence and Digital Science Research Center, Technology Innovation Institute, Abu Dhabi, UAE. His scholarly output includes over 140 papers published in international journals and conference proceedings. He has spearheaded numerous projects in research and development, fostering collaborative ties with academic institutions in the U.K., Australia, USA, Canada, and China. His contributions to the field include the creation of two cybersecurity datasets, namely, the Edge-IIoT dataset and the FormAI dataset, which have become essential resources for AI researchers worldwide. His research is primarily focused on a spectrum of topics within the cyber security domain, including wireless network security, network coding security, applied cryptography, blockchain technology, generative AI, software security, and the application of AI in cyber security. His academic contributions have been recognized with the IEEE TEM Best Paper Award in 2021, the Scopus Algeria Award in 2022, and many best paper conference awards. He has consistently been named on Stanford University's list of the world's top 2% of scientists four times from 2020 to 2023. He also contributes to the academic community as an Associate Editor for prestigious journals, such as the IEEE INTERNET OF THINGS JOURNAL and *ICT Express* (Elsevier).



YI SHI (Senior Member, IEEE) is a Research Associate Professor with Commonwealth Cyber Initiative, Virginia Tech, where he is also a Research Associate Professor of Electrical and Computer Engineering (by Courtesy). Prior to joining Virginia Tech, he was a Senior Lead Research Scientist with BlueHalo Inc. His research focuses on algorithm design, optimization, and machine learning for NextG wireless networks. He was a recipient of Test of the Time Paper Award at IEEE INFOCOM 2023, the Best Paper Award

at IEEE INFOCOM 2008, the only Best Paper Award Runner-Up at IEEE INFOCOM 2011, the Best Student Paper Award at ACM WUWNet 2014, and the Best Paper Award at IEEE HST 2018. He was an Exemplary Editor of IEEE COMMUNICATIONS SURVEYS AND TUTORIALS in 2014, and a Distinguished TPC Member for IEEE INFOCOM in 2021. He is an Editor of IEEE COMMUNICATIONS SURVEYS AND TUTORIALS and IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING. He served as the TPC chair for IEEE and ACM symposiums, tracks, and workshops.



NORBERT TIHANYI (Member, IEEE) received the B.Sc. degree in security engineering, the first M.Sc. degree in safety engineering, the second M.Sc. degree in IT engineering (with Hons.), and the Ph.D. degree in information science and technology from Eötvös Loránd University, Budapest, Hungary, in 2020. He is a Public Body member of the Hungarian Academy of Science. He is currently a Lead Researcher with Technology Innovation Institute, Abu Dhabi, UAE. His research interests include cryptanalysis, security of embedded devices, and cryptography-related prime number theory.